MOCCA 2025
Fundamentals of Character Animation and Motion Control

Lecture 07

Physics-based Simulation and Articulated Rigid Bodies 铰接刚体

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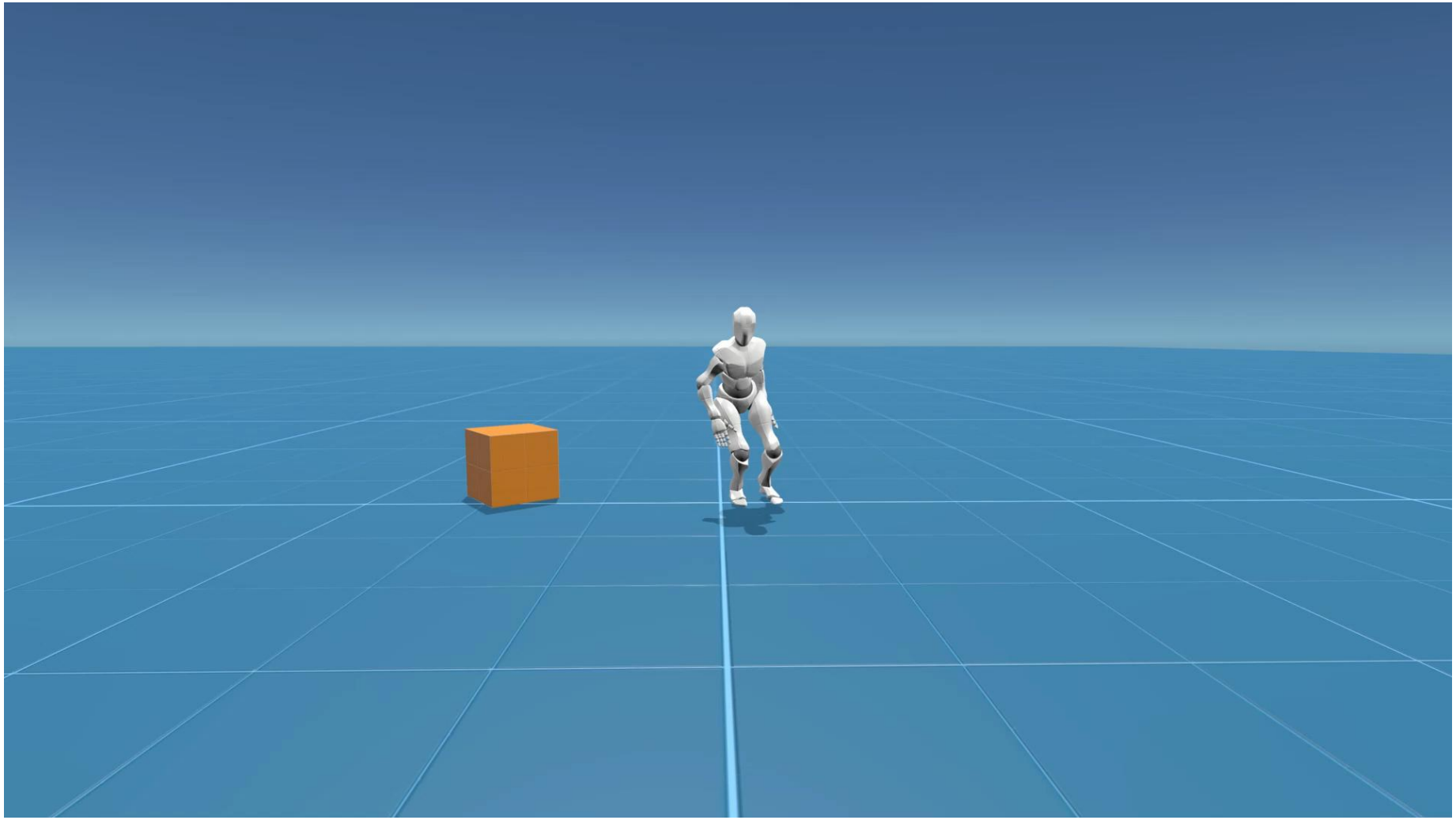
Problems of Kinematic Methods

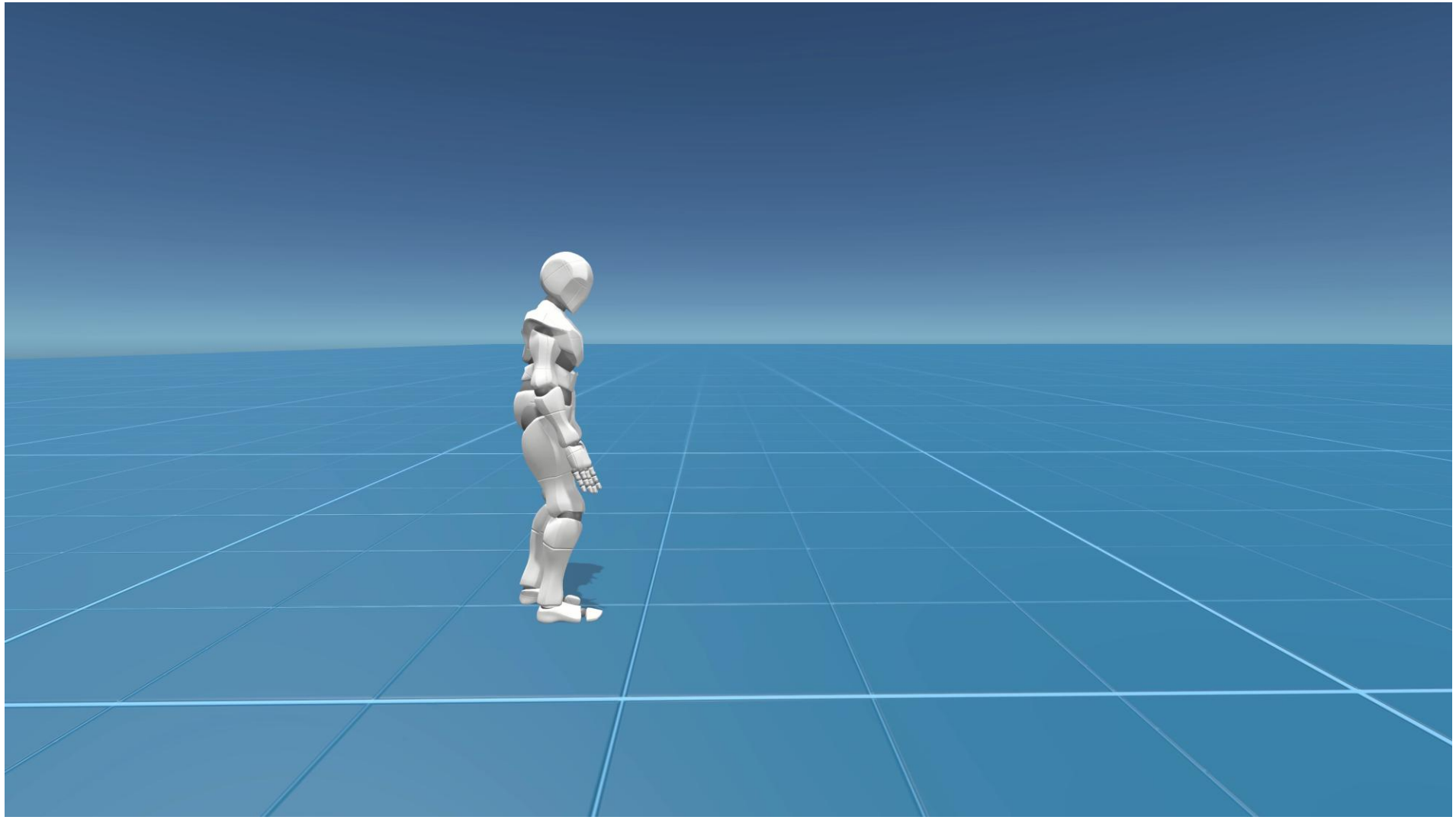
- Interaction with the environment

Physics-based Character Animation



[ControlVAE – Yao et al. 2022]

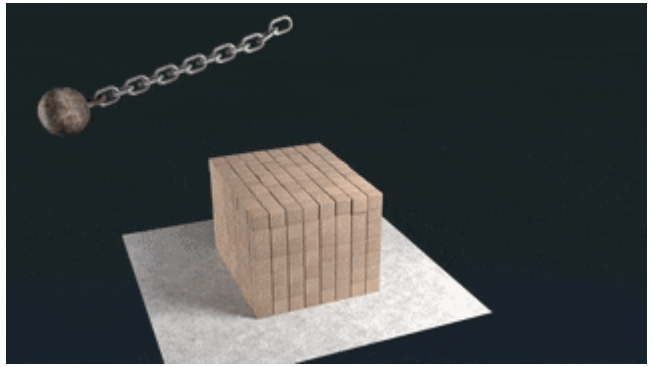
Physics-based Character Animation

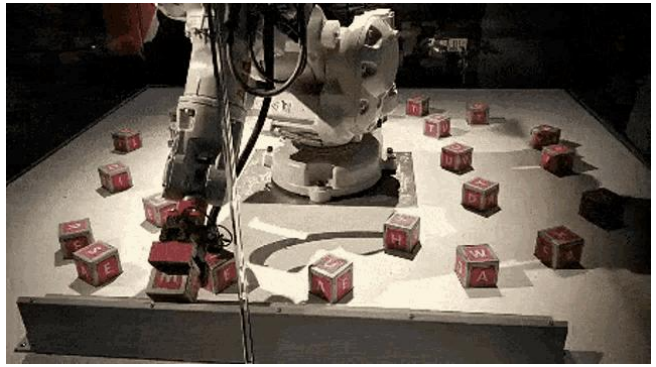


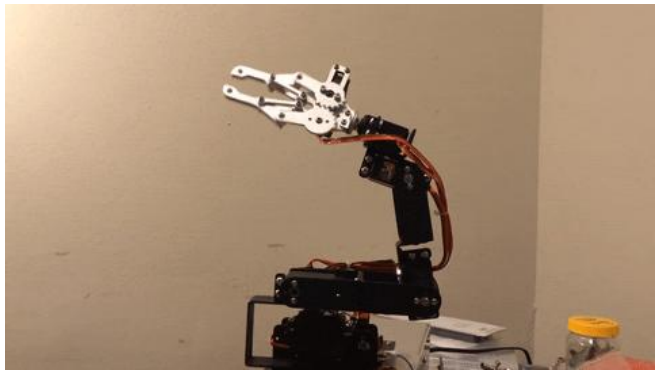
[ControlVAE – Yao et al. 2022]

Outline

- Simulation Basis
  - Numerical Integration: Euler methods
- Equations of Rigid Bodies
  - Rigid Body Kinematics
  - Newton-Euler equations
- Articulated Rigid Bodies
  - Joints and constraints
- Contact Models
  - Penalty-based contact
  - Constraint-based contact

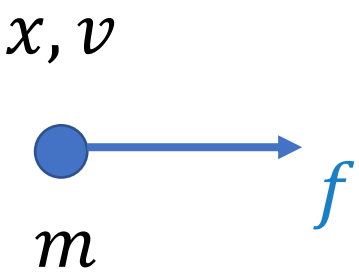






https://www.cs.cmu.edu/~baraff/sigcourse/

Dynamics of a Particle



Dynamics of a Particle

$x(t = 0)$
$v(t = 0)$

$$x(t = 10) = ?$$

Dynamics of a Particle

$x(t), v(t)$

$$f = ma$$

Dynamics of a Particle

$x(t), v(t)$

$$f = ma$$

$$a = \dot{v}$$

$$v = \dot{x}$$

Dynamics of a Particle

$x(t), v(t)$

$$f = ma$$

$$a = \dot{v}$$

$$v = \dot{x}$$

$$a = f/m$$

$$v = v_0 + \int_{t_0}^{t} a dt$$

$$x = x_0 + \int_{t_0}^{t} v dt$$

Dynamics of a Particle

$x(t), v(t)$

$$f = ma$$

$$a = \dot{v}$$

$$v = \dot{x}$$

$$\Rightarrow$$

$$a = f/m$$

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2} a t^2$$

Dynamics of a Particle

$x(t), v(t)$

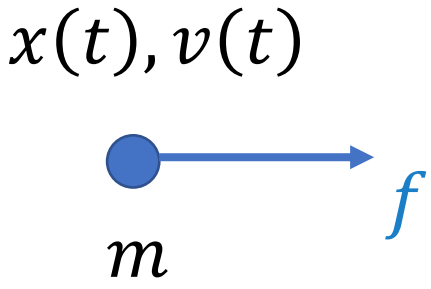
$$x(t=10) = x_0 + 10v_0 + 50\frac{f}{m}$$

$$f = ma \qquad a = f/m$$

$$a = \dot{v} \qquad \Rightarrow \qquad v = v_0 + at$$

$$v = \dot{x} \qquad x = x_0 + v_0 t + \frac{1}{2}at^2$$

Dynamics of a Particle



$x(t = 10) = ?$

$$f = ma$$

$$a = \dot{v}$$

$$v = \dot{x}$$

$$a = f(x, v, t)/m$$

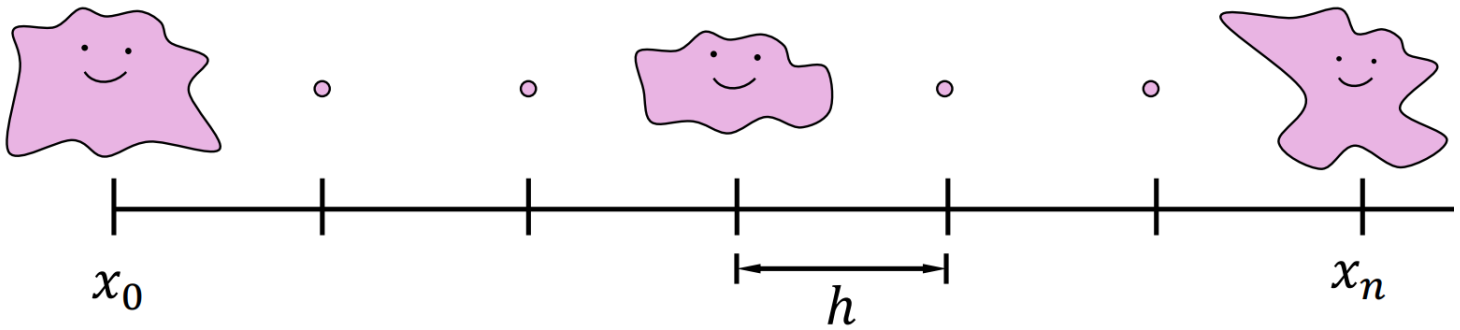
$$v = v_0 + \int_{t_0}^{t} a dt$$

$$x = x_0 + \int_{t_0}^{t} v dt$$

Temporal Discretization 时间离散化

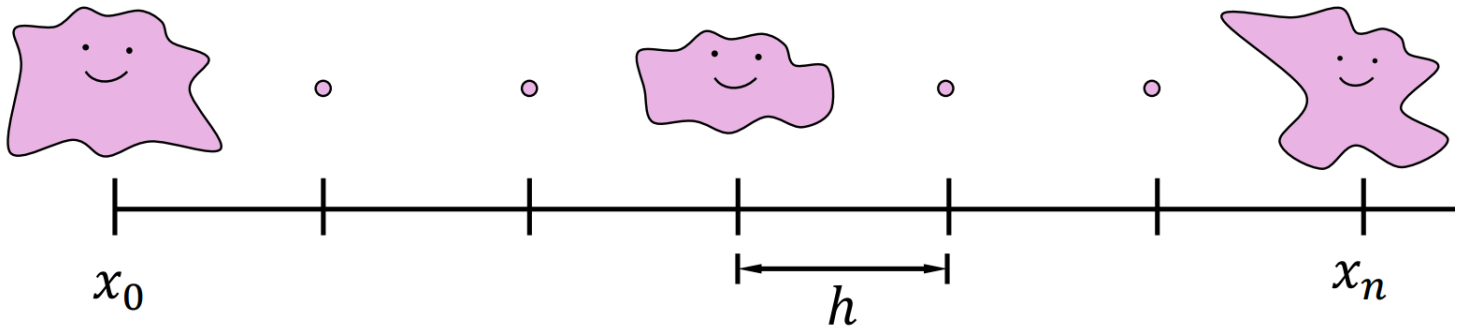
$$x = x(t)$$

$$x_n = x(t_n) \qquad t_n = nh$$



Simulation time step

Temporal Discretization



$$a = f(x, v, t)/m$$

$$v = v_0 + \int_{t_0}^{t} a dt$$

$$x = x_0 + \int_{t_0}^{t} v dt$$

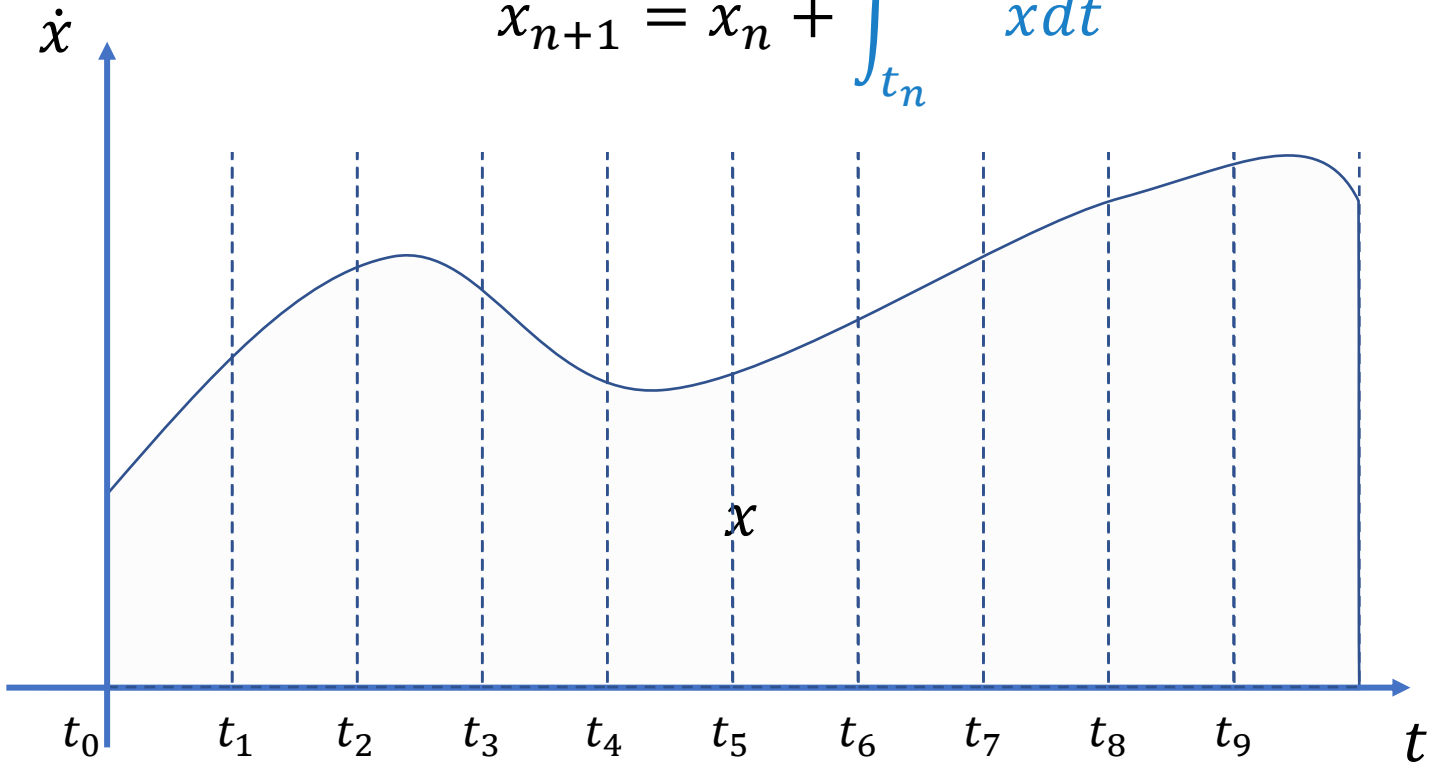
$$a = f(x, v, t)/m$$

$$v_{n+1} = v_n + \int_{t_n}^{t_{n+1}} a dt$$

$$x_{n+1} = x_n + \int_{t_n}^{t_{n+1}} \mathrm{v} dt$$

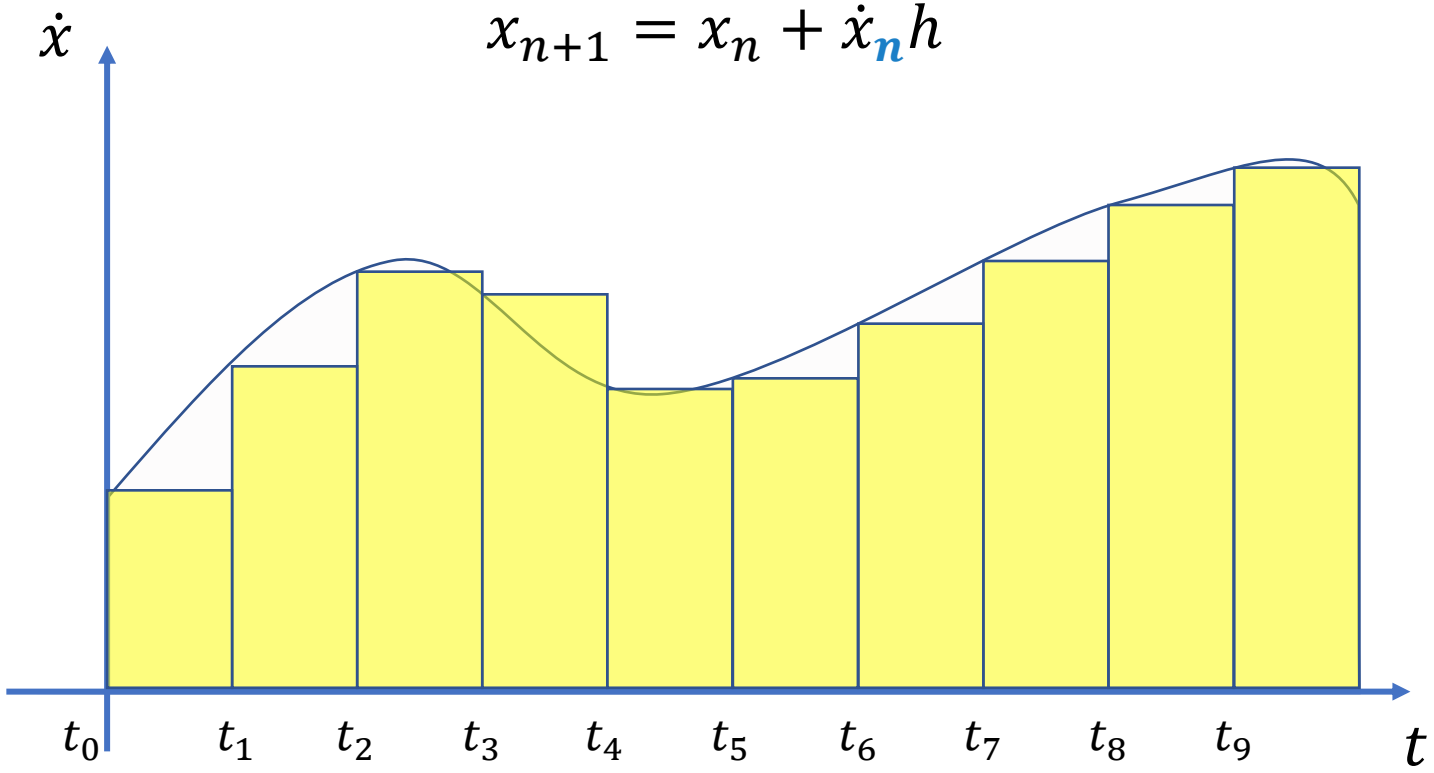
Numerical Integration

$$x_{n+1} = x_n + \int_{t_n}^{t_{n+1}} \dot{x} dt$$



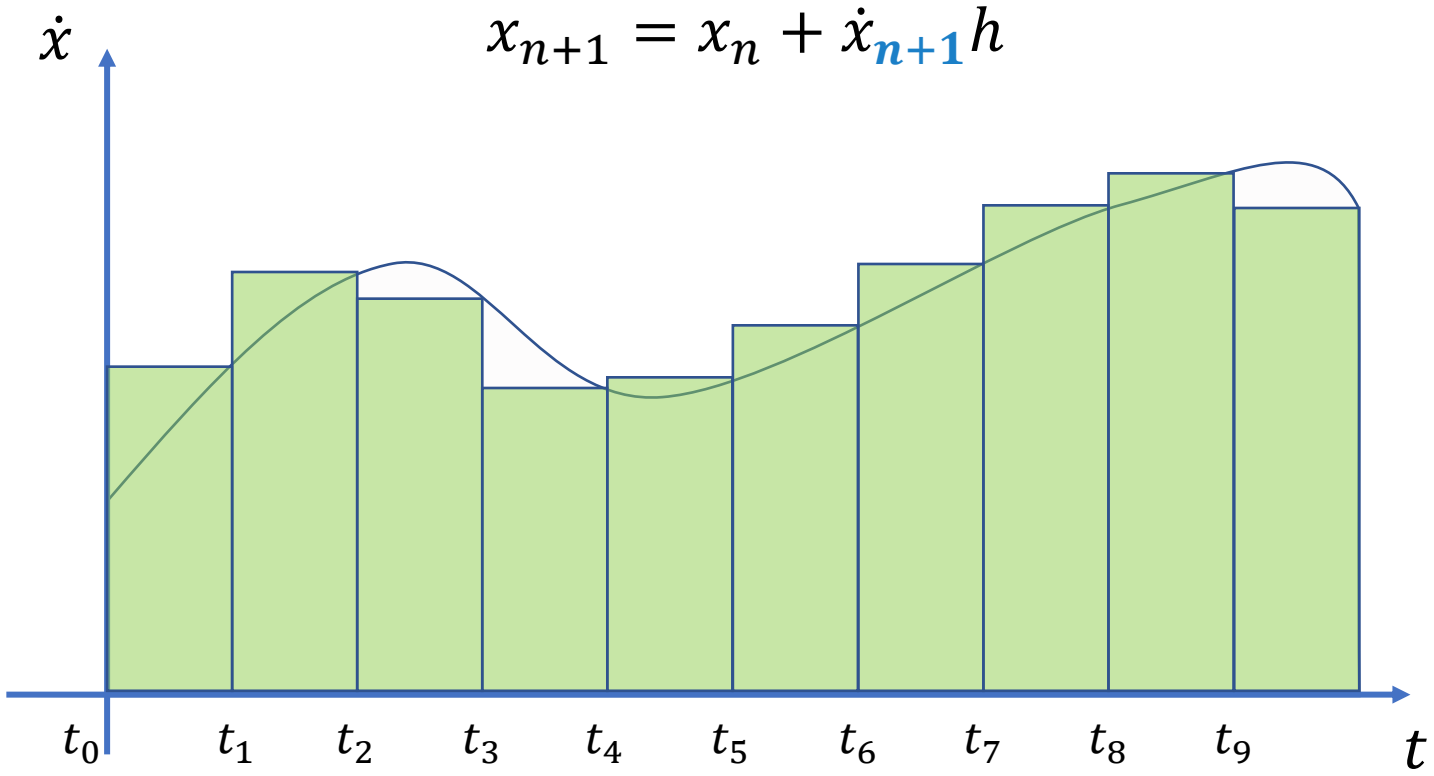
Numerical Integration

$$x_{n+1} = x_n + \dot{x}_{\boldsymbol{n}} h$$



Numerical Integration

$$x_{n+1} = x_n + \dot{x}_{\boldsymbol{n+1}} h$$



Numerical Integration

- Explicit/Forward Euler Integration

$$v_{n+1} = v_n + a_{\boldsymbol{n}} h$$
$$x_{n+1} = x_n + v_{\boldsymbol{n}} h$$

- Implicit/Backward Euler Integration

$$v_{n+1} = v_n + a_{\boldsymbol{n+1}} h$$
$$x_{n+1} = x_n + v_{\boldsymbol{n+1}} h$$

Numerical Integration

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Requires *"future"* information

Numerical Integration

- Explicit/Forward Euler Integration

$$v_{n+1} = v_n + a_{\boldsymbol{n}} h$$
$$x_{n+1} = x_n + v_{\boldsymbol{n}} h$$

- Implicit/Backward Euler Integration

$$v_{n+1} = v_n + f(x_{n+1}, v_{n+1}) h$$
$$x_{n+1} = x_n + v_{\boldsymbol{n+1}} h$$

Requires *"future"* information

Numerical Integration

- Explicit/Forward Euler Integration

$$v_{n+1} = v_n + a_{\boldsymbol{n}} h$$
$$x_{n+1} = x_n + v_{\boldsymbol{n}} h$$

- Implicit/Backward Euler Integration

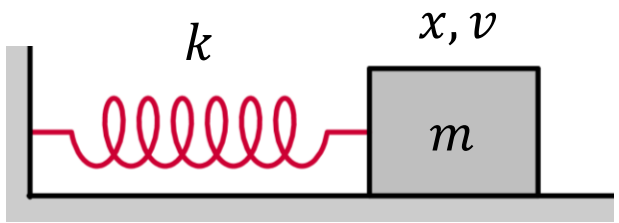
$$v_{n+1} = v_n + a_{\boldsymbol{n+1}} h$$ ⟵ Requires *"future"* information
$$x_{n+1} = x_n + v_{\boldsymbol{n+1}} h$$

- Symplectic / Semi-implicit Euler Integration

$$v_{n+1} = v_n + a_{\boldsymbol{n}} h$$ ⟵ All information is current
$$x_{n+1} = x_n + v_{\boldsymbol{n+1}} h$$

Mass on a Spring

一个简单的例子



$$f = -kx$$

Explicit Euler Integration

$$v_{n+1} = v_n - \frac{kx_{\boldsymbol{n}}}{m}h$$
$$x_{n+1} = x_n + v_{\boldsymbol{n}}h$$

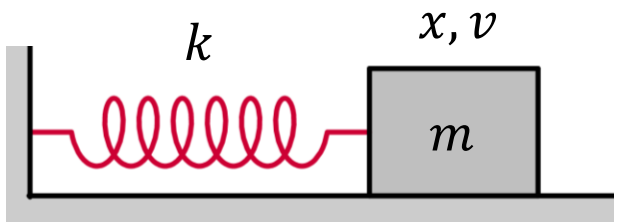
Semi-implicit Euler Integration

$$v_{n+1} = v_n - \frac{kx_{\boldsymbol{n}}}{m}h$$
$$x_{n+1} = x_n + v_{\boldsymbol{n+1}}h$$

Implicit Euler Integration

$$v_{n+1} = v_n - \frac{kx_{\boldsymbol{n+1}}}{m}h$$
$$x_{n+1} = x_n + v_{\boldsymbol{n+1}}h$$

Mass on a Spring



$$f = -kx$$
$$\hat{k} = k/m$$

Explicit Euler Integration

$$\begin{bmatrix} v_{n+1} \\ x_{n+1} \end{bmatrix} = \begin{bmatrix} 1 & -\hat{k}h \\ h & 1 \end{bmatrix} \begin{bmatrix} v_n \\ x_n \end{bmatrix}$$

Semi-implicit Euler Integration

$$\begin{bmatrix} v_{n+1} \\ x_{n+1} \end{bmatrix} = \begin{bmatrix} 1 & -\hat{k}h \\ h & 1 - \hat{k}h^2 \end{bmatrix} \begin{bmatrix} v_n \\ x_n \end{bmatrix}$$

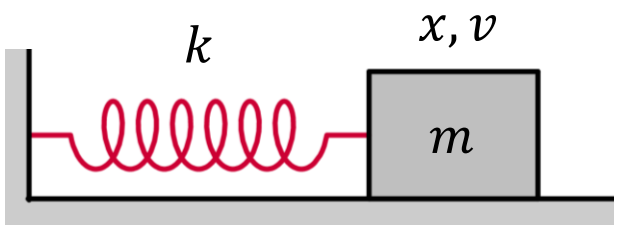
Implicit Euler Integration

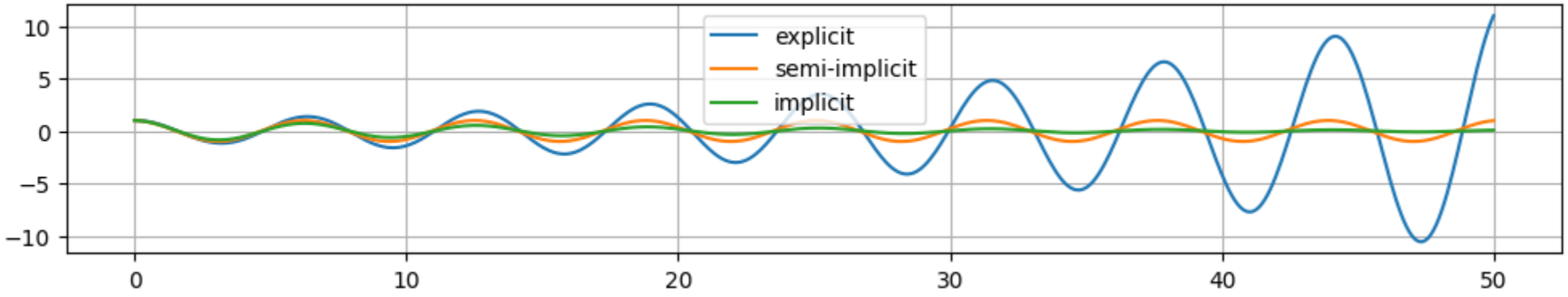
$$\begin{bmatrix} 1 & \hat{k}h \\ -h & 1 \end{bmatrix} \begin{bmatrix} v_{n+1} \\ x_{n+1} \end{bmatrix} = \begin{bmatrix} v_n \\ x_n \end{bmatrix}$$

$$\begin{bmatrix} v_{n+1} \\ x_{n+1} \end{bmatrix} = \frac{1}{1 + \hat{k}h^2} \begin{bmatrix} 1 & -\hat{k}h \\ h & 1 \end{bmatrix} \begin{bmatrix} v_n \\ x_n \end{bmatrix}$$

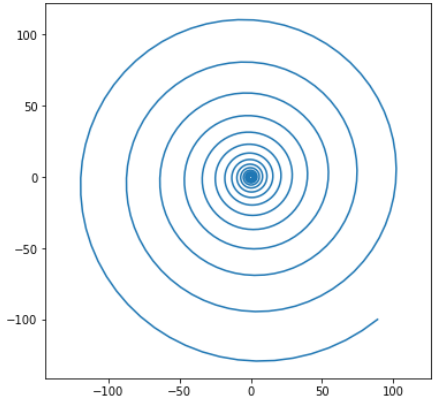
Mass on a Spring

$$f = -kx$$

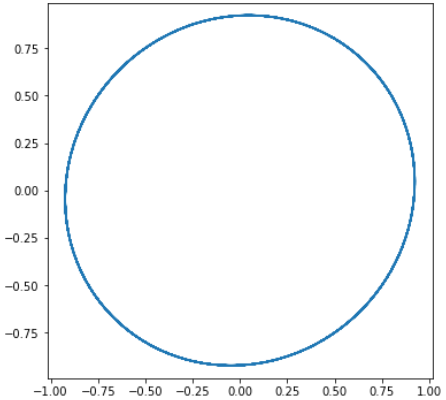




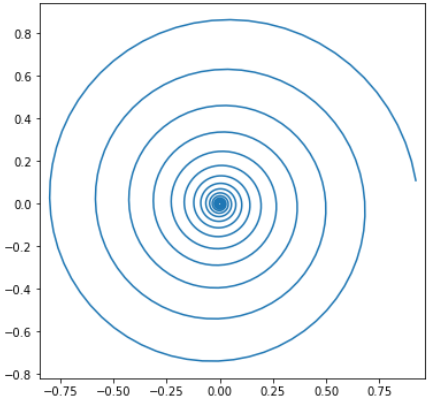
Explicit Euler Integration



Semi-implicit Euler Integration



Implicit Euler Integration



Numerical Integration

- Explicit/Forward Euler
  Symplectic/Semi-implicit Euler
  - Fast, no need to solve equations
  - Can be unstable under large time step

$$v_{n+1} = v_n + f(x_n, v_n)h$$
$$x_{n+1} = x_n + v_{\boldsymbol{n}}h$$

$$v_{n+1} = v_n + f(x_n, v_n)h$$
$$x_{n+1} = x_n + v_{\boldsymbol{n}+1}h$$

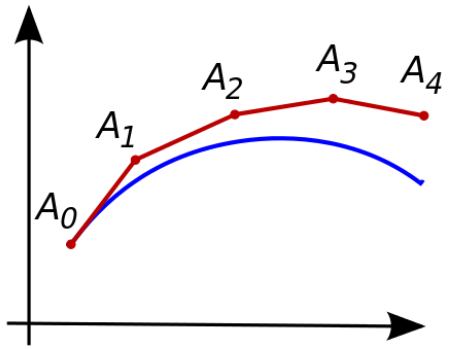
复杂系统下，半隐式欧拉也不稳定

- Implicit/Backward Euler
  - Rock stable (unconditionally)
  - Slow, need to solve a large problem

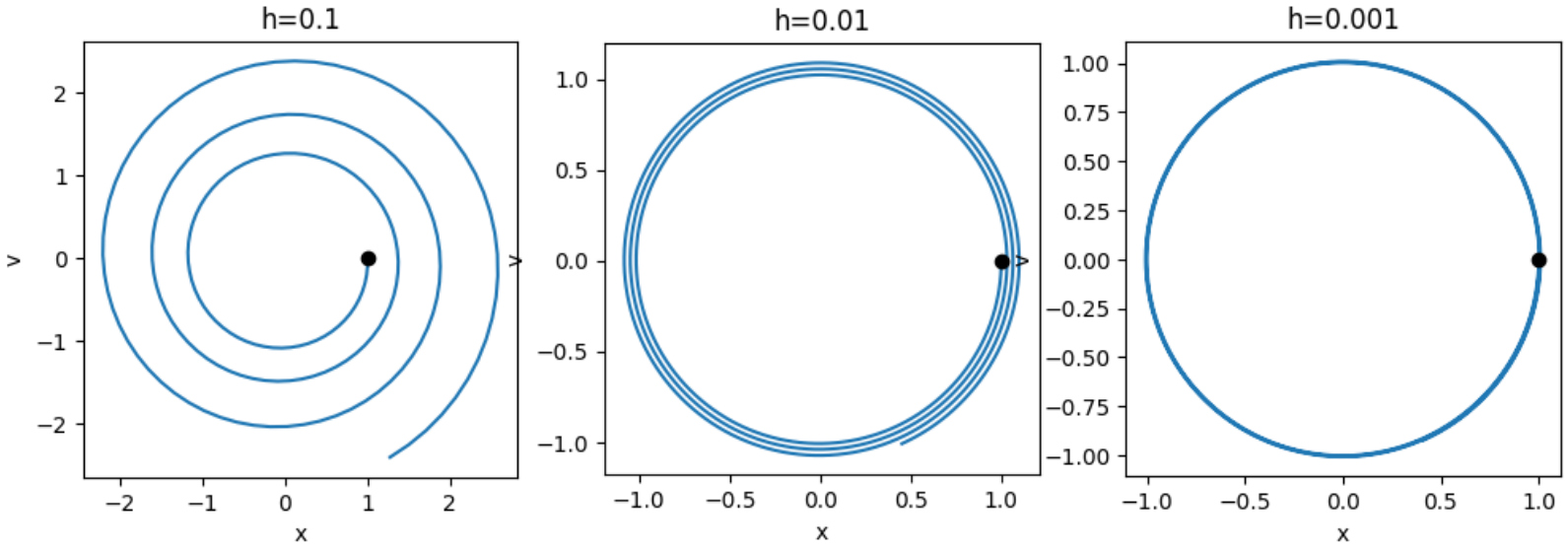
$$v_{n+1} = v_n + f(x_{n+1}, v_{n+1})h$$
$$x_{n+1} = x_n + v_{\boldsymbol{n+1}}h$$

Accuracy Issues

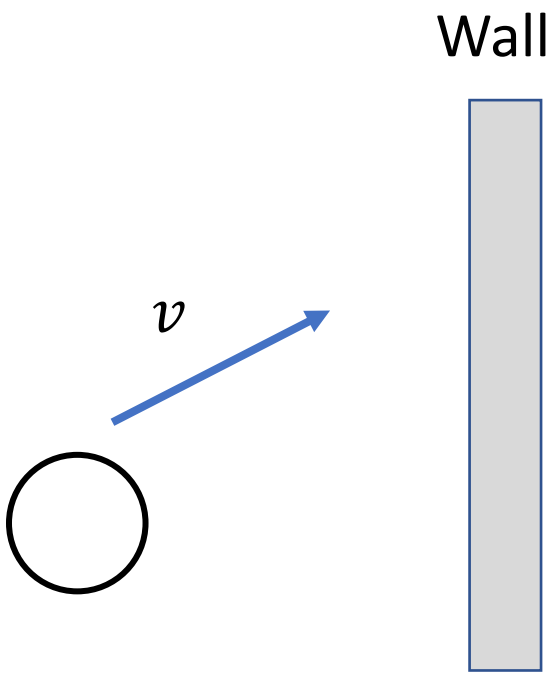
$$x_{n+1} = x_n + \int_{t_n}^{t_{n+1}} f(t)dt \approx x_n + f(t)h$$



We need a smaller h to achieve higher accuracy
(and stability, we will see later)

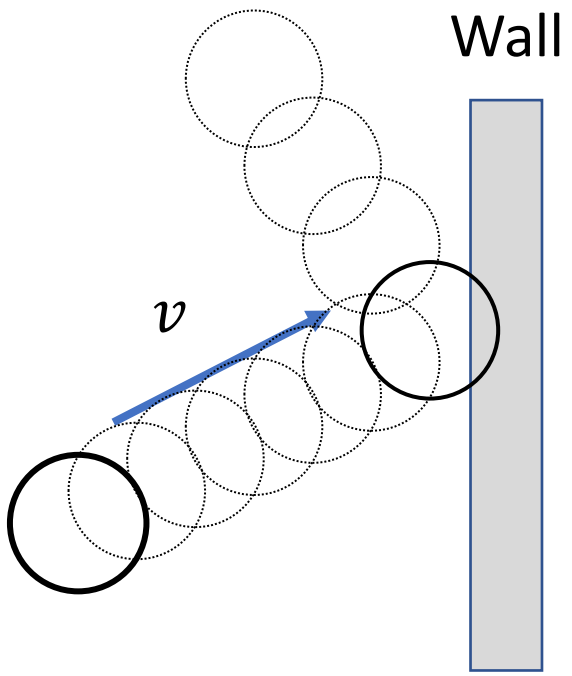


Accuracy Issues

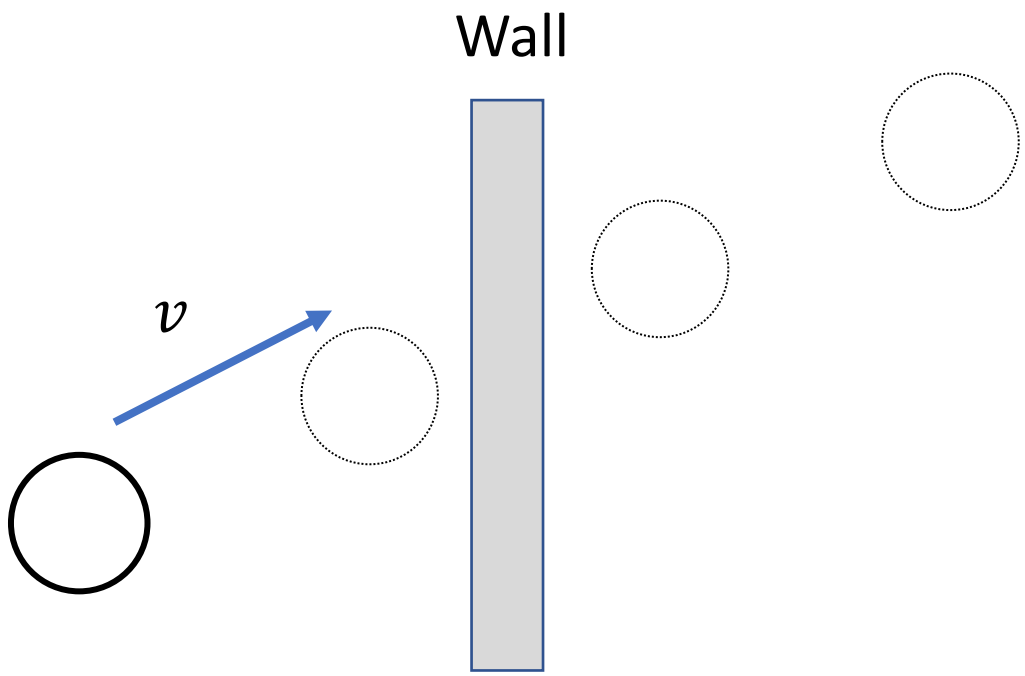


Accuracy Issues

$$x_{n+1} = x_n + v_n h$$



A small h



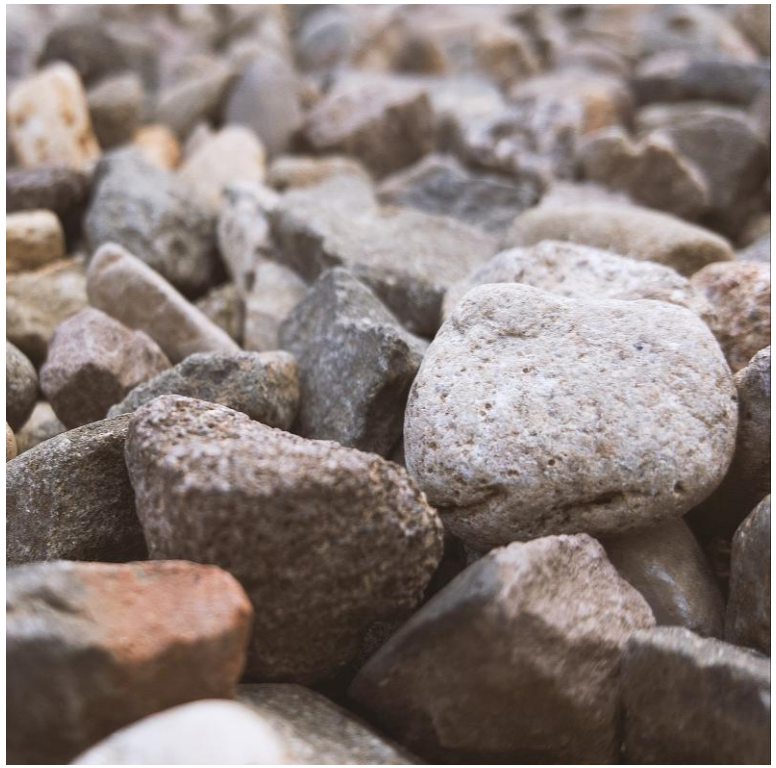
A large h

More Advanced Integration

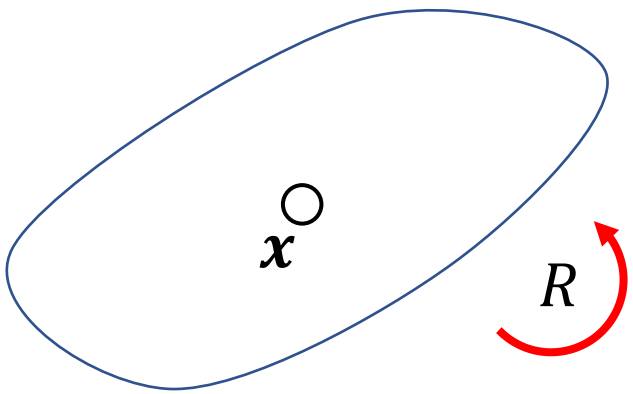
- Runge–Kutta methods
- Variational integration
- Position-based dynamics (PBD)
- ……

Rigid Bodies

- They are rigid….



Position and Orientation



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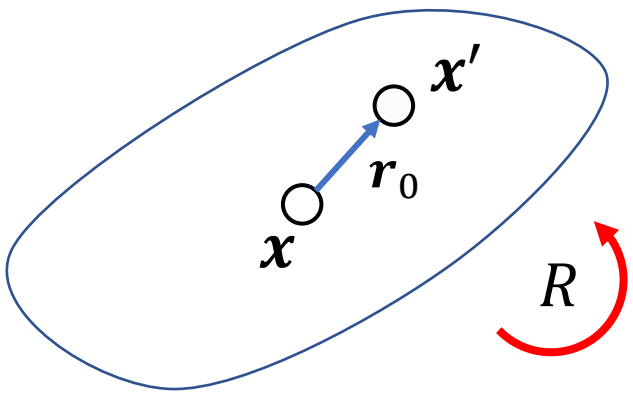
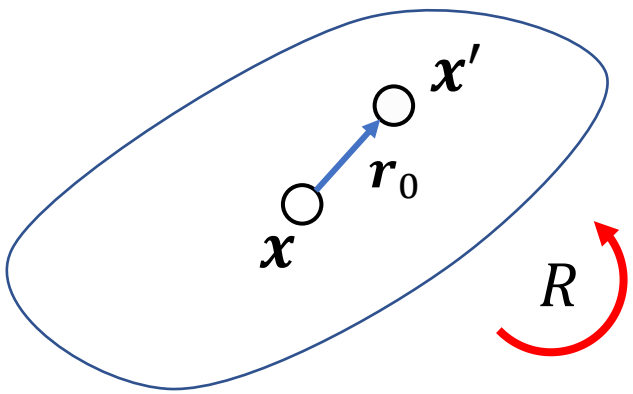


$$\boldsymbol{x}' = \boldsymbol{x} + \boldsymbol{R}\boldsymbol{r}_0$$

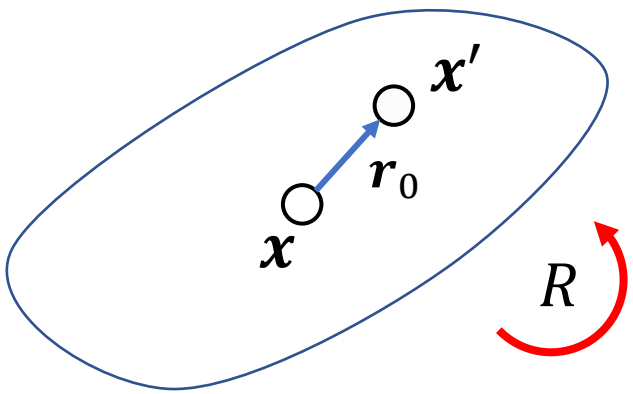
r0是刚体的局部坐标系下的

Position and Orientation



$$\boldsymbol{x}' = \boldsymbol{x} + \boldsymbol{R}\boldsymbol{r}_0 = \boldsymbol{x} + \boldsymbol{r}$$

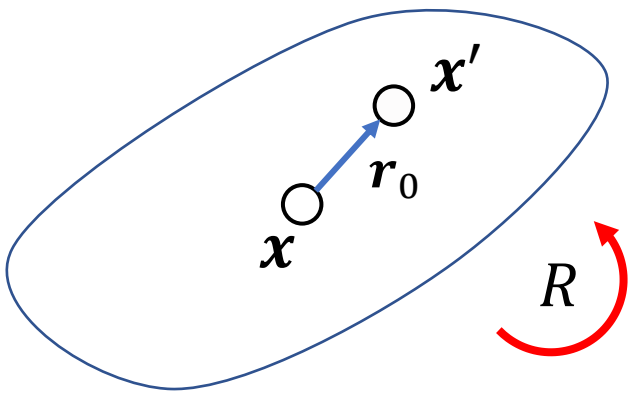
Linear and Angular Velocity 线速度和角速度



$$\boldsymbol{x}' = \boldsymbol{x} + R\boldsymbol{r}_0 = \boldsymbol{x} + \boldsymbol{r}$$

$$\frac{d\boldsymbol{x}'}{dt} = ?$$

Linear and Angular Velocity

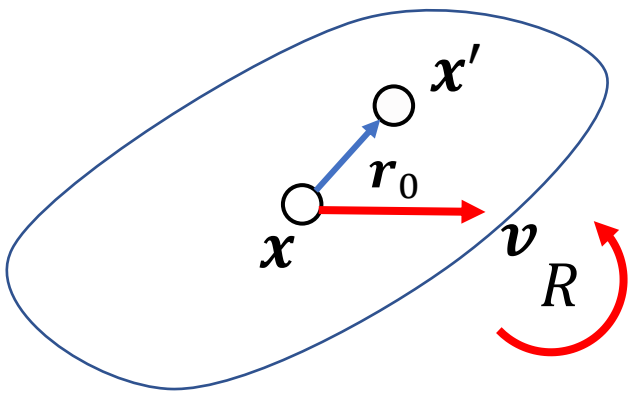


$$\frac{d\boldsymbol{x}'}{dt} \Leftrightarrow \dot{\boldsymbol{x}}' = \dot{\boldsymbol{x}} + \dot{R}\boldsymbol{r}_0$$

$$\boldsymbol{x}' = \boldsymbol{x} + R\boldsymbol{r}_0 = \boldsymbol{x} + \boldsymbol{r}$$

$$\frac{d\boldsymbol{x}'}{dt} = ?$$

Linear and Angular Velocity

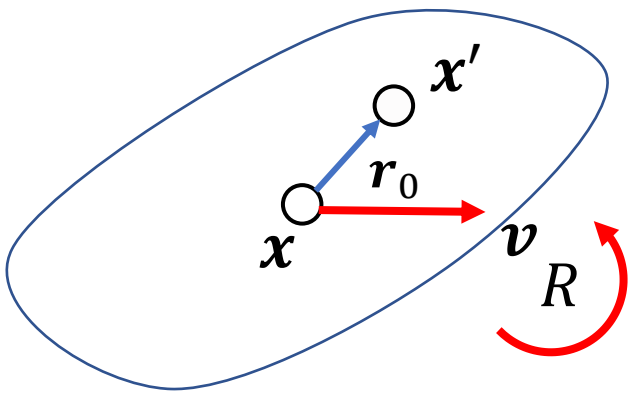


$$\frac{d\boldsymbol{x}'}{dt} \Leftrightarrow \dot{\boldsymbol{x}}' = \underbrace{\dot{\boldsymbol{x}}}_{v} + \dot{R}\boldsymbol{r}_0$$

$$\boldsymbol{x}' = \boldsymbol{x} + R\boldsymbol{r}_0 = \boldsymbol{x} + \boldsymbol{r}$$

$$\frac{d\boldsymbol{x}'}{dt} = ?$$

Linear and Angular Velocity

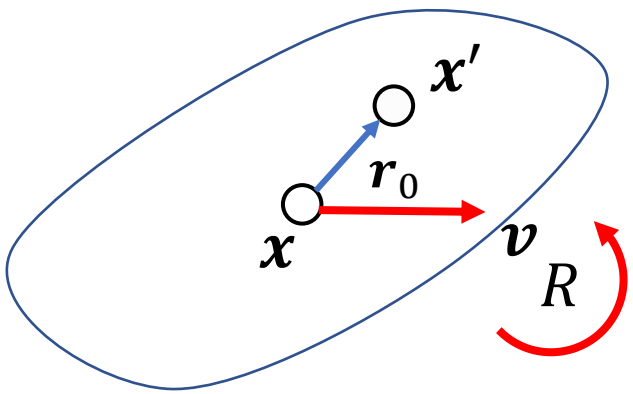


$$\frac{d\boldsymbol{x}'}{dt} \Leftrightarrow \dot{\boldsymbol{x}}' = \underbrace{\dot{\boldsymbol{x}}}_{\boldsymbol{v}} + \overbrace{\dot{R}\boldsymbol{r}_0}^{???}$$

$$\boldsymbol{x}' = \boldsymbol{x} + R\boldsymbol{r}_0 = \boldsymbol{x} + \boldsymbol{r}$$

$$\frac{d\boldsymbol{x}'}{dt} = ?$$

Linear and Angular Velocity



$$\boldsymbol{x}' = \boldsymbol{x} + R\boldsymbol{r}_0 = \boldsymbol{x} + \boldsymbol{r}$$

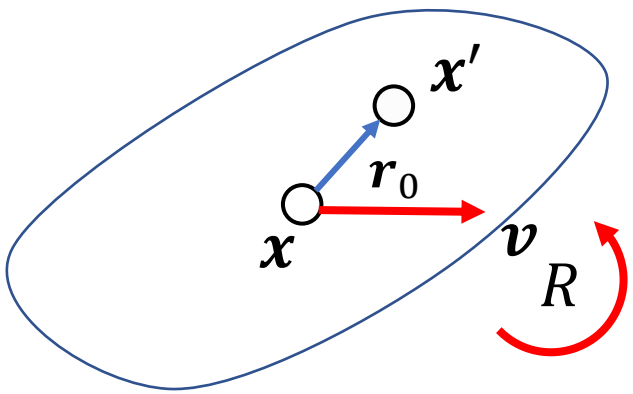
$$\frac{d\boldsymbol{x}'}{dt} = ?$$

???

$$\frac{d\boldsymbol{x}'}{dt} \Leftrightarrow \dot{\boldsymbol{x}}' = \underbrace{\dot{\boldsymbol{x}}}_{\boldsymbol{v}} + \dot{R}\boldsymbol{r}_0$$

$$RR^T = I$$

Linear and Angular Velocity



$$\boldsymbol{x}' = \boldsymbol{x} + R\boldsymbol{r}_0 = \boldsymbol{x} + \boldsymbol{r}$$

$$\frac{d\boldsymbol{x}'}{dt} = ?$$

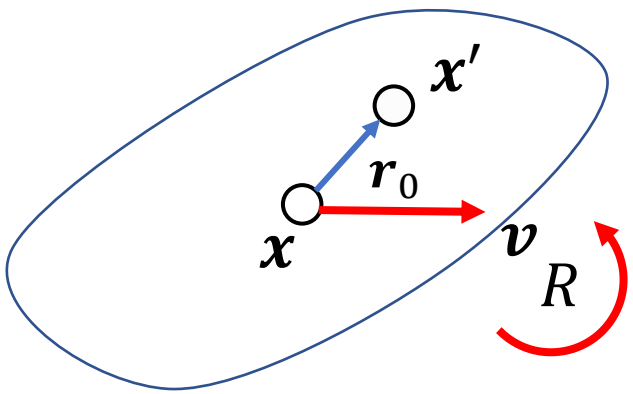
???

$$\frac{d\boldsymbol{x}'}{dt} \Leftrightarrow \dot{\boldsymbol{x}}' = \underbrace{\dot{\boldsymbol{x}}}_{\boldsymbol{v}} + \dot{R}\boldsymbol{r}_0$$

$$RR^T = I$$

$$\frac{d(RR^T)}{dt} = 0$$

Linear and Angular Velocity



$$\boldsymbol{x}' = \boldsymbol{x} + R\boldsymbol{r}_0 = \boldsymbol{x} + \boldsymbol{r}$$

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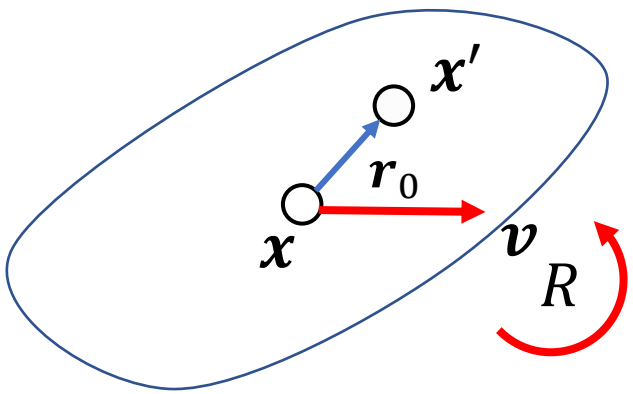
$$\frac{d\boldsymbol{x}'}{dt} \Leftrightarrow \dot{\boldsymbol{x}}' = \underbrace{\dot{\boldsymbol{x}}}_{\boldsymbol{v}} + \dot{R}\boldsymbol{r}_0$$

$$RR^T = I$$

$$\dot{R}R^T + R\dot{R}^T = 0$$

Linear and Angular Velocity



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$$RR^T = I$$

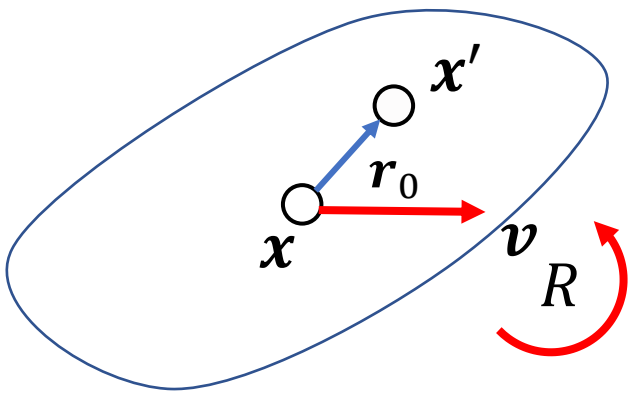
$$\Downarrow$$

$$\dot{R}R^T + \left(\dot{R}R^T\right)^T = 0$$

$$\Downarrow$$

$\dot{R}R^T$ is a Skew-Symmetric Matrix

Linear and Angular Velocity



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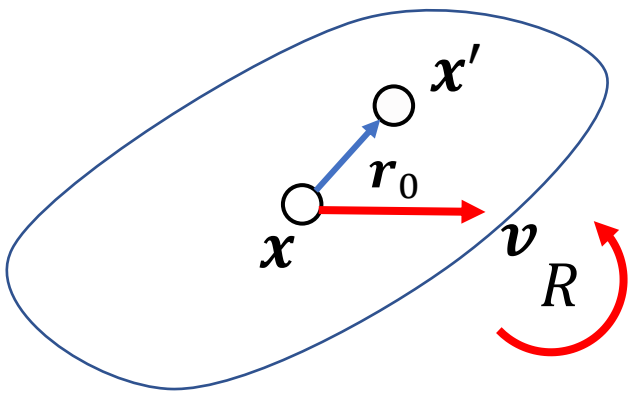
$$RR^T = I$$

$$\dot{R}R^T + \left(\dot{R}R^T\right)^T = 0$$

$$\dot{R}R^T = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} = [\omega]_\times$$

Linear and Angular Velocity



$$\boldsymbol{x}' = \boldsymbol{x} + R\boldsymbol{r}_0 = \boldsymbol{x} + \boldsymbol{r}$$

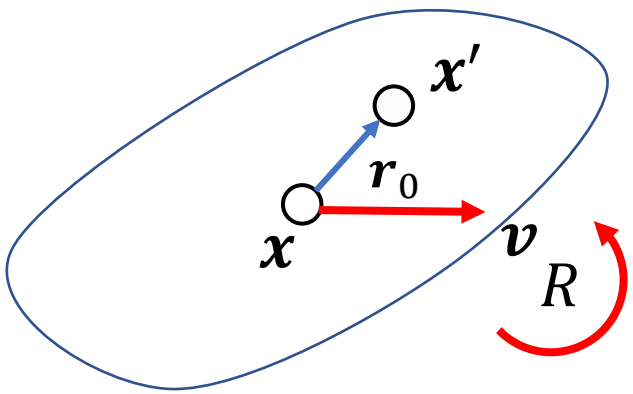
$$\frac{d\boldsymbol{x}'}{dt} = ?$$

$$\frac{d\boldsymbol{x}'}{dt} \Leftrightarrow \dot{\boldsymbol{x}}' = \underbrace{\dot{\boldsymbol{x}}}_{\boldsymbol{v}} + [\omega]_\times R\boldsymbol{r}_0$$

$$\dot{R} = [\omega]_\times R$$

$$\dot{R}R^T = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} = [\omega]_\times$$

Linear and Angular Velocity



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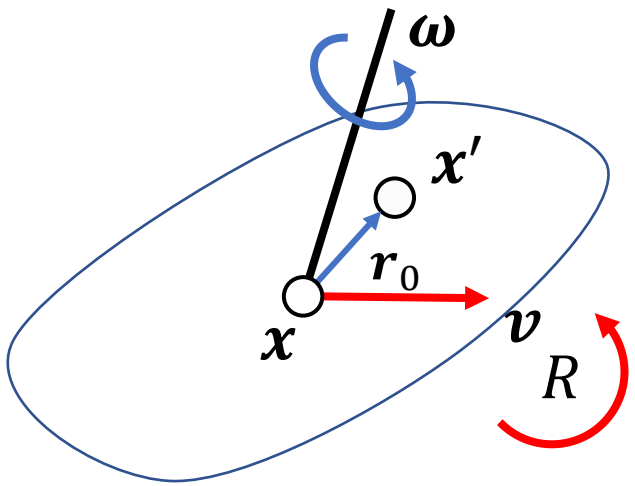
$$\frac{d\boldsymbol{x}'}{dt} = ?$$

$$\frac{d\boldsymbol{x}'}{dt} \Leftrightarrow \dot{\boldsymbol{x}}' = \underbrace{\dot{\boldsymbol{x}}}_{\boldsymbol{v}} + \boldsymbol{\omega} \times (R\boldsymbol{r}_0)$$

$$\dot{R} = [\omega]_\times R$$

$$\dot{R}R^T = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} = [\omega]_\times$$

Linear and Angular Velocity



$$\dot{\boldsymbol{x}} = \boldsymbol{v}$$

$$\dot{R} = [\omega]_\times R$$

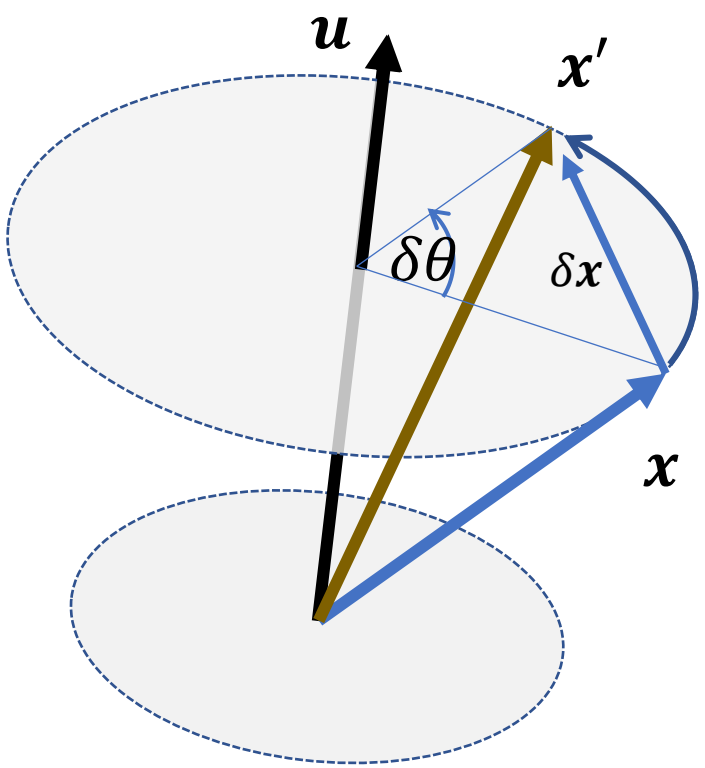
$\boldsymbol{v}$: linear velocity

$\boldsymbol{\omega}$: angular velocity

$$\boldsymbol{x}' = \boldsymbol{x} + R\boldsymbol{r}_0 = \boldsymbol{x} + \boldsymbol{r}$$

$$\boldsymbol{v}' = \boldsymbol{v} + \boldsymbol{\omega} \times \boldsymbol{r}$$

Angular Velocity and Rotation Matrix



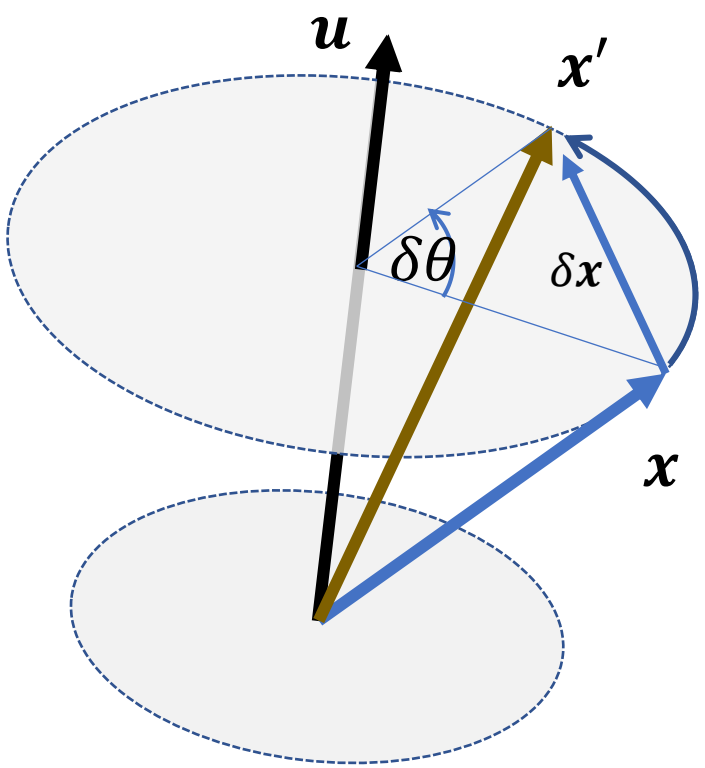
$$\|\boldsymbol{u}\| = \mathbf{1}$$

Rodrigues' rotation formula

$$\delta\boldsymbol{x} = \boldsymbol{x}' - \boldsymbol{x}$$

$$= (\sin\delta\theta\,)\,\boldsymbol{u}\times\boldsymbol{x} + (1-\cos\delta\theta\,)\,\boldsymbol{u}\times(\boldsymbol{u}\times\boldsymbol{x})$$

Angular Velocity and Rotation Matrix



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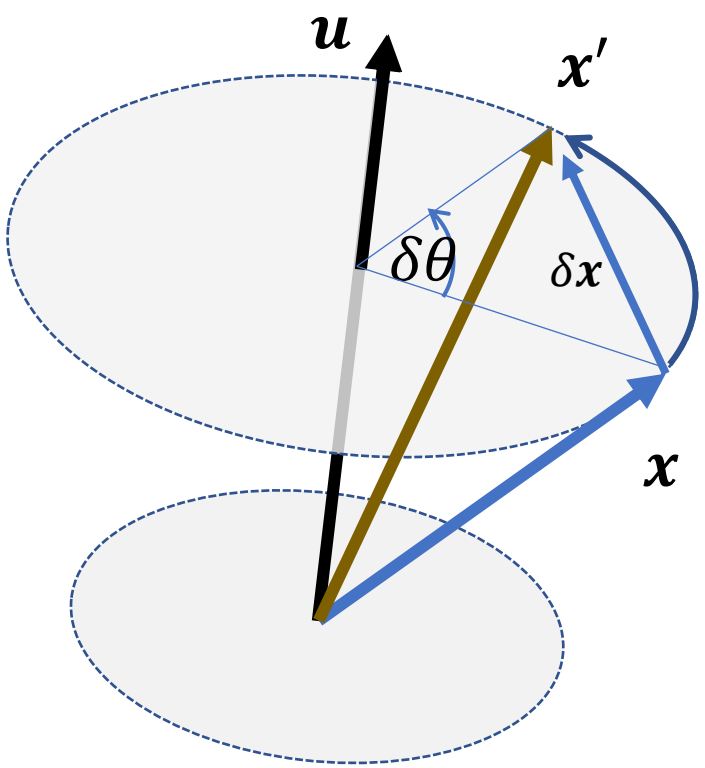
$$\dot{\boldsymbol{x}} = \frac{d\boldsymbol{x}}{dt} = \frac{d\boldsymbol{x}}{d\theta}\cdot\frac{d\theta}{dt} = \dot{\theta}\boldsymbol{u}\times\boldsymbol{x}$$

Angular Velocity and Rotation Matrix



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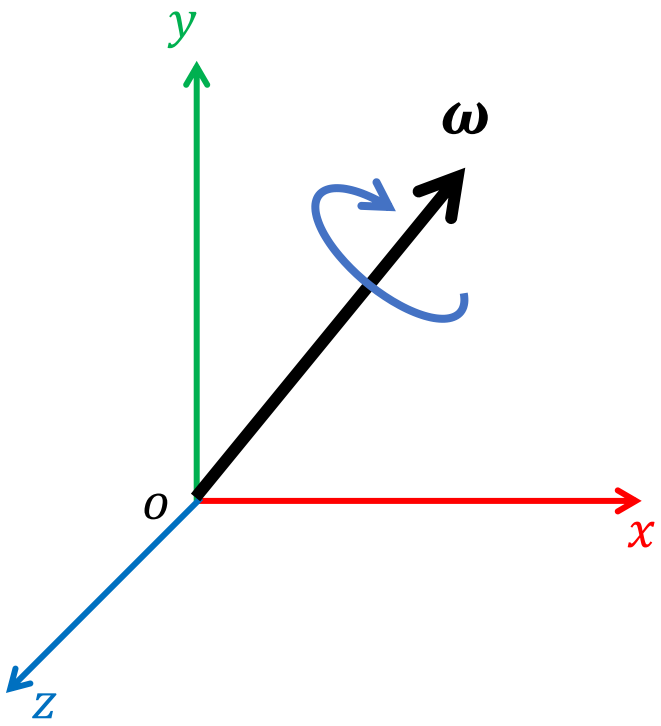
定义旋转速度乘旋转轴
=角速度

$$\dot{\boldsymbol{x}} = \boldsymbol{\omega}\times\boldsymbol{x}$$

Angular Velocity and Rotation Matrix

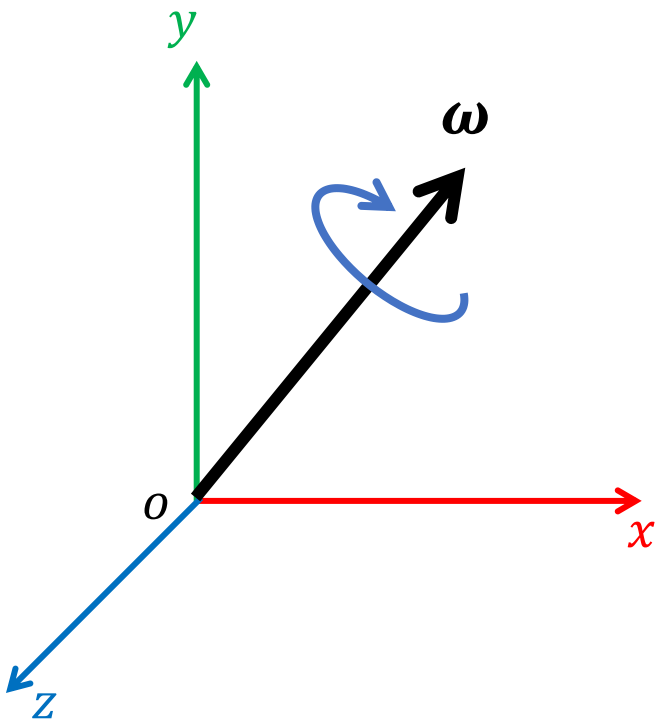
$\boldsymbol{\omega}$: angular velocity $\Longleftrightarrow$ $\dot{R} = [\omega]_\times R$



$$R = \begin{bmatrix} | & | & | \\ \boldsymbol{e}_x & \boldsymbol{e}_y & \boldsymbol{e}_z \\ | & | & | \end{bmatrix}$$

Angular Velocity and Rotation Matrix

$\boldsymbol{\omega}$: angular velocity $\Longleftrightarrow$ $\dot{R} = [\omega]_\times R$



$$R = \begin{bmatrix} | & | & | \\ \boldsymbol{e}_x & \boldsymbol{e}_y & \boldsymbol{e}_z \\ | & | & | \end{bmatrix}$$

$$\Downarrow$$

$$\dot{\boldsymbol{e}}_x = \boldsymbol{\omega} \times \boldsymbol{e}_x$$

$$\dot{\boldsymbol{e}}_y = \boldsymbol{\omega} \times \boldsymbol{e}_y$$

$$\dot{\boldsymbol{e}}_z = \boldsymbol{\omega} \times \boldsymbol{e}_z$$

Angular Velocity and Rotation Matrix

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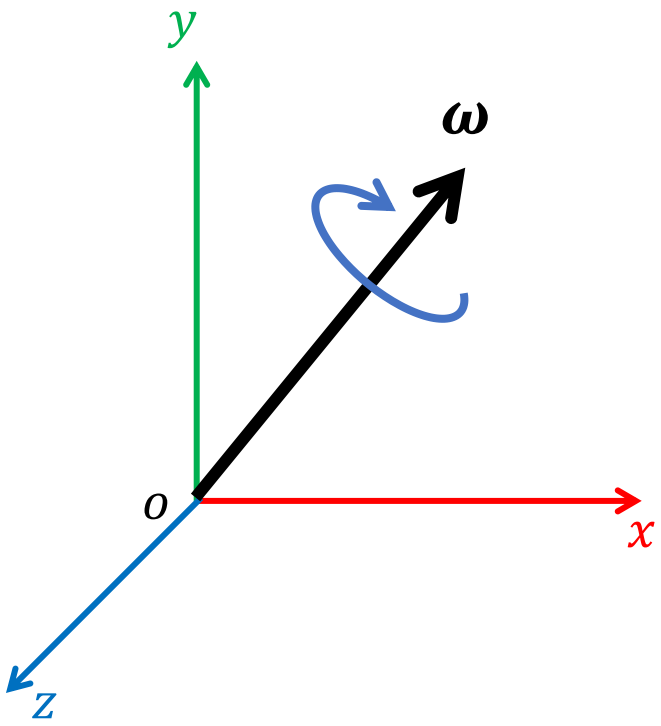
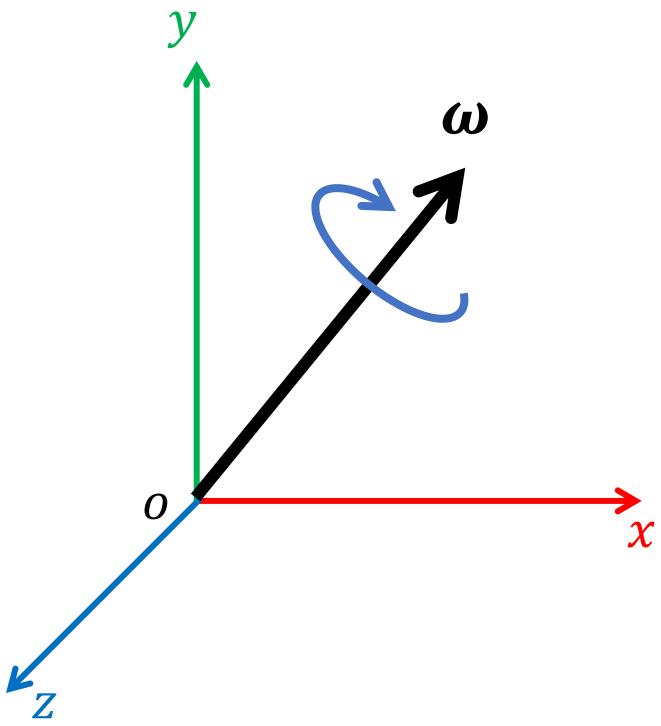


$$R = \begin{bmatrix} | & | & | \\ \boldsymbol{e}_x & \boldsymbol{e}_y & \boldsymbol{e}_z \\ | & | & | \end{bmatrix}$$

$$\dot{R} = \begin{bmatrix} | & | & | \\ \dot{\boldsymbol{e}}_x & \dot{\boldsymbol{e}}_y & \dot{\boldsymbol{e}}_z \\ | & | & | \end{bmatrix} = \begin{bmatrix} | & | & | \\ \boldsymbol{\omega} \times \overset{e_x}{\cancel{\dot{\boldsymbol{e}}_x}} & \boldsymbol{\omega} \times \overset{e_y}{\cancel{\dot{\boldsymbol{e}}_y}} & \boldsymbol{\omega} \times \overset{e_z}{\cancel{\dot{\boldsymbol{e}}_z}} \\ | & | & | \end{bmatrix}$$

Angular Velocity and Rotation Matrix

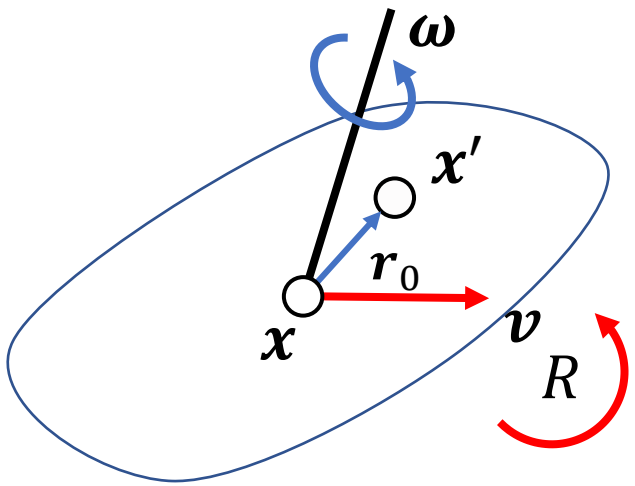
$\boldsymbol{\omega}$: angular velocity $\Longleftrightarrow$ $\dot{R} = [\omega]_\times R$



$$R = \begin{bmatrix} | & | & | \\ \boldsymbol{e}_x & \boldsymbol{e}_y & \boldsymbol{e}_z \\ | & | & | \end{bmatrix}$$

$$\dot{R} = \begin{bmatrix} | & | & | \\ \dot{\boldsymbol{e}}_x & \dot{\boldsymbol{e}}_y & \dot{\boldsymbol{e}}_z \\ | & | & | \end{bmatrix} = [\boldsymbol{\omega}]_\times \begin{bmatrix} | & | & | \\ \boldsymbol{e}_x & \boldsymbol{e}_y & \boldsymbol{e}_z \\ | & | & | \end{bmatrix}$$

Linear and Angular Velocity



$$\dot{\boldsymbol{x}} = \boldsymbol{v}$$

$$\dot{R} = [\omega]_\times R$$

$\boldsymbol{v}$: linear velocity

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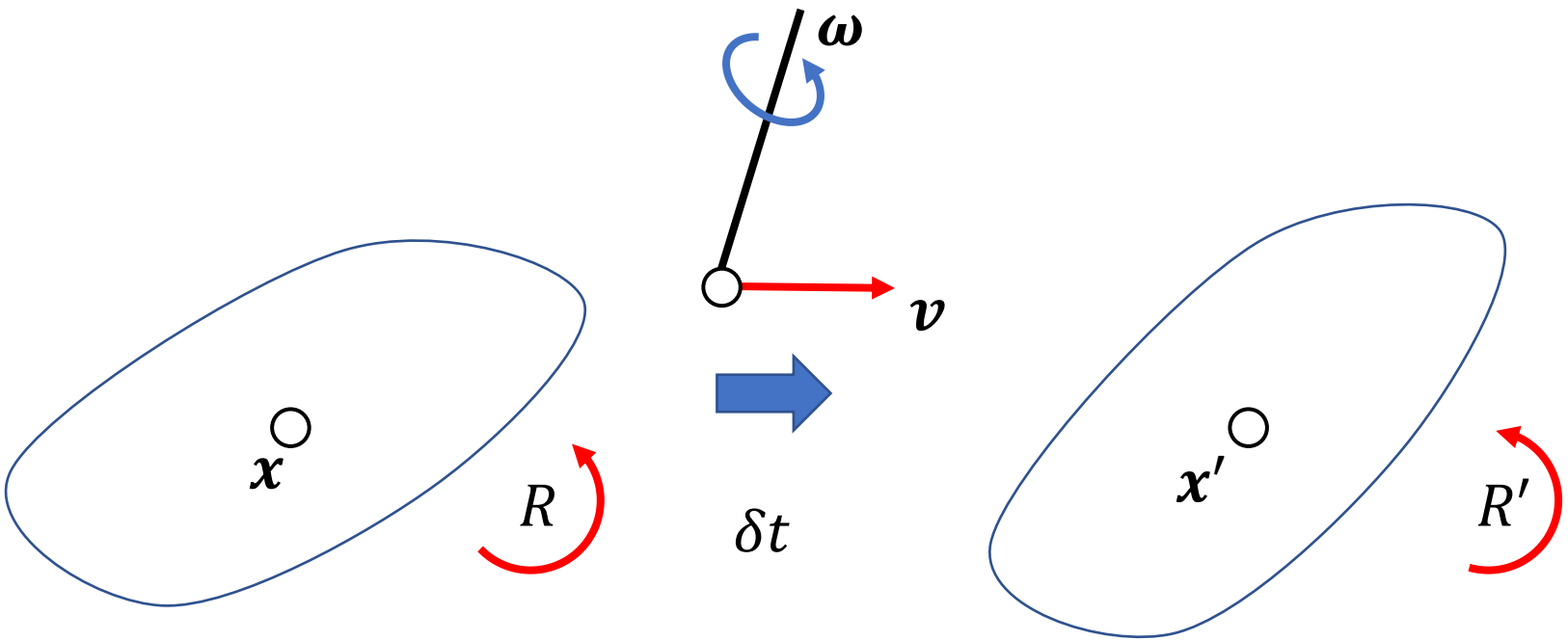
$$\boldsymbol{x}' = \boldsymbol{x} + R\boldsymbol{r}_0 = \boldsymbol{x} + \boldsymbol{r}$$

$$\boldsymbol{v}' = \boldsymbol{v} + \boldsymbol{\omega} \times \boldsymbol{r}$$

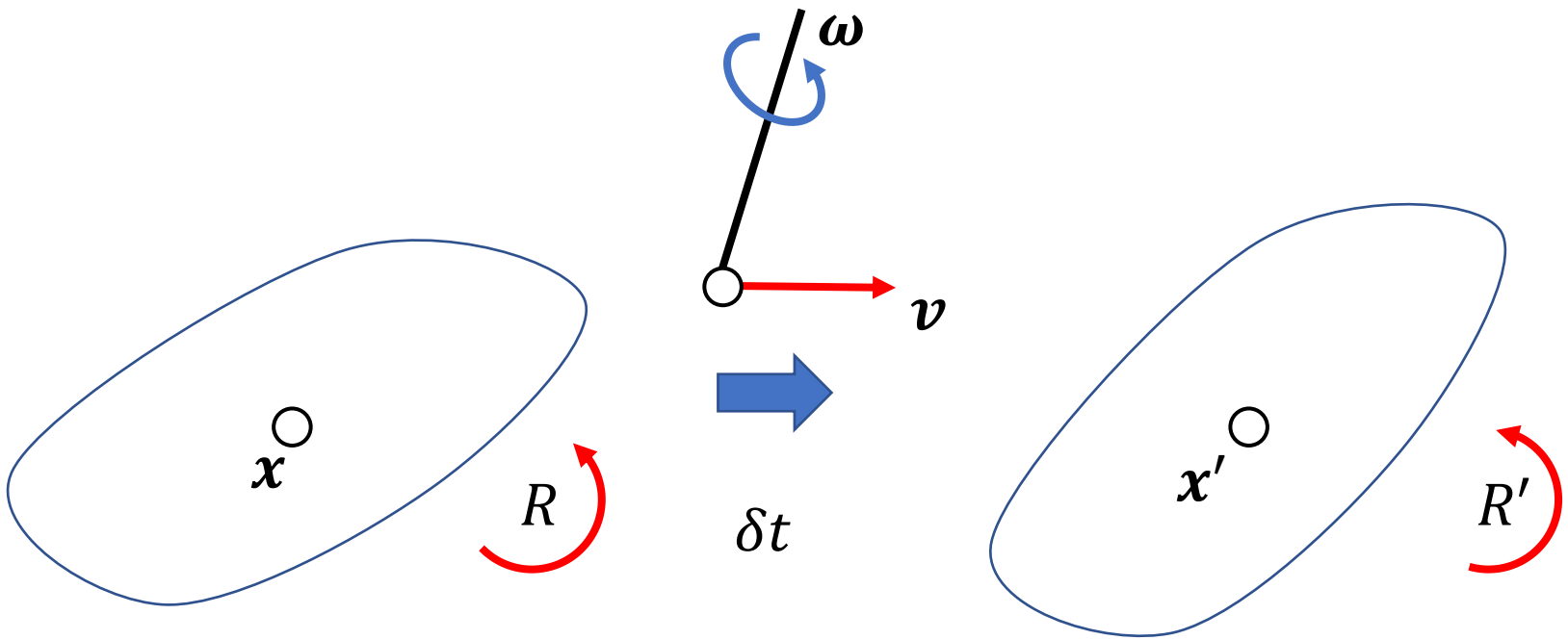
Numerical Integration



$$\boldsymbol{x}' = ?$$

$$R' = ?$$

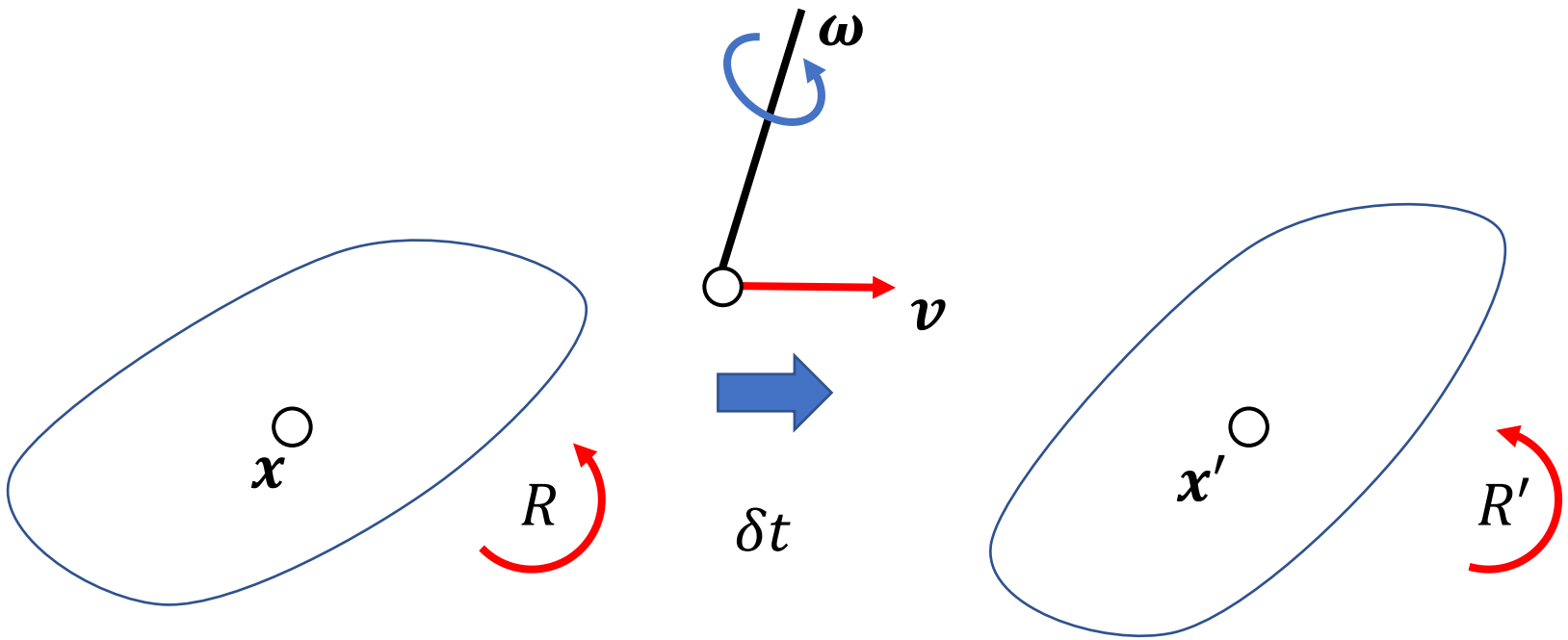
Numerical Integration



$$\dot{\boldsymbol{x}} = \boldsymbol{v}$$

$$\dot{R} = [\omega]_\times R$$

Numerical Integration



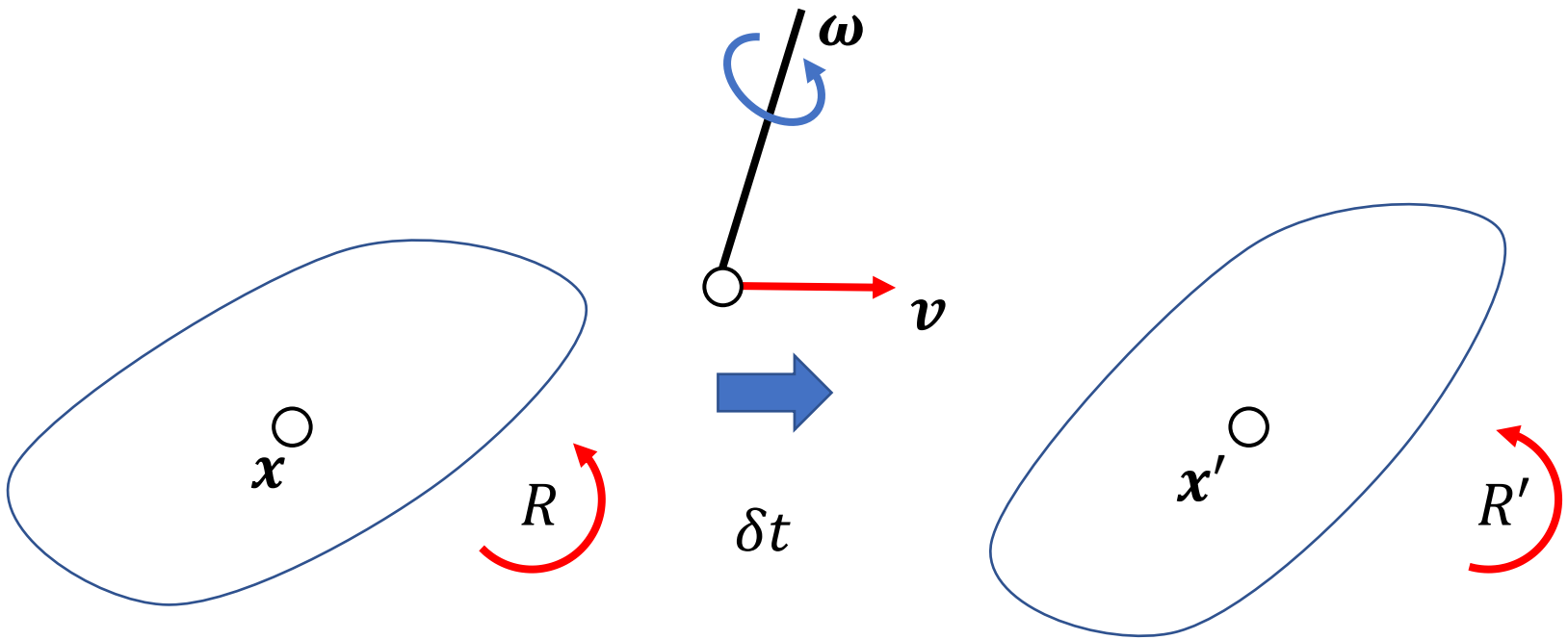
$$\dot{\boldsymbol{x}} = \boldsymbol{v} \qquad \boldsymbol{x}' = \boldsymbol{x} + \delta t \cdot \boldsymbol{v}$$

$$\dot{R} = [\omega]_\times R \qquad R' = R + \delta t \cdot [\omega]_\times R$$

积分过程中要对R做正交化，保证R依然是一个旋转矩阵

Need orthogonalization!

Numerical Integration: Quaternion

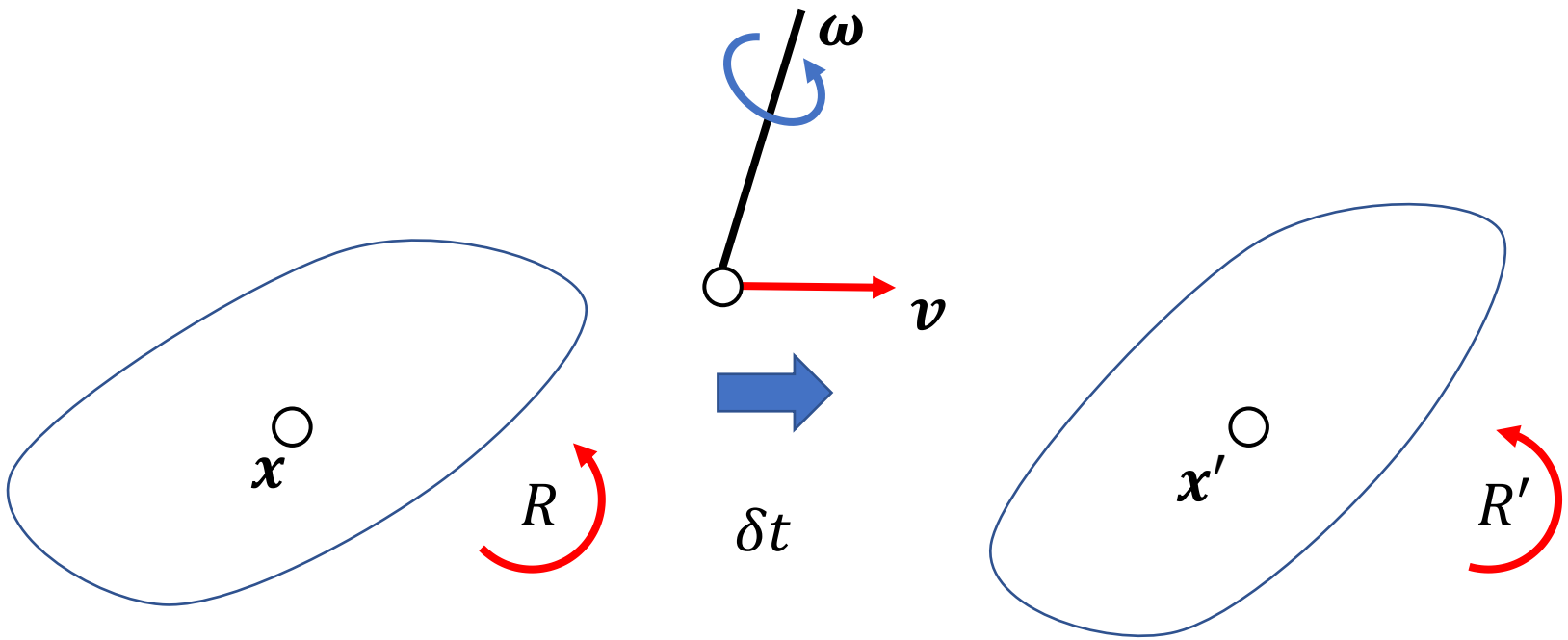


$$\dot{\boldsymbol{x}} = \boldsymbol{v} \qquad \boldsymbol{x}' = \boldsymbol{x} + \delta t \cdot \boldsymbol{v}$$

$$\dot{\boldsymbol{q}} = ? \qquad \boldsymbol{q}' = \boldsymbol{q} + \delta t \cdot \dot{\boldsymbol{q}}$$

单位化要比正交化简单的多

Numerical Integration: Quaternion



$$\dot{\boldsymbol{x}} = \boldsymbol{v} \qquad \boldsymbol{x}' = \boldsymbol{x} + \delta t \cdot \boldsymbol{v}$$

直接给出结论： $\dot{\boldsymbol{q}} = \frac{1}{2}\bar{\boldsymbol{\omega}}\boldsymbol{q}$ $\qquad \boldsymbol{q}' = \boldsymbol{q} + \delta t \cdot \dot{\boldsymbol{q}}$

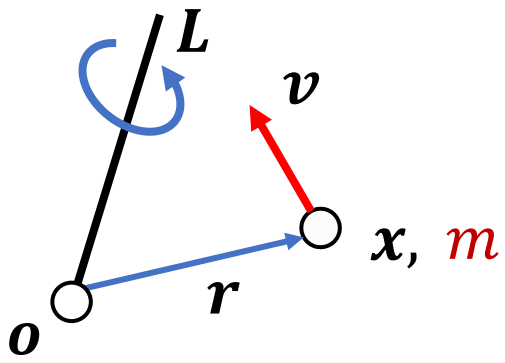
角速度的纯四元数 $\bar{\boldsymbol{\omega}} = (0, \boldsymbol{\omega})$

Need Normalization!

Kinematics vs. Dynamics

Kinematics		Dynamics
	二者差的就是质量	
$\boldsymbol{x}, R$	m, I	
$\boldsymbol{v}, \boldsymbol{\omega}$		$\boldsymbol{p}, \boldsymbol{L}$
$\boldsymbol{a}, \boldsymbol{\alpha}$	$\Longleftrightarrow$	
$\dddot{\boldsymbol{x}}, \dddot{\boldsymbol{\omega}}$		$\boldsymbol{F}, \boldsymbol{\tau}$
…		

Linear and Angular Momentum of a Particle

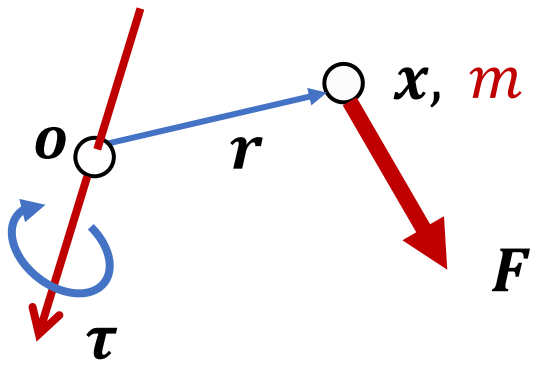


$\boldsymbol{p} = m\,\boldsymbol{v}$ Linear momentum of x

$\boldsymbol{L} = m\,\boldsymbol{r} \times \boldsymbol{v}$ Angular momentum of x w.r.t. o

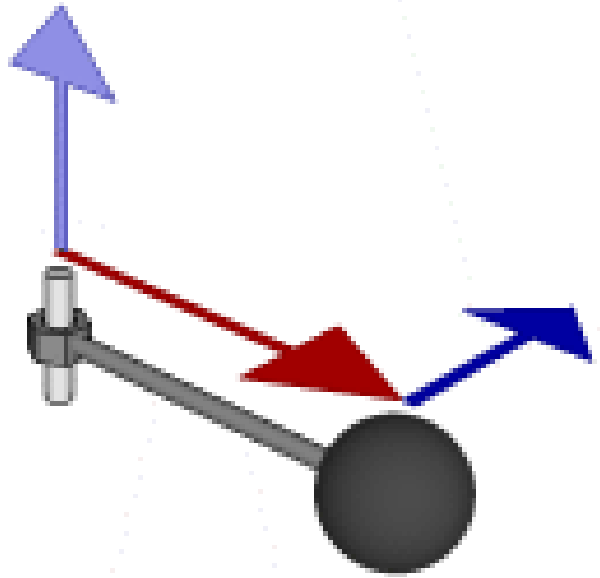
角动量是相对于一个参考点的

Force and Torque 力和力矩



$$\boldsymbol{\tau} = \boldsymbol{r} \times \boldsymbol{F}$$

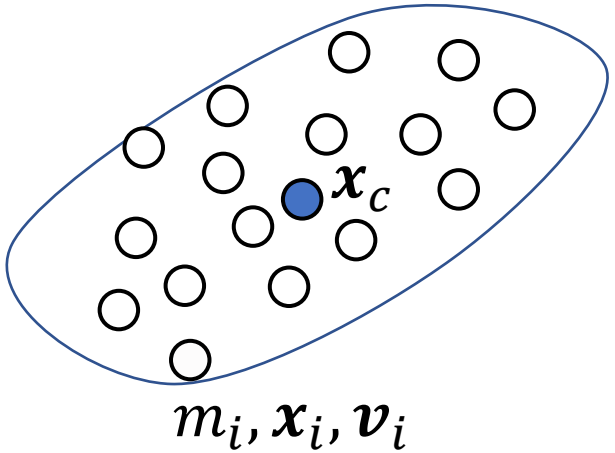
Torque and Angular Momentum



$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$$
$$\mathbf{L} = \mathbf{r} \times \mathbf{p}$$

https://en.wikipedia.org/wiki/Torque

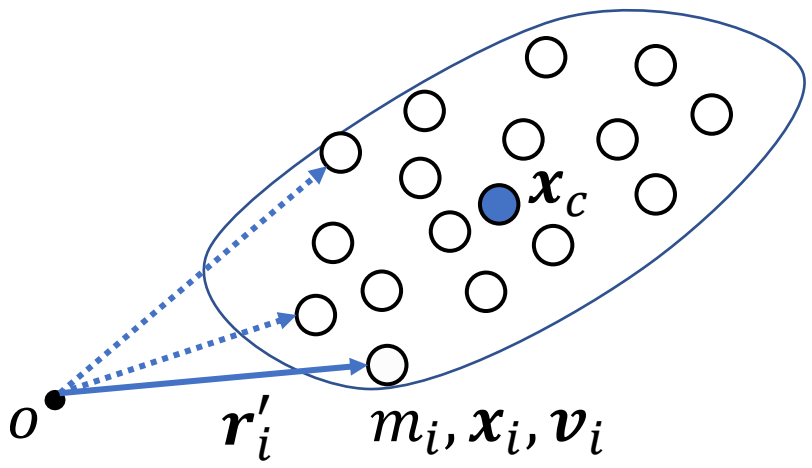
Rigid Body as a Collection of Particles



$$M = \sum_i m_i$$

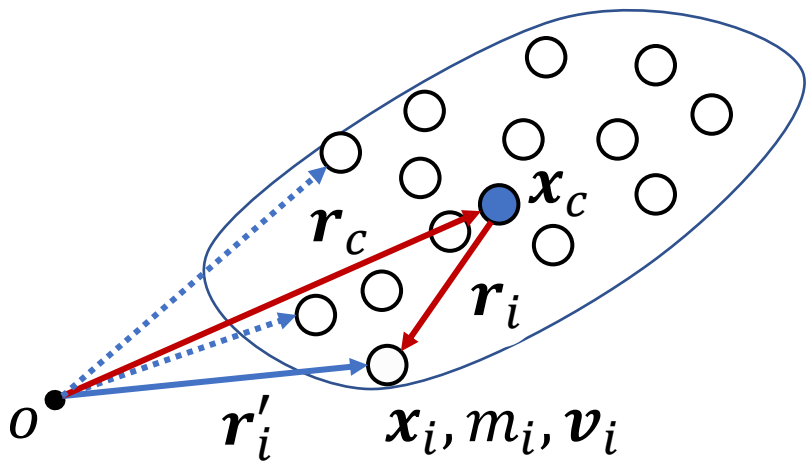
$$\boldsymbol{x}_c = \frac{\sum_i m_i \boldsymbol{x}_i}{\sum_i m_i} \qquad \boldsymbol{v}_c = \frac{\sum_i m_i \boldsymbol{v}_i}{\sum_i m_i}$$

Moments of a Rigid Body



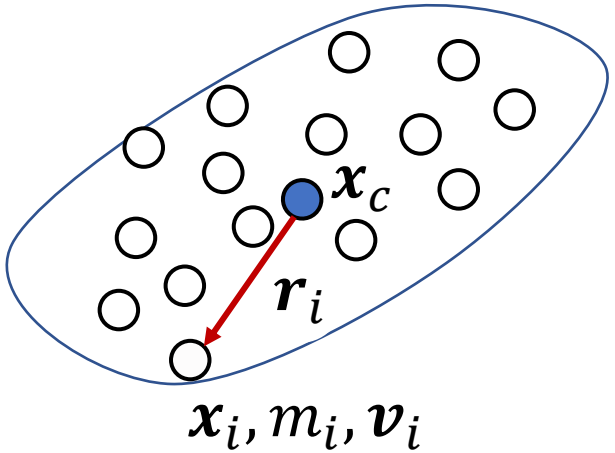
$$\boldsymbol{p} = \sum_i m_i \boldsymbol{v}_i \qquad \boldsymbol{L}_o = \sum_i m_i \boldsymbol{r}_i' \times \boldsymbol{v}_i$$

Angular Moment of a Rigid Body



$$\boldsymbol{L}_o = \sum_i m_i \boldsymbol{r}'_i \times \boldsymbol{v}_i = M\boldsymbol{r}_c \times \boldsymbol{v}_c + \sum_i m_i \boldsymbol{r}_i \times \boldsymbol{v}_i$$

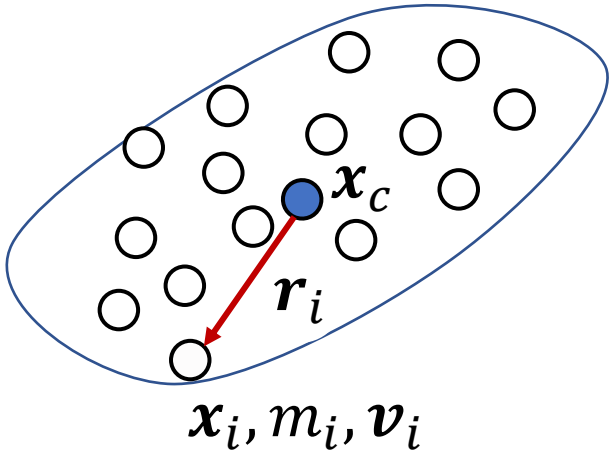
Angular Moment of a Rigid Body



$$\boldsymbol{L}_{x_c} = \sum_i m_i \boldsymbol{r}_i \times \boldsymbol{v}_i$$

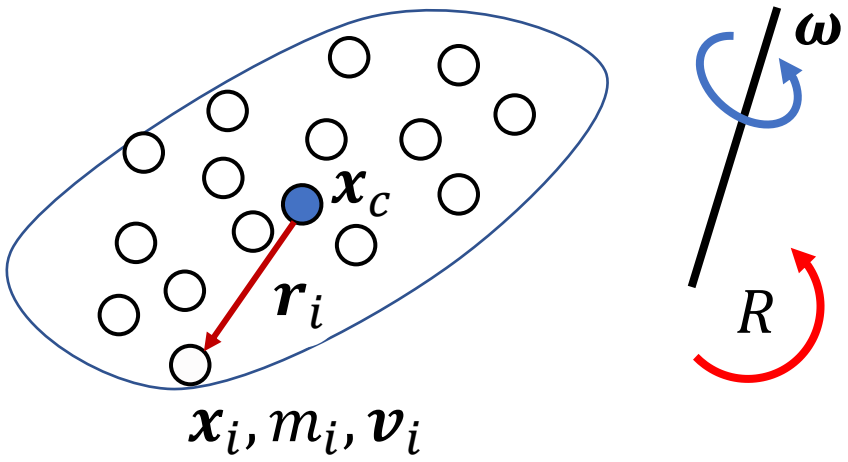
以质心为参考点

Angular Moment of a Rigid Body



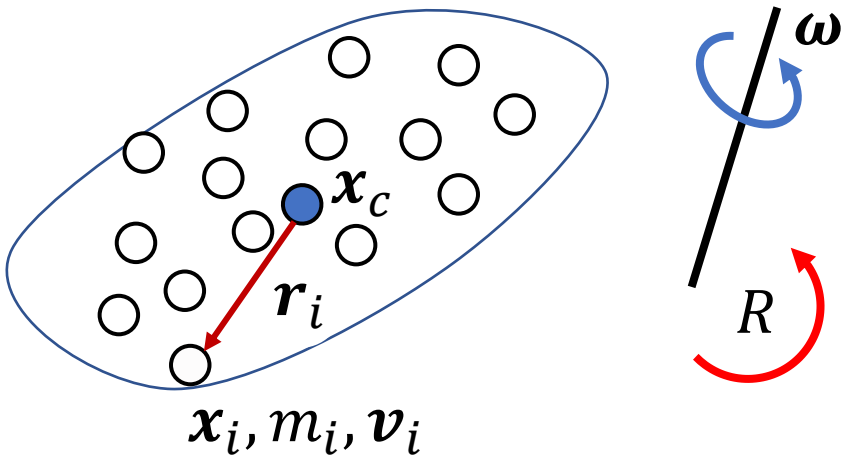
$$\boldsymbol{L} = \sum_i m_i \boldsymbol{r}_i \times \boldsymbol{v}_i$$

Angular Moment of a Rigid Body



$$\boldsymbol{L} = \sum_i m_i \boldsymbol{r}_i \times \boldsymbol{v}_i$$

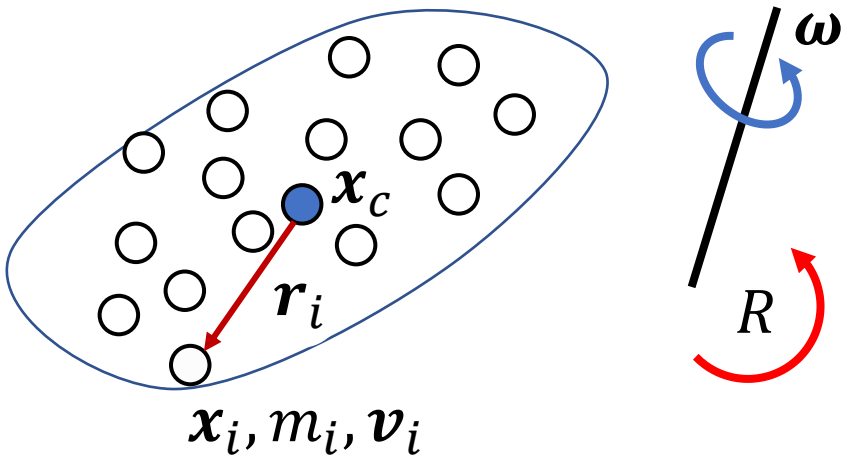
Angular Moment of a Rigid Body



$$\boldsymbol{L} = \sum_i m_i \boldsymbol{r}_i \times (\omega \times \boldsymbol{r}_i) + \sum_i m_i \boldsymbol{r}_i \times \boldsymbol{v}_c$$

展开： vi = w × ri + vc

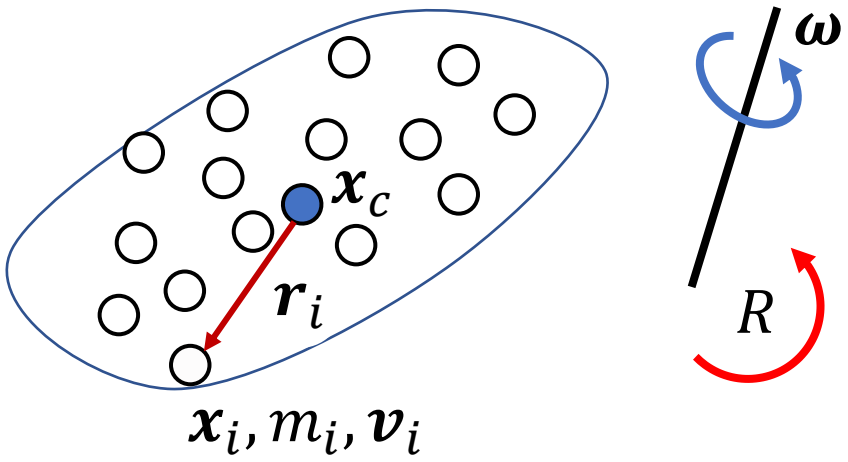
Angular Moment of a Rigid Body



$$\boldsymbol{L} = \sum_i -m_i [r_i]_\times^2 \, \boldsymbol{\omega}$$

$$[a]_\times = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$$

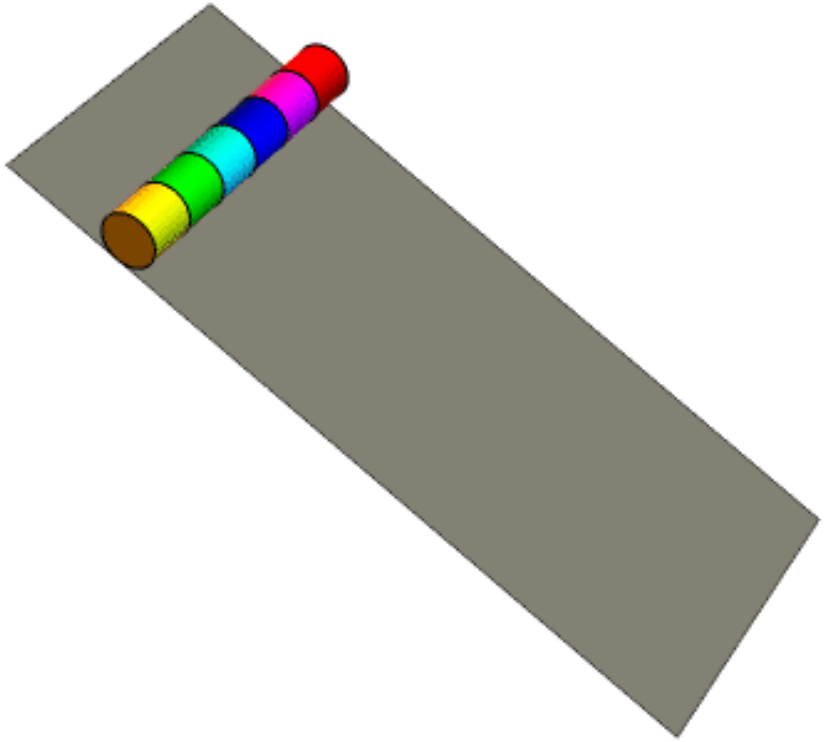
Angular Moment of a Rigid Body



$$\boldsymbol{L} = I\boldsymbol{\omega}$$

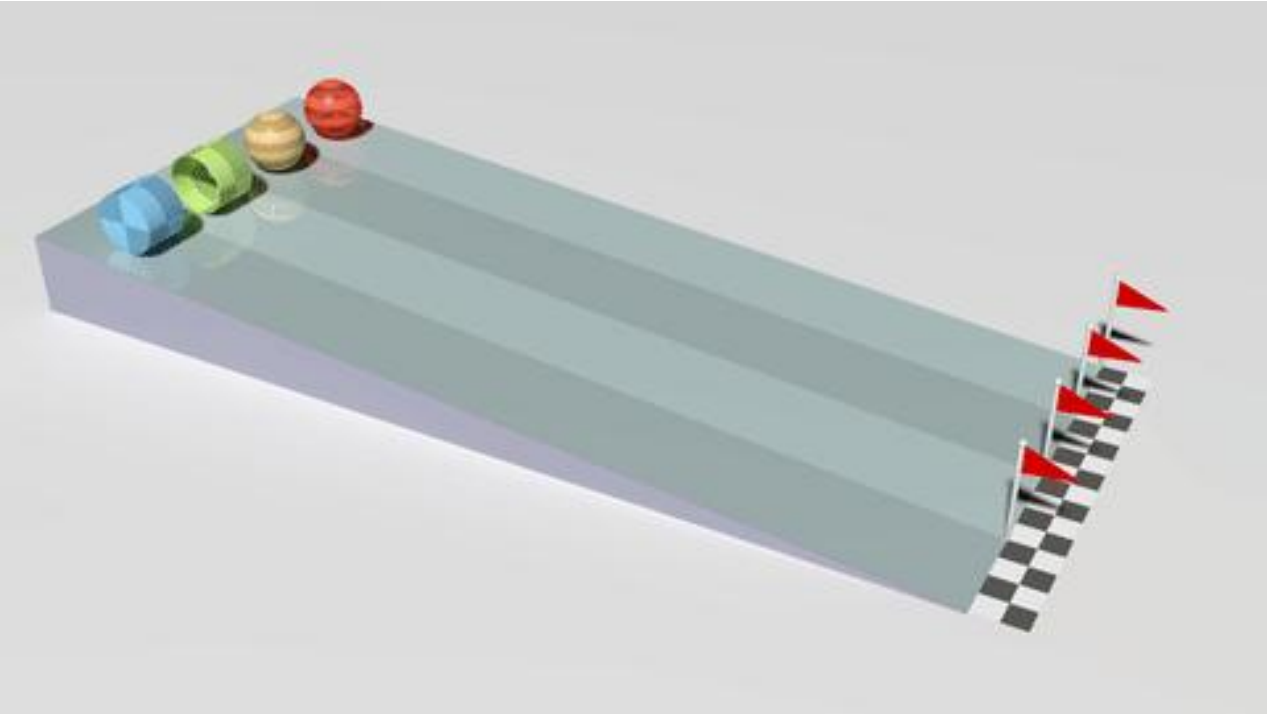
Moment of Inertia: $\boldsymbol{I} = \sum_i -m_i[r_i]_\times^2$ 转动惯量

Moment of Inertia



- $m = m_0$ $I = 1\,I_0$
- $m = m_0$ $I = 2\,I_0$
- $m = m_0$ $I = 3\,I_0$
- $m = m_0$ $I = 4\,I_0$
- $m = m_0$ $I = 5\,I_0$
- $m = m_0$ $I = 6\,I_0$

Moment of Inertia



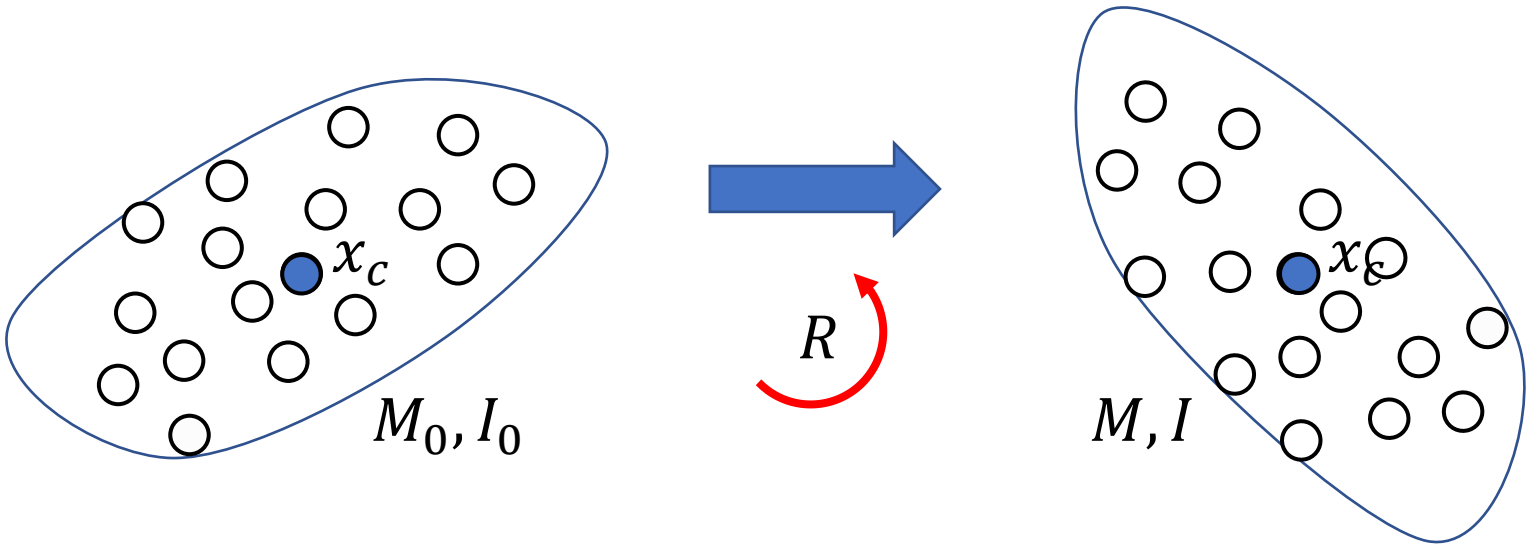
Same mass, different shapes

Different Moments of Inertia

Moment of Inertia

https://en.wikipedia.org/wiki/Moment_of_inertia

Rotation of Moment of Inertia



$$(Rr) \times x = R\big(r \times (R^T x)\big)$$

$$M = M_0$$

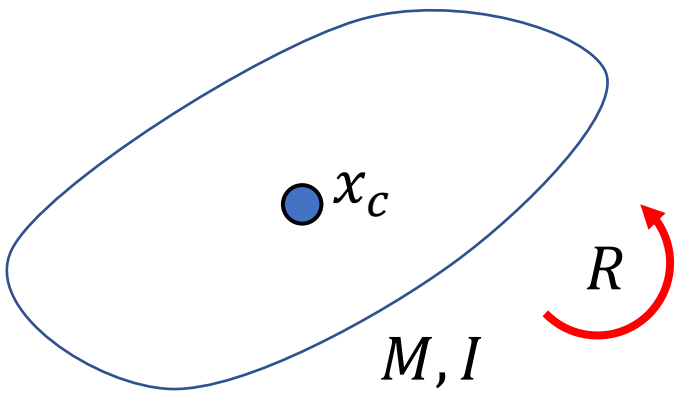
$$[Rr]_\times = R[r]_\times R^T$$

$$I = R I_0 R^T$$

$$[Rr]_\times^2 = R[r]_\times^2 R^T$$

$$\boldsymbol{I} = \sum_i -m_i [r_i]_\times^2$$

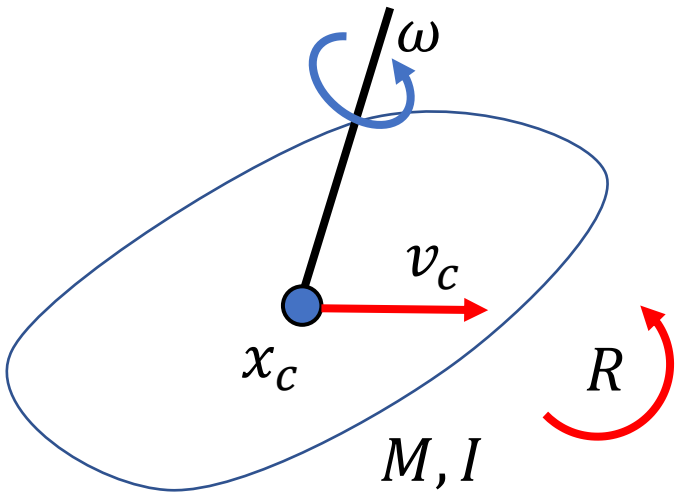
Principal Axes of Moment of Inertia



Eigendecomposition ⇒ $I = R I_0 R^T$

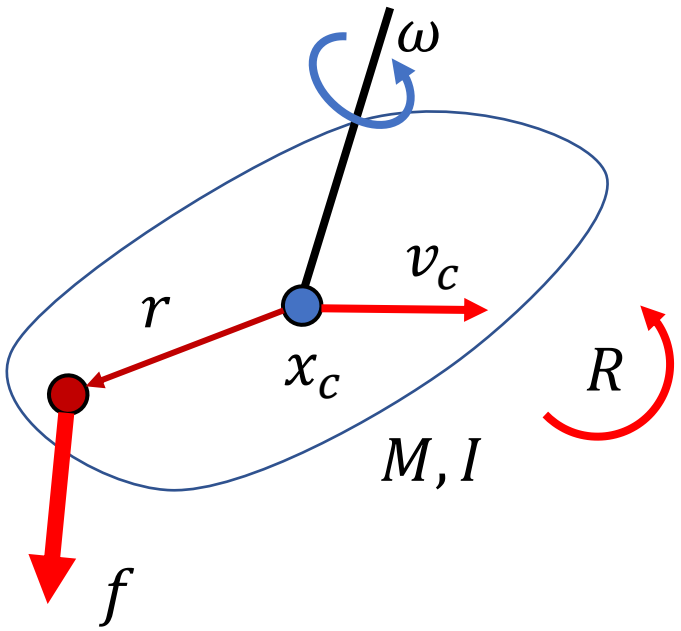
$$I_0 = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} = \mathrm{diag}(I_1, I_2, I_3)$$

Center of Momentum (CoM) Frame

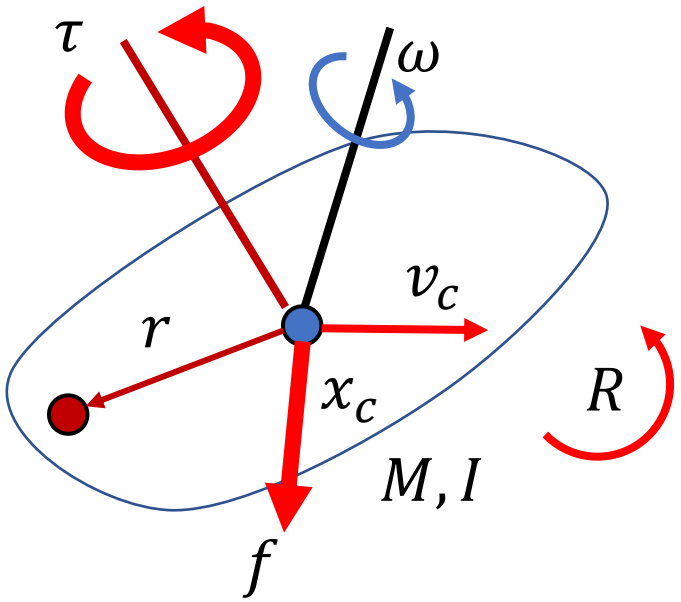


$$p = Mv_c \qquad\qquad L = I\omega$$

Force on a Rigid Body



Force on a Rigid Body

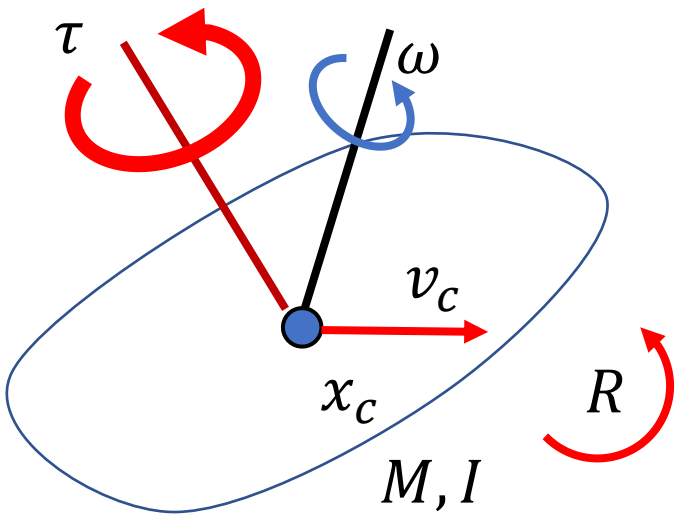


加了一个力

等效为质心上的力和力矩

$$\tau = r \times f$$

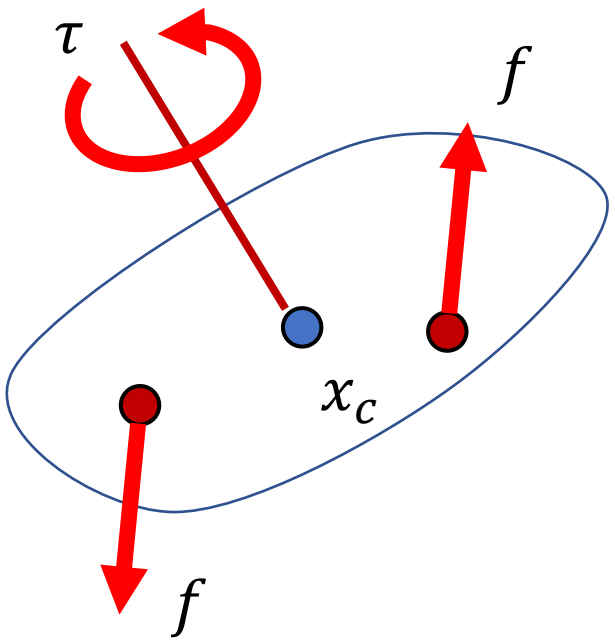
Torque on a Rigid Body



力偶，等效为只加一个力矩

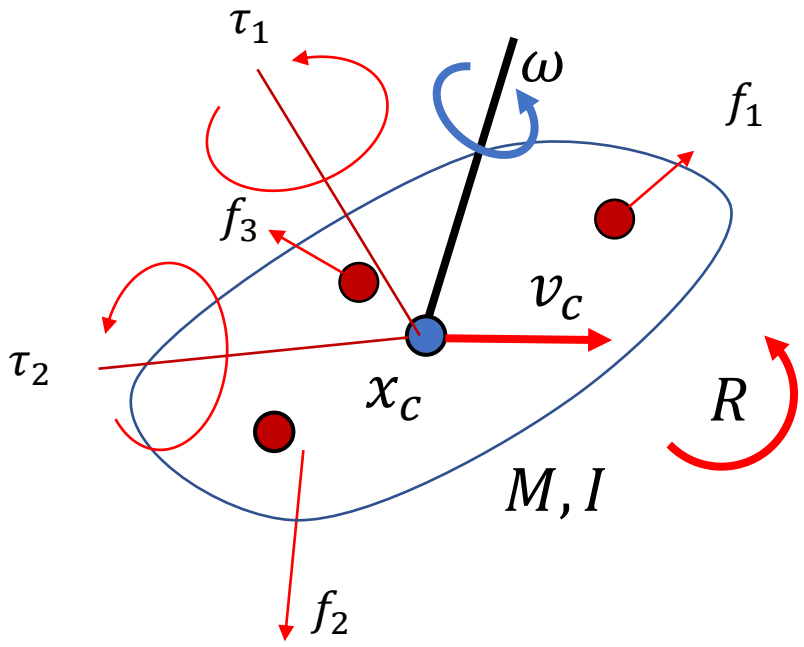
$$\tau = ???$$

Parallel Forces and Torques



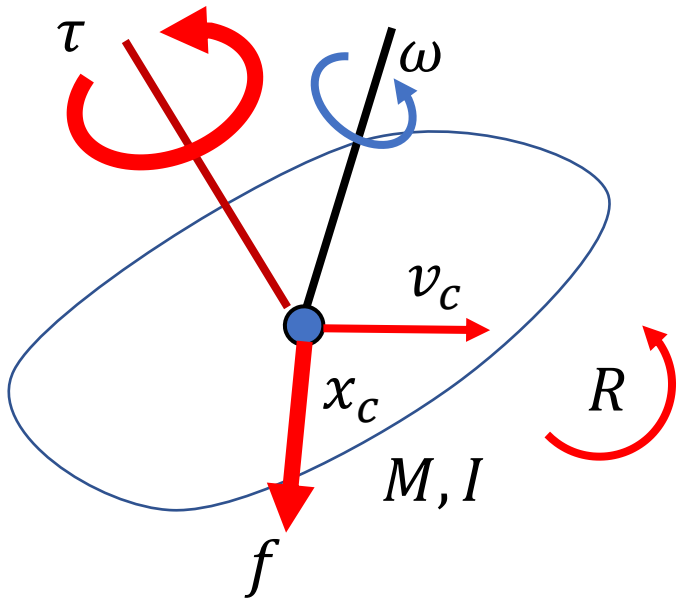
$$\tau = ???$$

Center of Momentum (CoM) Frame



$$p = Mv_c \qquad L = I\omega$$

Center of Momentum (CoM) Frame

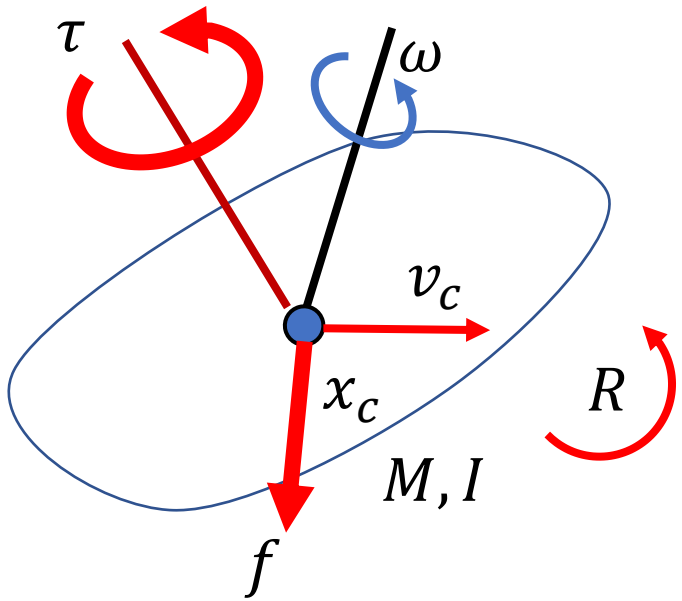


$$p = Mv_c \qquad L = I\omega$$

$$f = \Sigma_i f_i \qquad \tau = \Sigma_i \tau_i$$

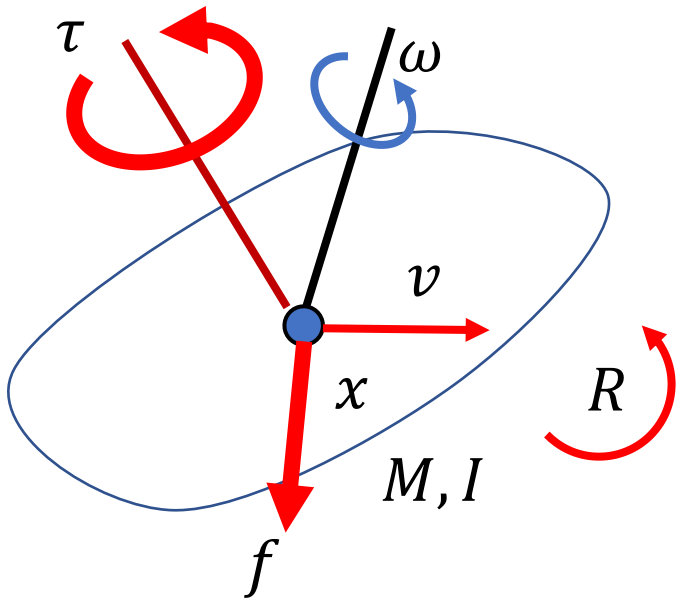
力带来动量的变化　　力矩带来角动量的变化

Equation of Motion of Rigid Body



Kinematics		Dynamics
	m, I	
x, R		p, L
v, ω		f, τ

Equation of Motion of Rigid Body



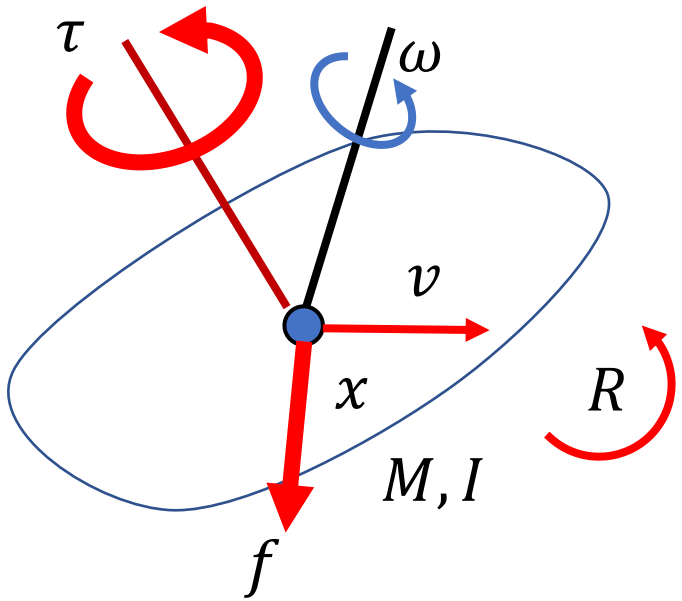
x, R, v, ω

$p = Mv$

$L = I\omega$

Newton's Second Law: $f = Ma$

Equation of Motion of Rigid Body



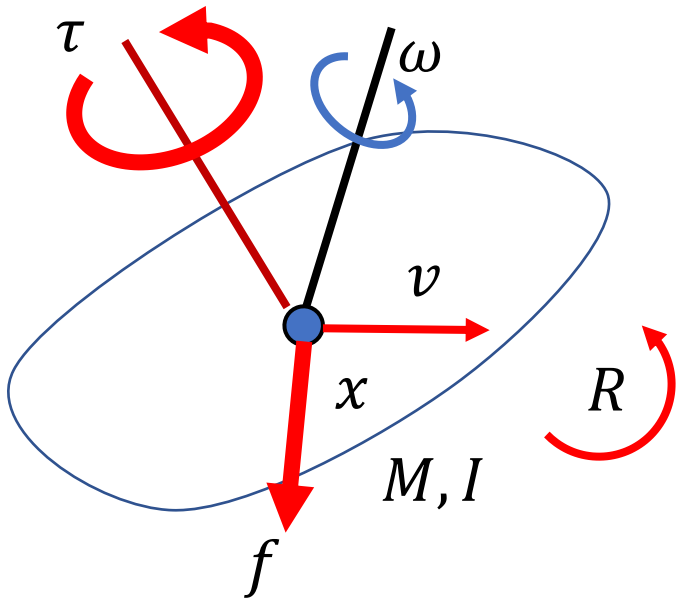
$$x, R, v, \omega$$

$$p = Mv$$

$$L = I\omega$$

Newton's Second Law: $$\frac{dp}{dt} = f$$

Equation of Motion of Rigid Body



x, R, v, ω

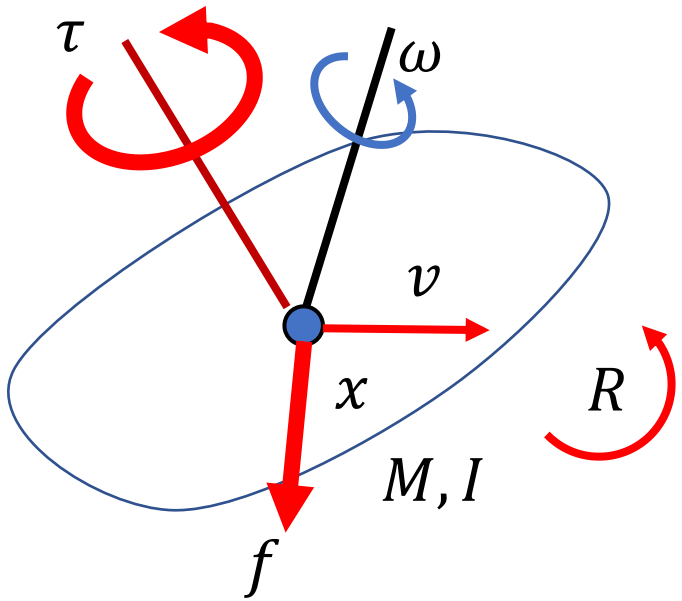
$p = Mv$

$L = I\omega$

Newton's Second Law: $$\frac{dp}{dt} = f$$

Euler's laws of motion: $$\frac{dL}{dt} = \tau$$

Equation of Motion of Rigid Body



$$x, R, v, \omega$$

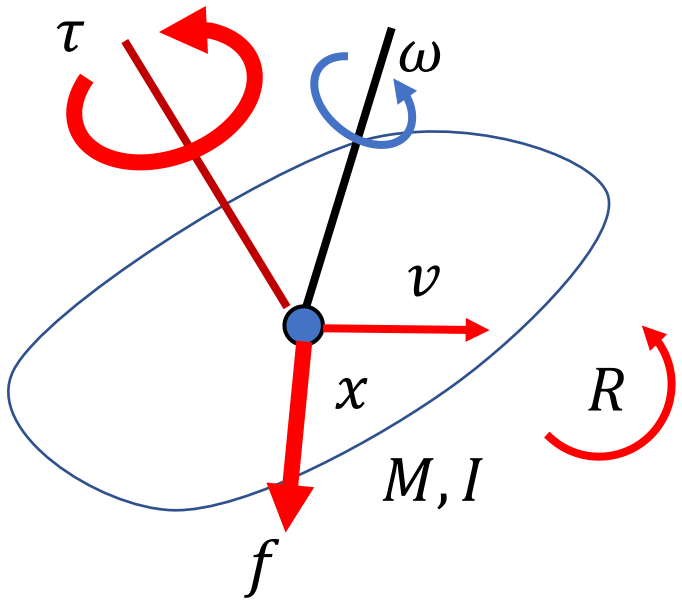
$$p = Mv$$

$$L = I\omega$$

Newton's Second Law: $\dfrac{dp}{dt} = f$ ➡ $M\dot{v} = f$

Euler's laws of motion: $\dfrac{dL}{dt} = \tau$ ➡ $I\dot{\omega} + \dot{I}\omega = \tau$

Equation of Motion of Rigid Body



$$x, R, v, \omega$$

$$p = Mv$$

$$L = I\omega$$

$$\dot{I} = \frac{d}{dt}(RI_0R^T)$$

$$= \dot{R}I_0R^T + RI_0\dot{R}^T$$

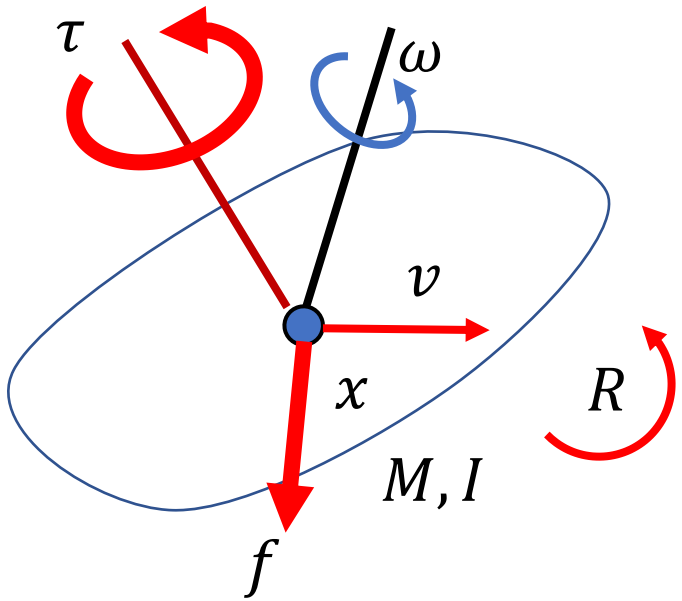
$$= [\omega]_\times RI_0R^T + RI_0R^T[\omega]_\times^T$$

$$\dot{I}\omega = \omega \times I\omega + I(-\omega \times \omega)$$

Newton's Second Law: $\frac{dp}{dt} = f$ ➡ $M\dot{v} = f$

Euler's laws of motion: $\frac{dL}{dt} = \tau$ ➡ $I\dot{\omega} + \omega \times I\omega = \tau$

Newton–Euler Equations



x, R, v, ω

$p = mv$

$L = I\omega$

$$\begin{bmatrix} m\mathrm{I}_3 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \dot{v} \\ \dot{\omega} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega \times I\omega \end{bmatrix} = \begin{bmatrix} f \\ \tau \end{bmatrix}$$

Numerical Integration

$$\begin{bmatrix} m\mathrm{I}_3 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \dot{v} \\ \dot{\omega} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega \times I\omega \end{bmatrix} = \begin{bmatrix} f \\ \tau \end{bmatrix}$$

$$\frac{1}{h} \begin{bmatrix} m\mathrm{I}_3 & 0 \\ 0 & I_n \end{bmatrix} \begin{bmatrix} v_{n+1} - v_n \\ \omega_{n+1} - \omega_n \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_n \times I_n\omega_n \end{bmatrix} = \begin{bmatrix} f \\ \tau \end{bmatrix}$$

隐式：把后两项换成n+1的时刻

Rigid Body Simulation

$$\frac{1}{h}\begin{bmatrix} m\mathrm{I}_3 & 0 \\ 0 & I_n \end{bmatrix}\begin{bmatrix} v_{n+1} - v_n \\ \omega_{n+1} - \omega_n \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_n \times I_n\omega_n \end{bmatrix} = \begin{bmatrix} f \\ \tau \end{bmatrix}$$

$$I_n = R_n I_0 R_n^T$$

$$v_{n+1} = \cdots$$

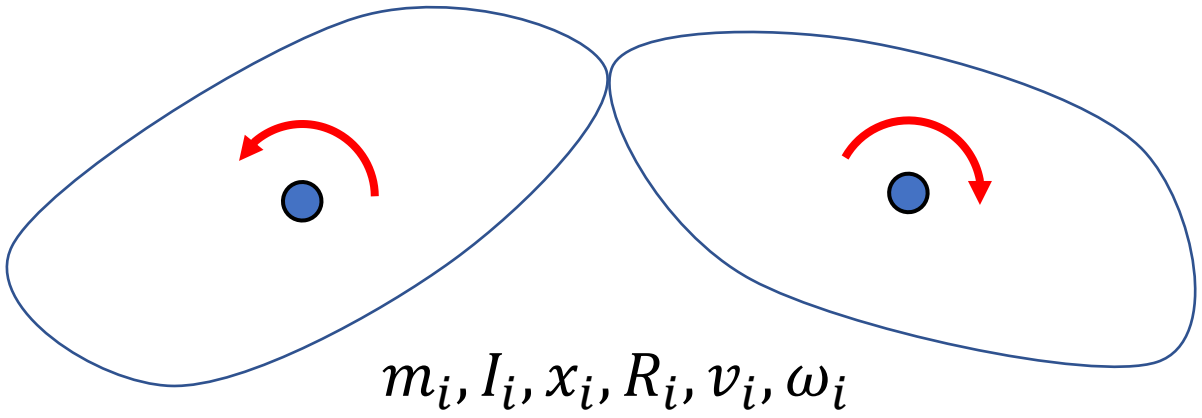
$$\omega_{n+1} = \cdots$$

再用半隐式欧拉

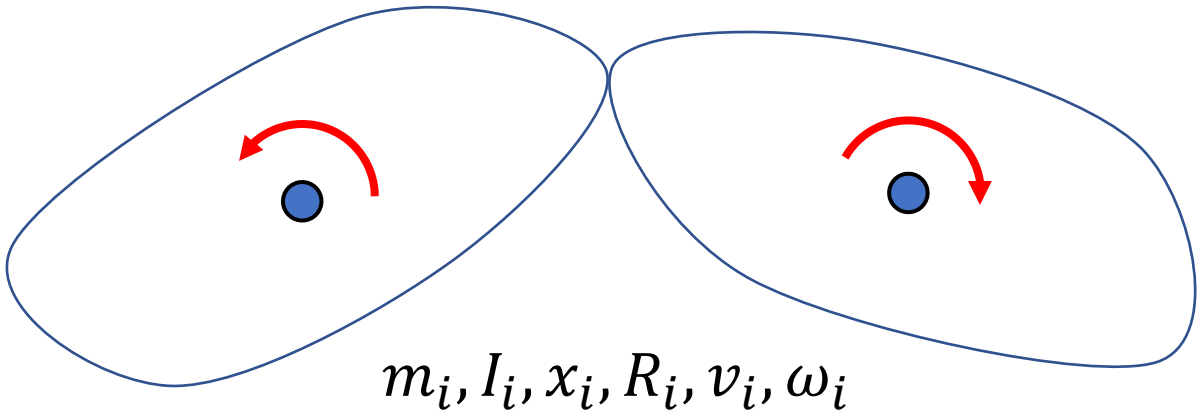
$$x_{n+1} = x_n + hv_{n+1}$$

$$q_{n+1} = q_n + \frac{1}{2}h\bar{\omega}_{n+1}q$$

A System with Two Links

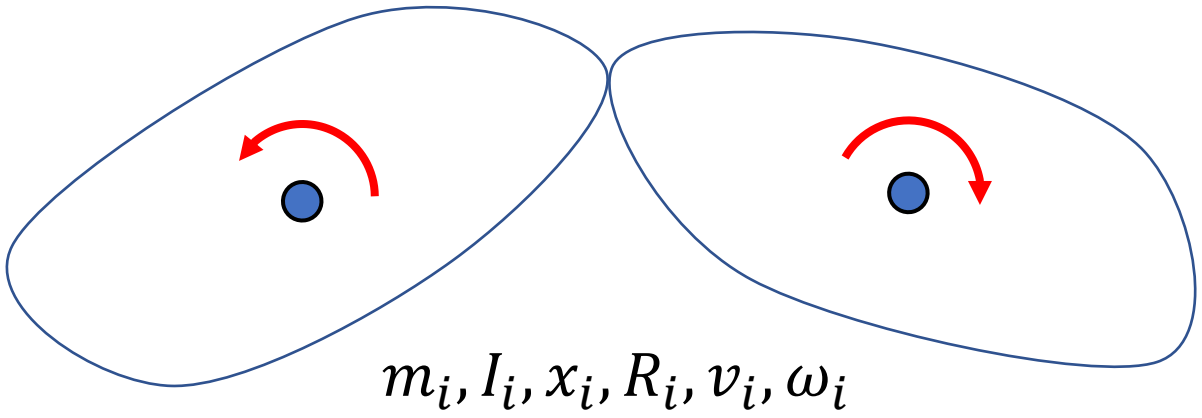


A System with Two Links



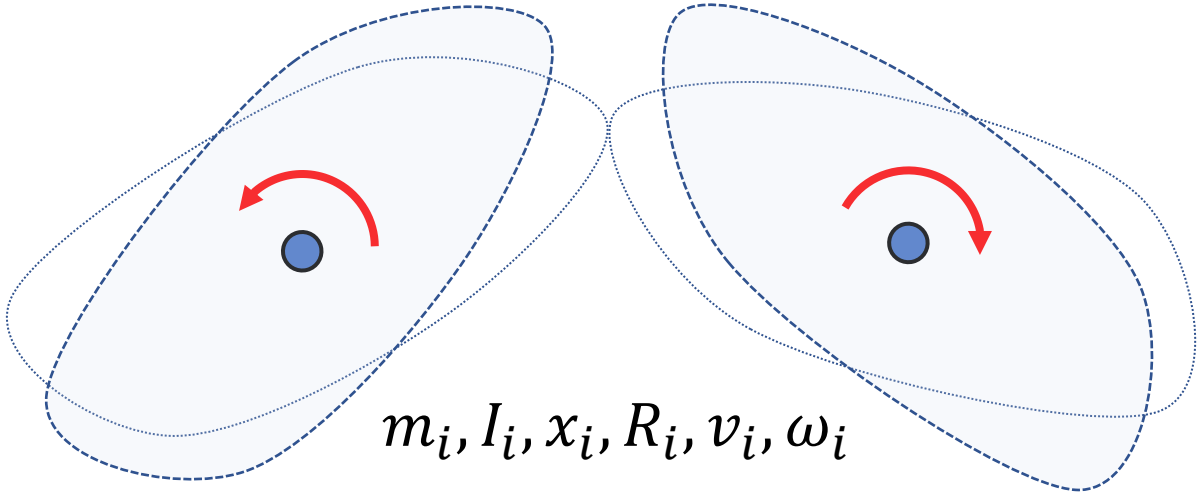
$$\begin{bmatrix} m_1 \mathrm{I}_3 & & & \\ & I_1 & & \\ & & m_2 \mathrm{I}_3 & \\ & & & I_2 \end{bmatrix} \begin{bmatrix} \dot{v}_1 \\ \dot{\omega}_1 \\ \dot{v}_2 \\ \dot{\omega}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_1 \times I_1 \omega_1 \\ 0 \\ \omega_2 \times I_2 \omega_2 \end{bmatrix} = \begin{bmatrix} f_1 \\ \tau_1 \\ f_2 \\ \tau_2 \end{bmatrix}$$

A System with Two Links



$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f}$$

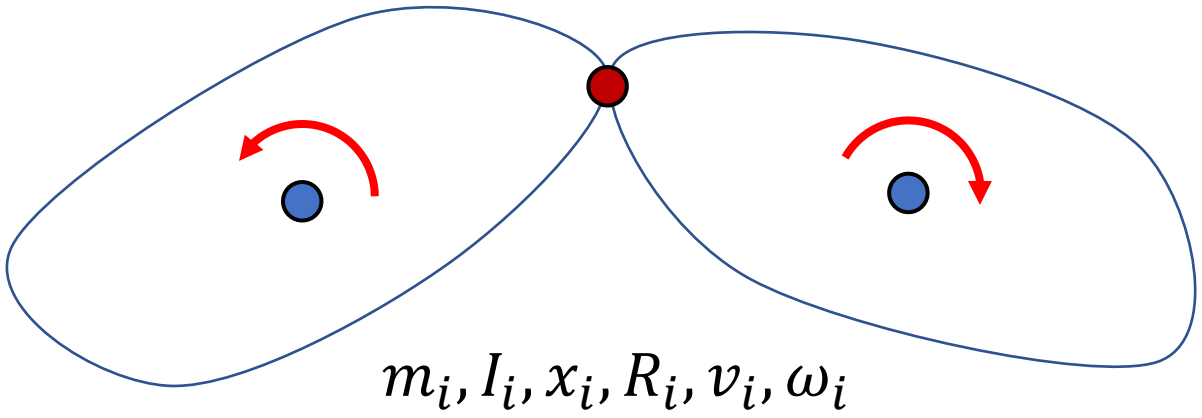
A System with Two Links



$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f}$$

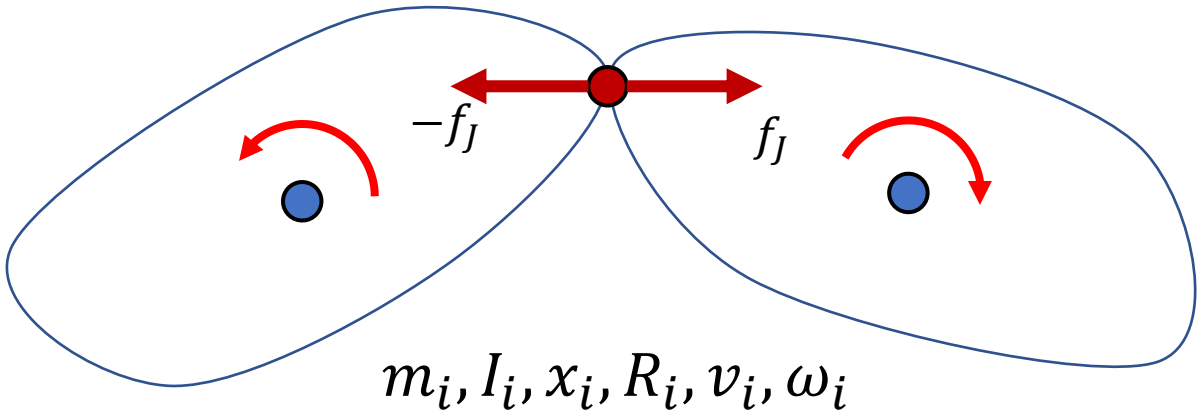
简写

A System with Two Links and a Joint



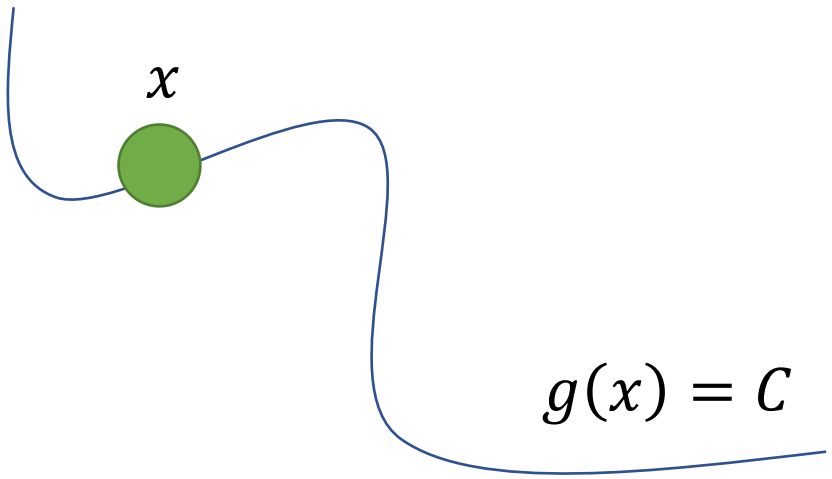
$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f}$$

A System with Two Links and a Joint



$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f} + \boldsymbol{f}_J$$

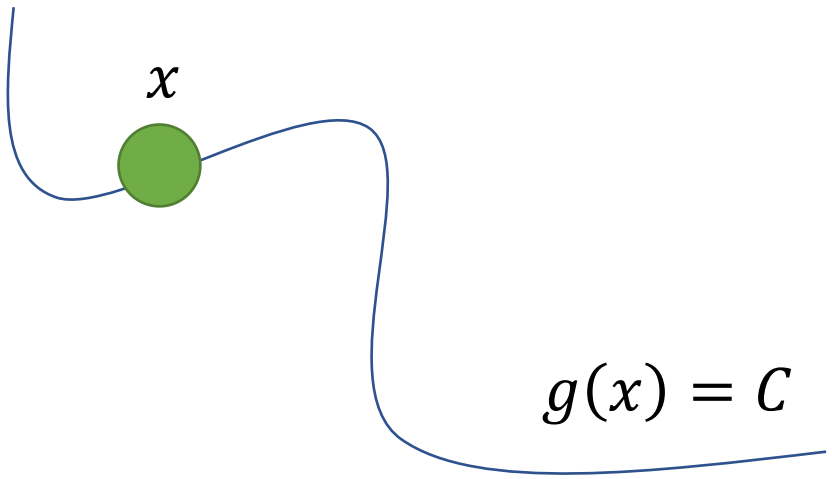
Constraints



$$g(\boldsymbol{x}) = C$$

$$\frac{d}{dt} g(\boldsymbol{x}) = 0$$

Constraints

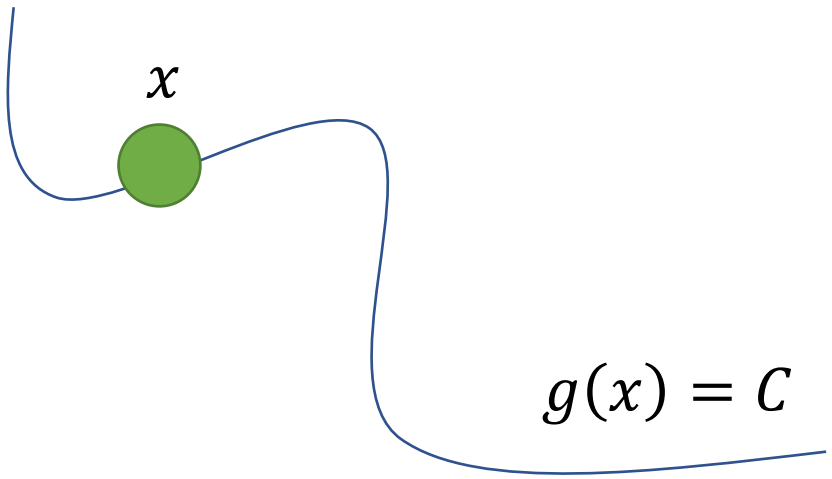


$$g(\boldsymbol{x}) = C$$

$$\frac{d}{dt} g(\boldsymbol{x}) = 0$$

$$\frac{\partial g}{\partial \boldsymbol{x}} \cdot \frac{d\boldsymbol{x}}{dt} = 0$$

Constraints



$$g(\boldsymbol{x}) = C$$

$$\frac{d}{dt} g(\boldsymbol{x}) = 0$$

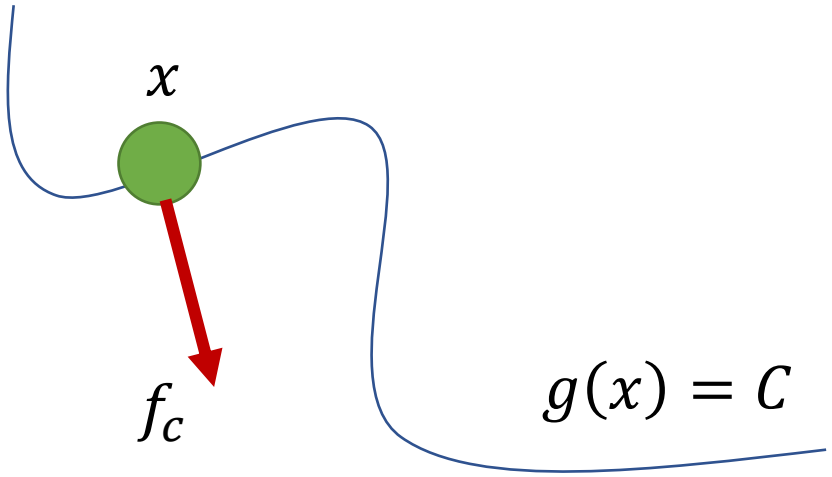
$$J\boldsymbol{v} = 0$$

$$J = [\nabla g]^T$$

Constraint Force

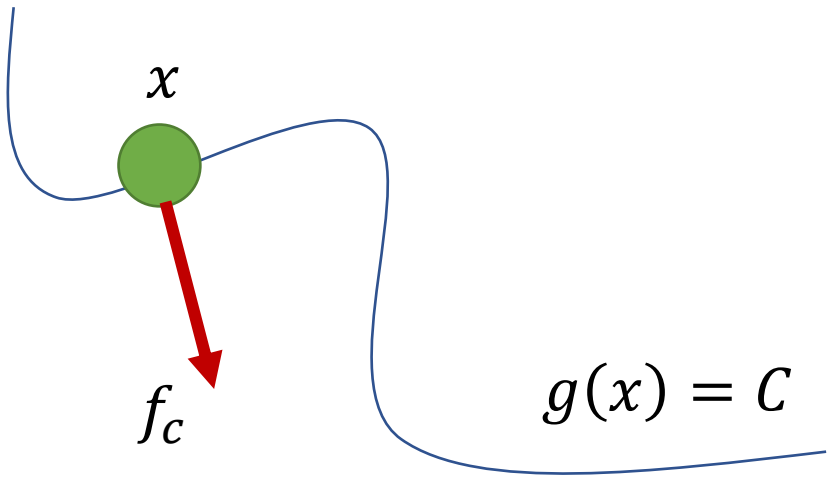
约束力不会给系统能量，也不会造成能量损失

* Constraint is passive
No energy gain or loss!!!



$$f_c \cdot v = 0$$

Constraint Force



* Constraint is passive
No energy gain or loss!!!

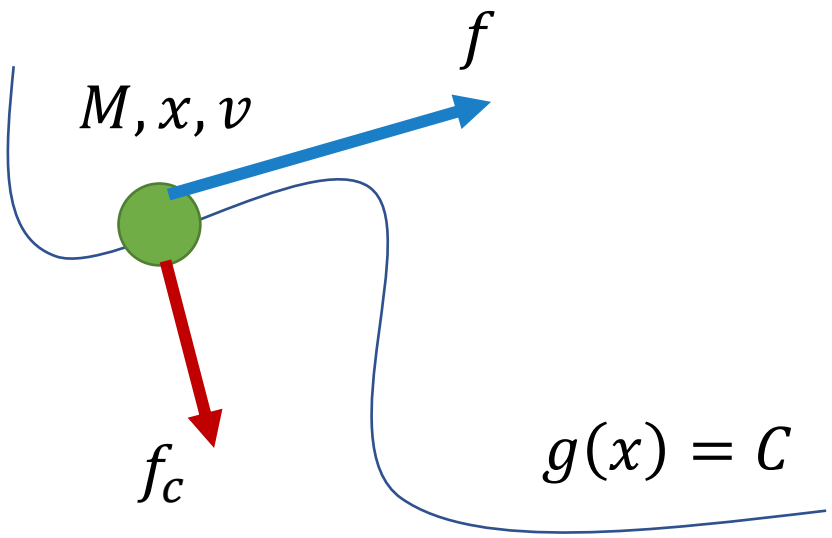
$$f_c \cdot v = 0 \Leftrightarrow f_c^T v = 0$$

$$Jv = 0$$

$$f_c = J^T \boldsymbol{\lambda}$$

unknown

Equation of Motion with Constraints



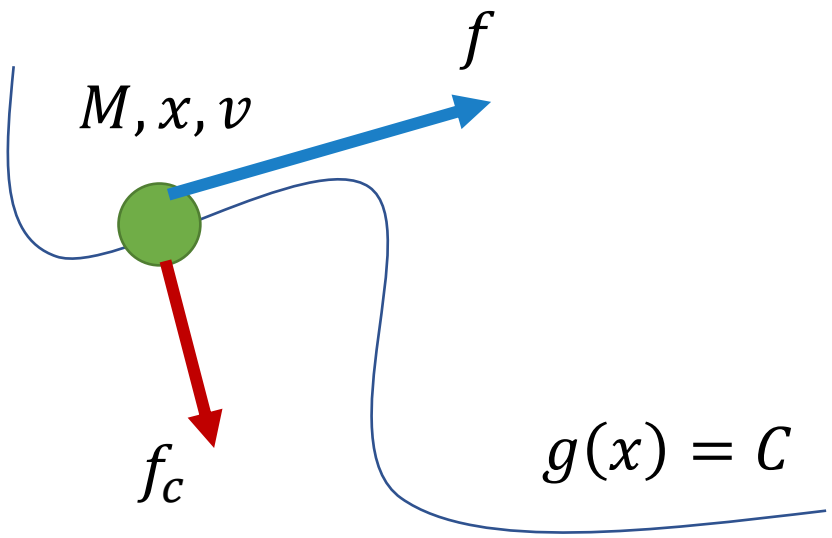
$$M\dot{v} = f + J^T\lambda$$

$$Jv = 0$$

$$M\frac{v_{n+1} - v_n}{h} = f + J^T\lambda$$

$$Jv_{n+1} = 0$$

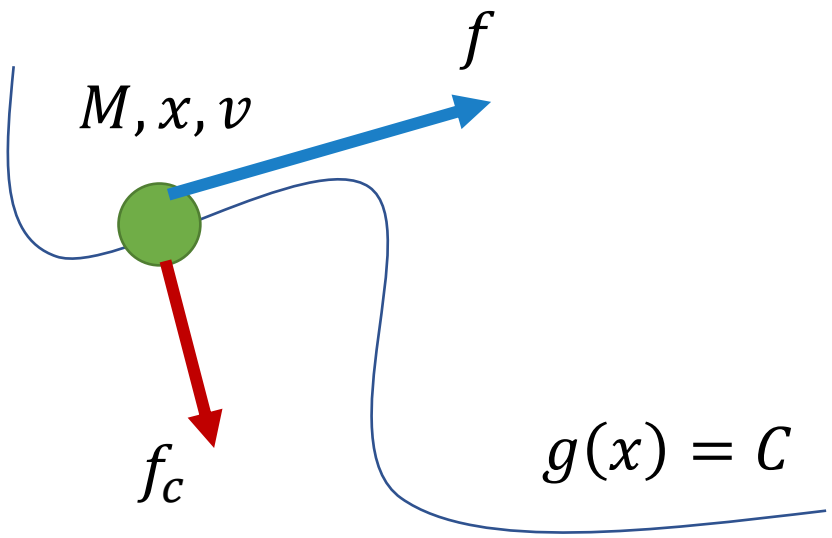
Numerical Solution



$$M\frac{v_{n+1} - v_n}{h} = f + J^T\lambda$$

$$Jv_{n+1} = 0$$

Numerical Solution



$$M\frac{v_{n+1} - v_n}{h} = f + J^T\lambda$$

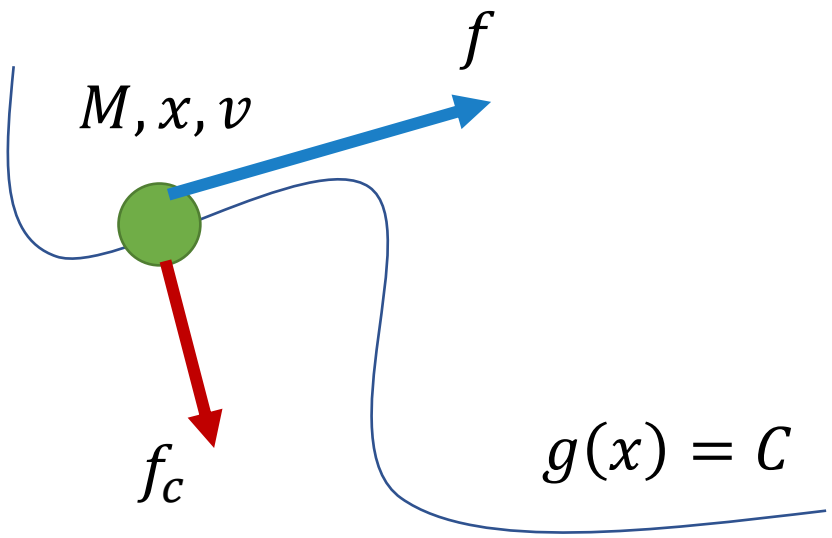
$$Jv_{n+1} = \mathbf{0}$$

给一个小的量

$$Jv_{n+1} = \alpha\frac{C - g(x_n)}{h}$$

Correction of numerical errors
α: error reduction parameter (ERP)

Numerical Solution



$$M\frac{v_{n+1} - v_n}{h} = f + J^T\lambda$$

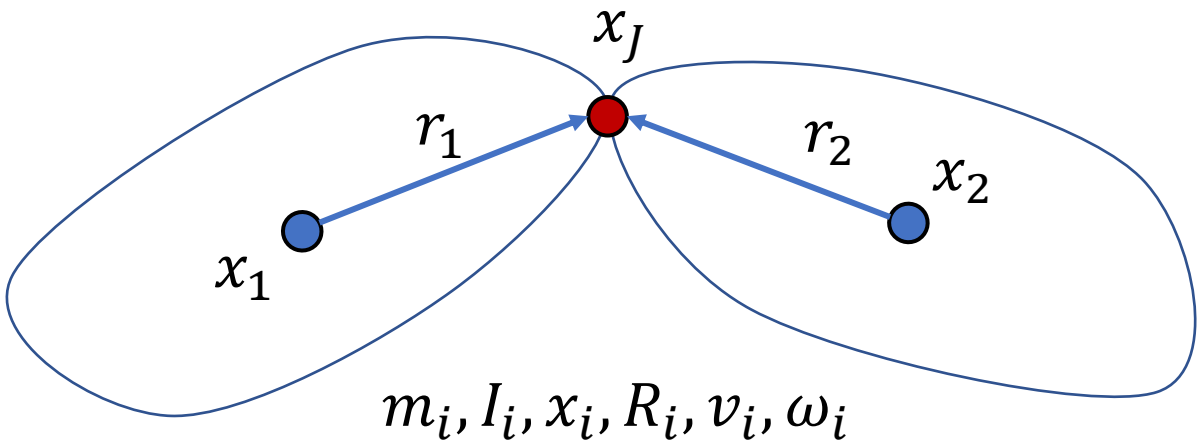
$$Jv_{n+1} = b_n$$

$$JM^{-1}J^T\lambda = c_n$$

$$(JM^{-1}J^T + \beta\mathbf{I})\lambda = c_n$$

β: constraint force mixing (CFM)

Joint Constraint

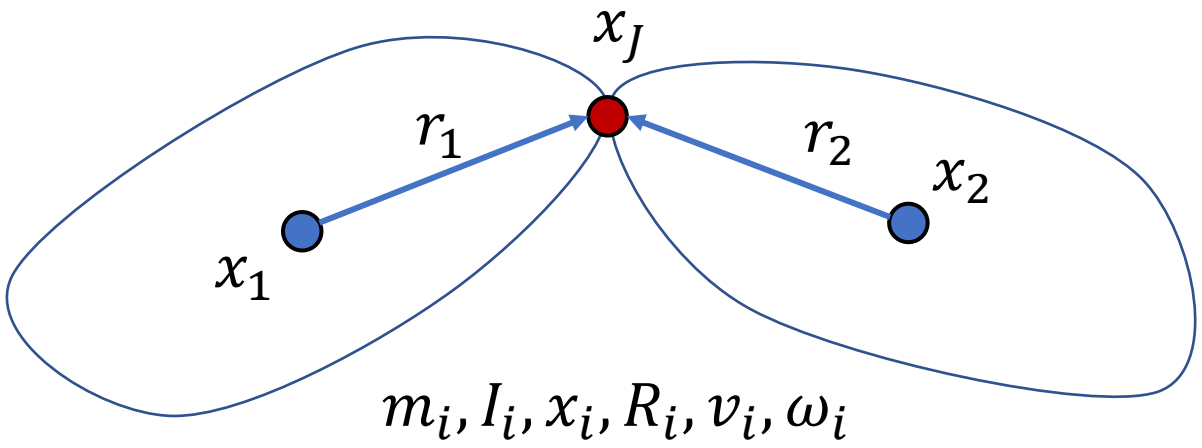


$$x_1 + R_1 r_1 = x_J = x_2 + R_2 r_2$$

$$\xrightarrow{d/dt}$$

$$v_1 + \omega_1 \times R_1 r_1 = v_2 + \omega_2 \times R_2 r_2$$

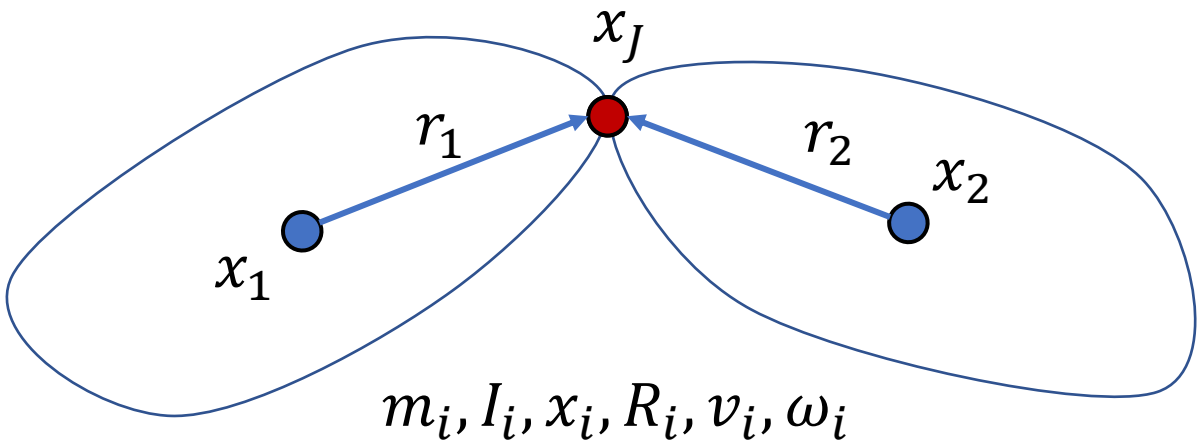
Joint Constraint



$$\begin{bmatrix} I_3 & -[r_1]_\times & -I_3 & [r_2]_\times \end{bmatrix} \begin{bmatrix} v_1 \\ w_1 \\ v_2 \\ w_2 \end{bmatrix} = 0$$

$$Jv = 0$$

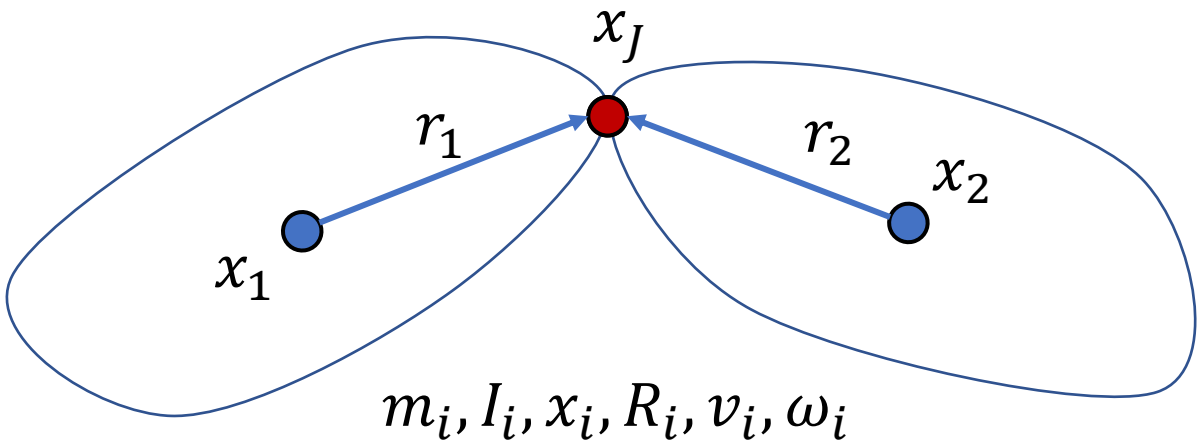
A System with Two Links and a Joint



$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f} + \boldsymbol{J}^{\boldsymbol{T}}\boldsymbol{\lambda}$$

$$Jv = 0$$

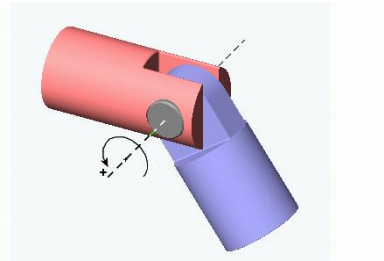
A System with Two Links and a Joint



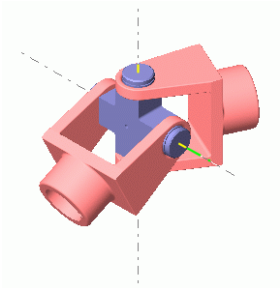
$$\begin{bmatrix} m_1 \mathrm{I}_3 & & & \\ & I_1 & & \\ & & m_2 \mathrm{I}_3 & \\ & & & I_2 \end{bmatrix} \begin{bmatrix} \dot{v}_1 \\ \dot{\omega}_1 \\ \dot{v}_2 \\ \dot{\omega}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_1 \times I_1 \omega_1 \\ 0 \\ \omega_2 \times I_2 \omega_2 \end{bmatrix} = \begin{bmatrix} f_1 \\ \tau_1 \\ f_2 \\ \tau_2 \end{bmatrix} + \begin{bmatrix} I_3 \\ [r_1]_\times \\ -I_3 \\ -[r_2]_\times \end{bmatrix} \lambda$$

$$Jv = 0$$

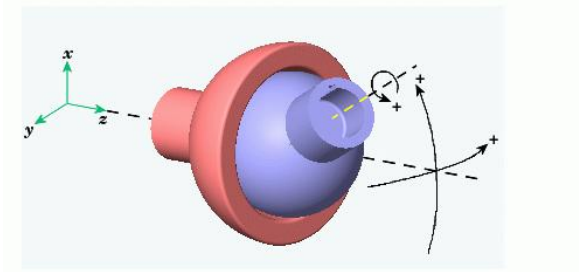
Different Types of Joints



Hinge joint
Revolute joint



Universal joint

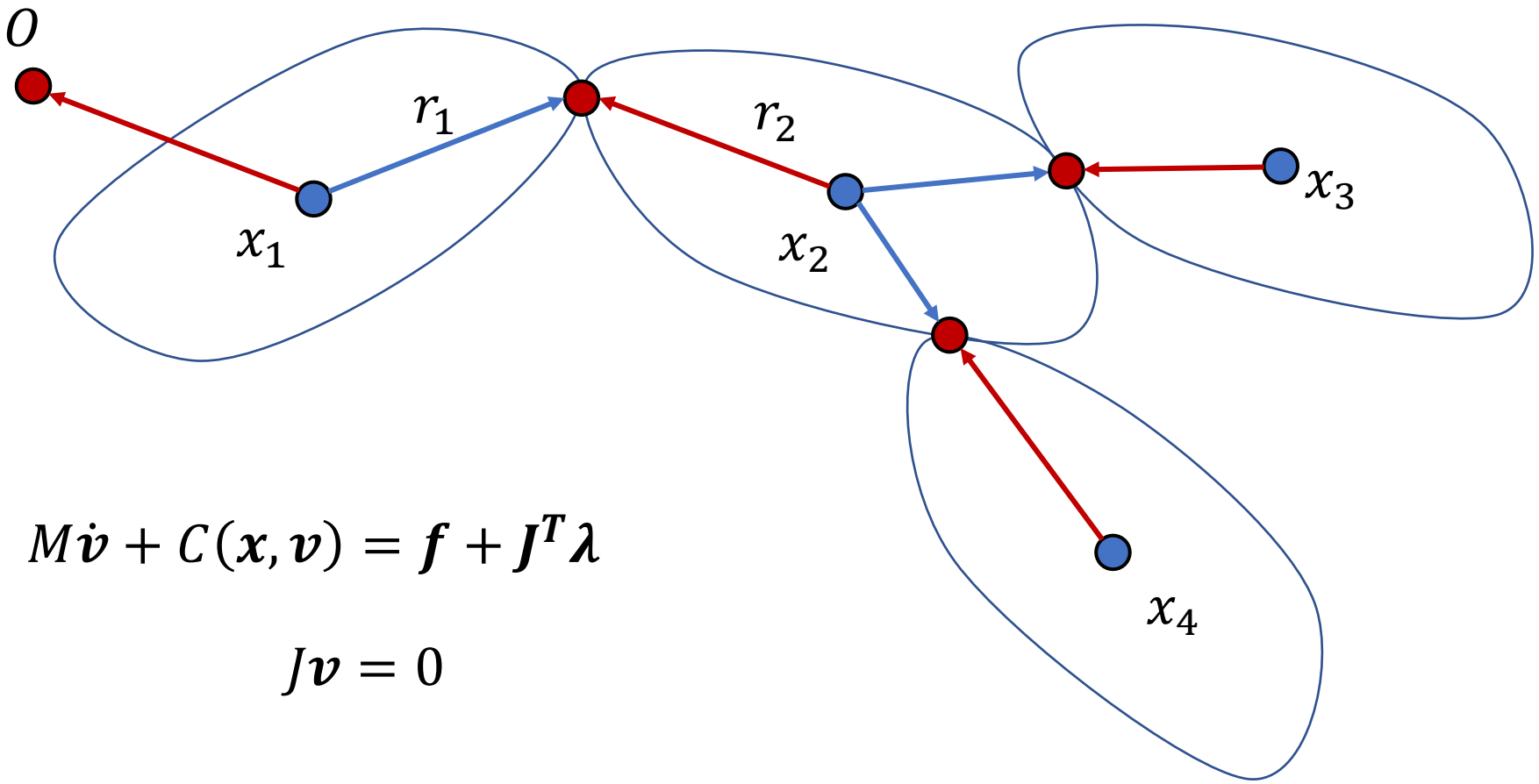


Ball-and-socket

$$\begin{bmatrix} I_3 & -[r_1]_\times & -I_3 & [r_2]_\times \\ ? & ? & ? & ? \end{bmatrix} \begin{bmatrix} v_1 \\ w_1 \\ v_2 \\ w_2 \end{bmatrix} = 0$$

A System with Many Links Joints

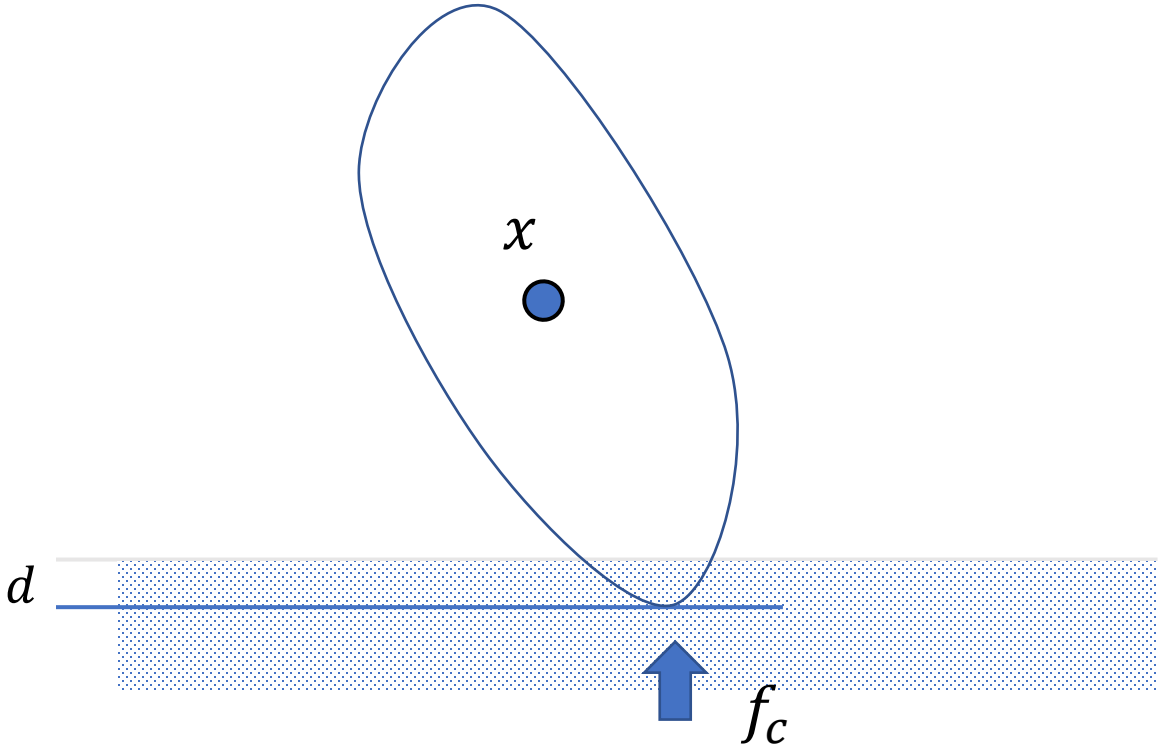
$$m_i, I_i, x_i, R_i, v_i, \omega_i$$



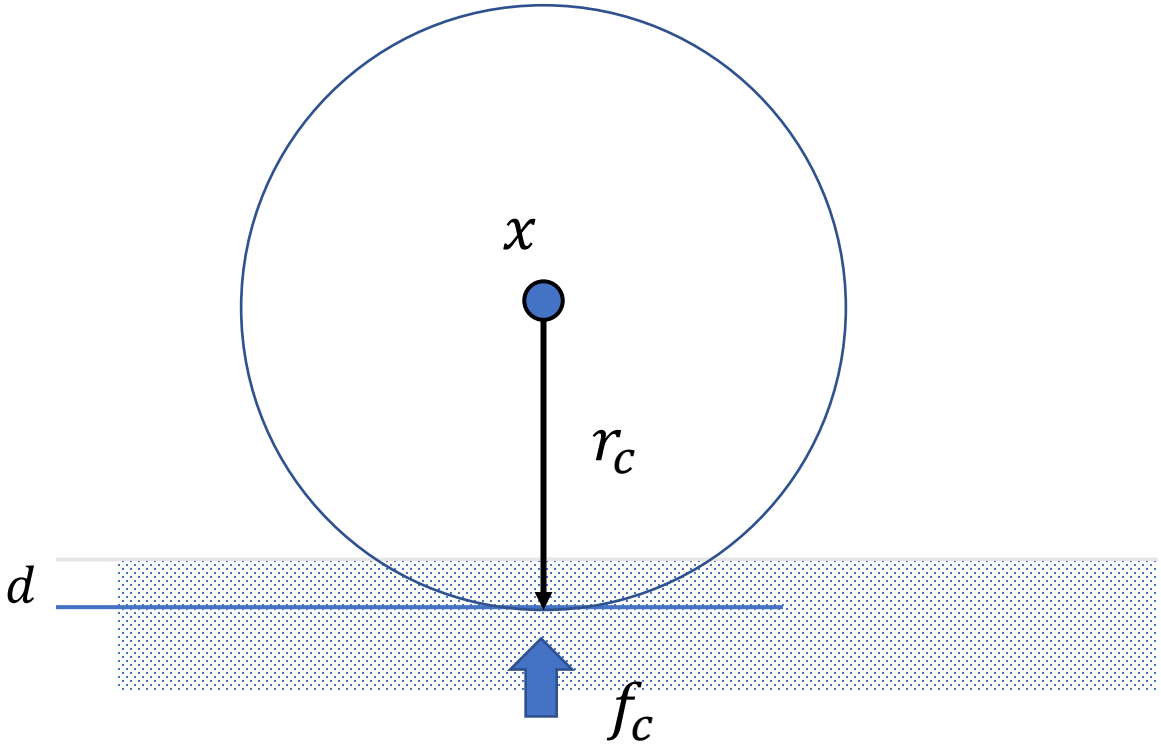
$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f} + \boldsymbol{J}^{\boldsymbol{T}}\boldsymbol{\lambda}$$

$$J\boldsymbol{v} = 0$$

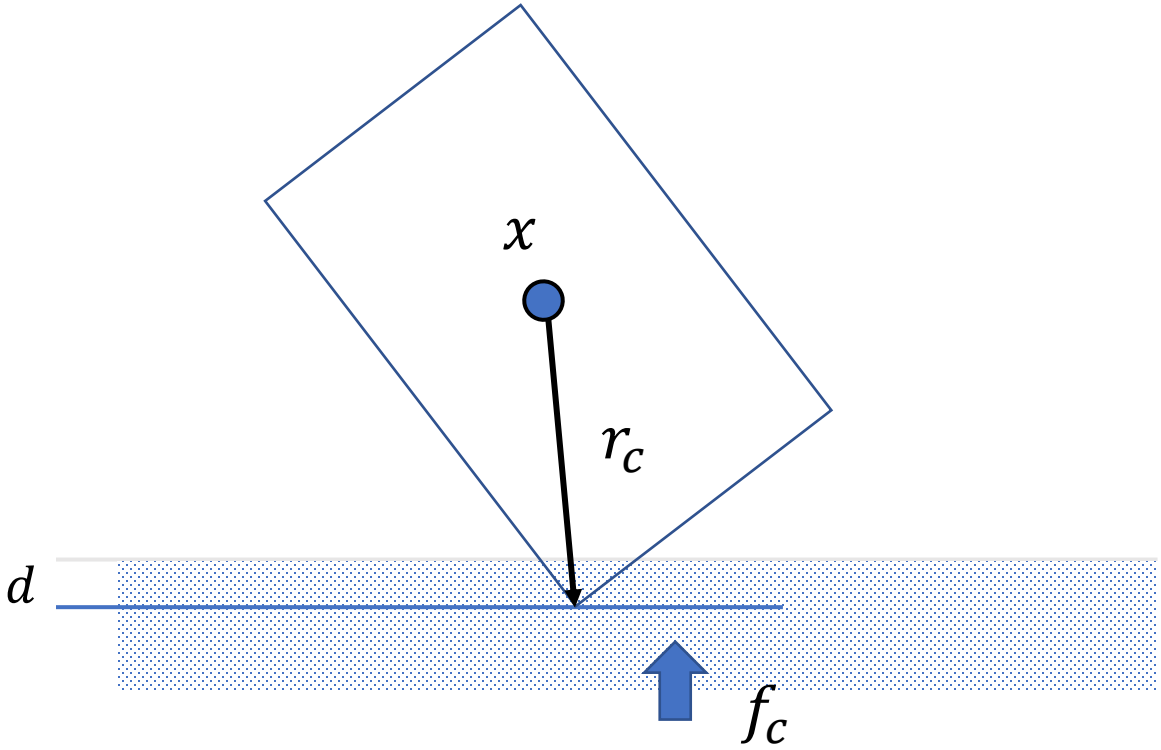
Contacts



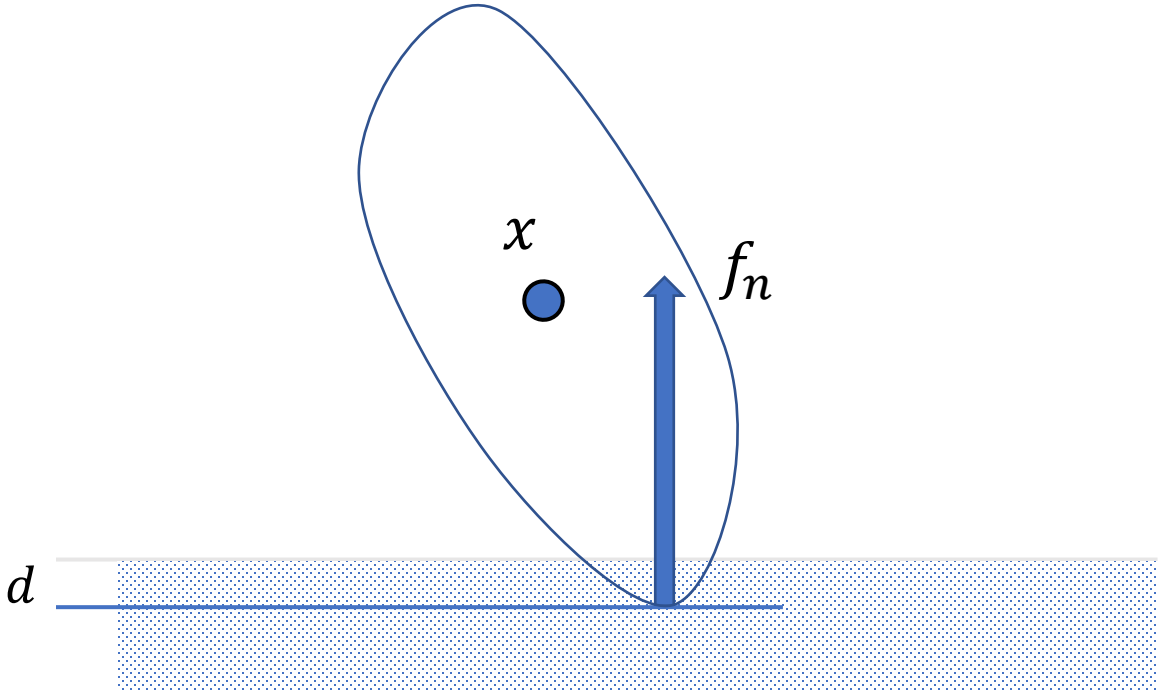
Contact Detection 碰撞检测



Contact Detection

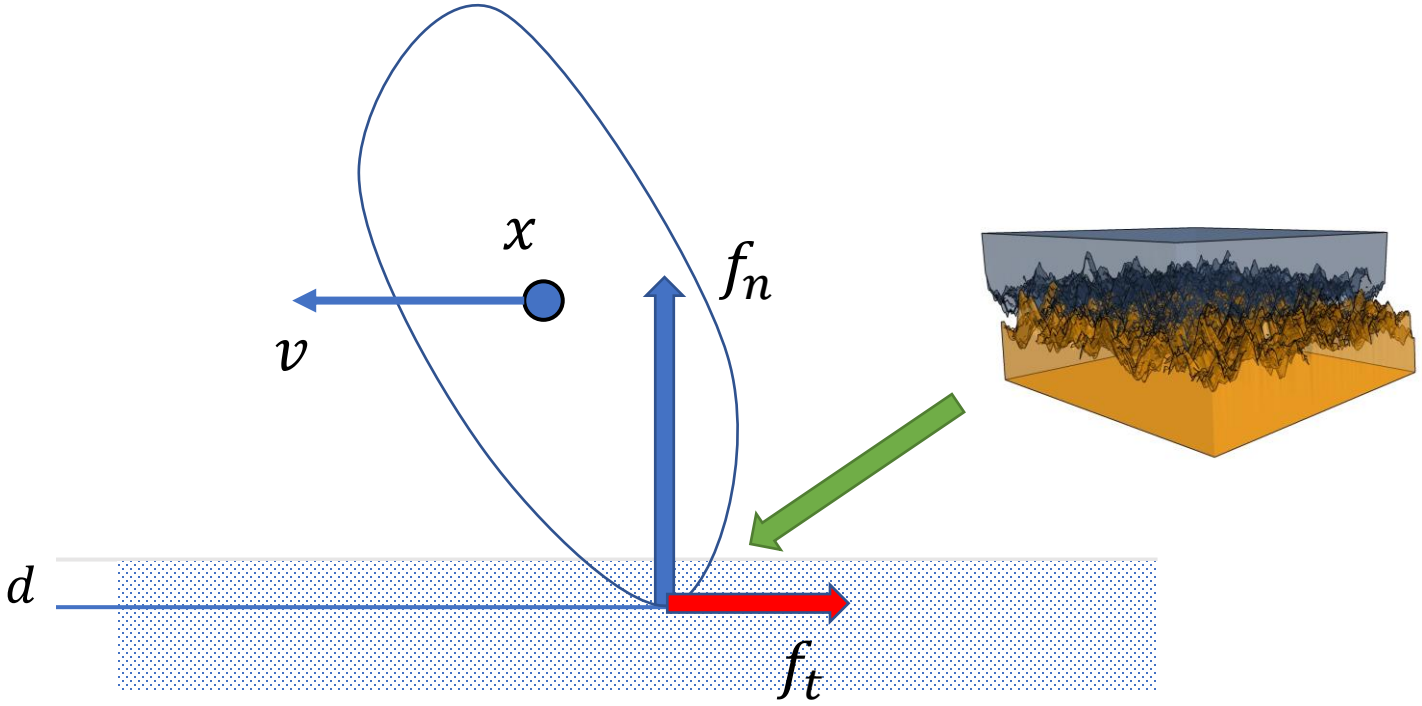


Penalty-based Contact Model



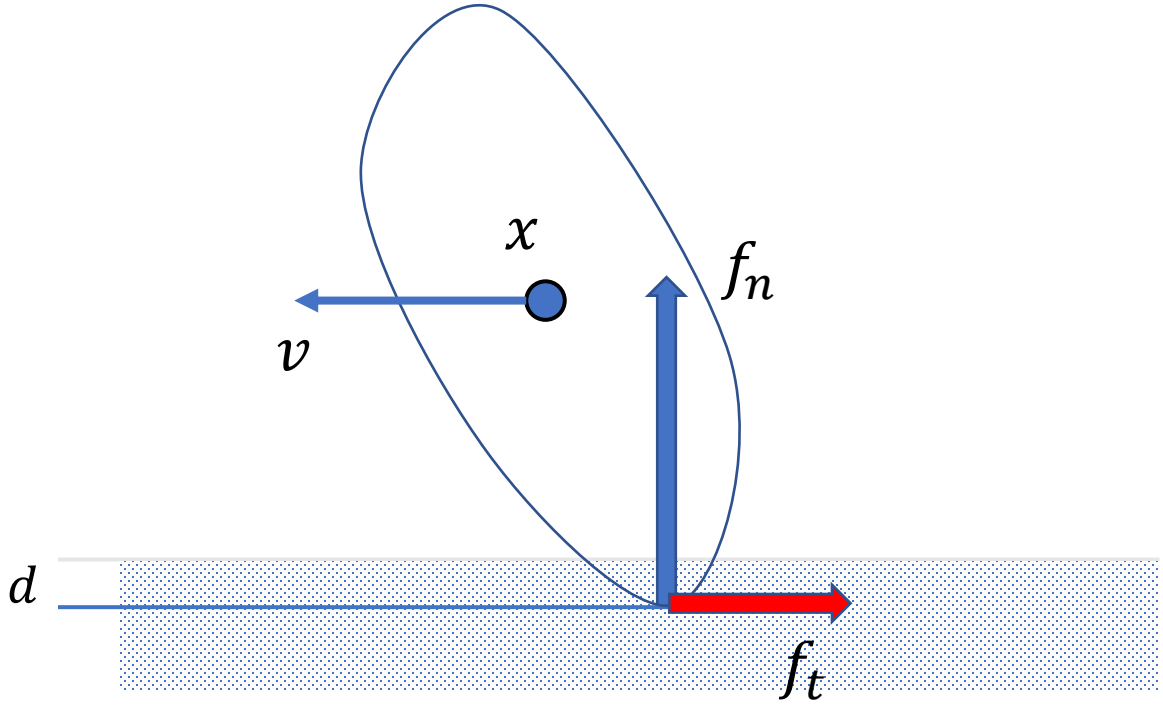
$$f_n = -k_p d - k_d v_{c,\perp}$$

Frictional Contact 摩擦



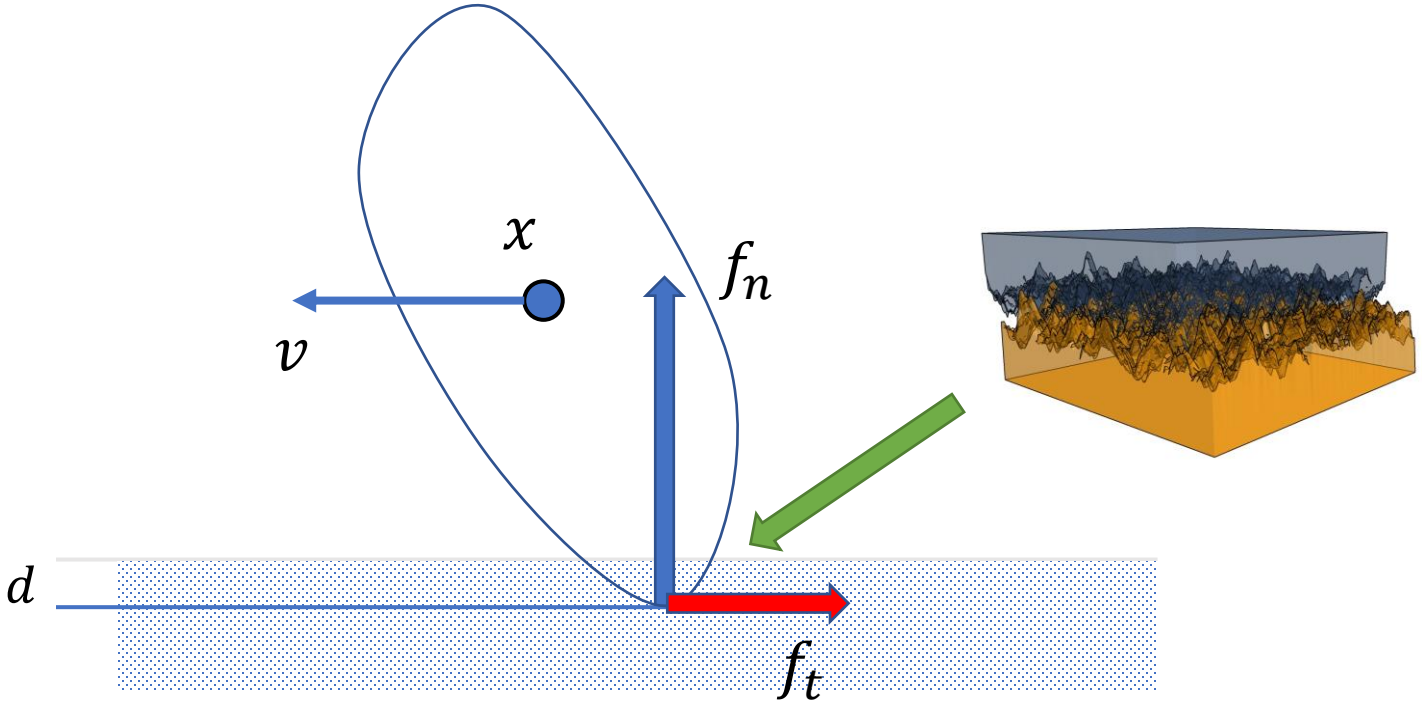
Coulomb's law of friction: $|f_t| = \mu f_n$

Frictional Contact



$$f_n = -k_p d - k_d v_{c,\perp} \qquad f_t = -\mu f_n \frac{v_{c,\parallel}}{\|v_{c,\parallel}\|}$$

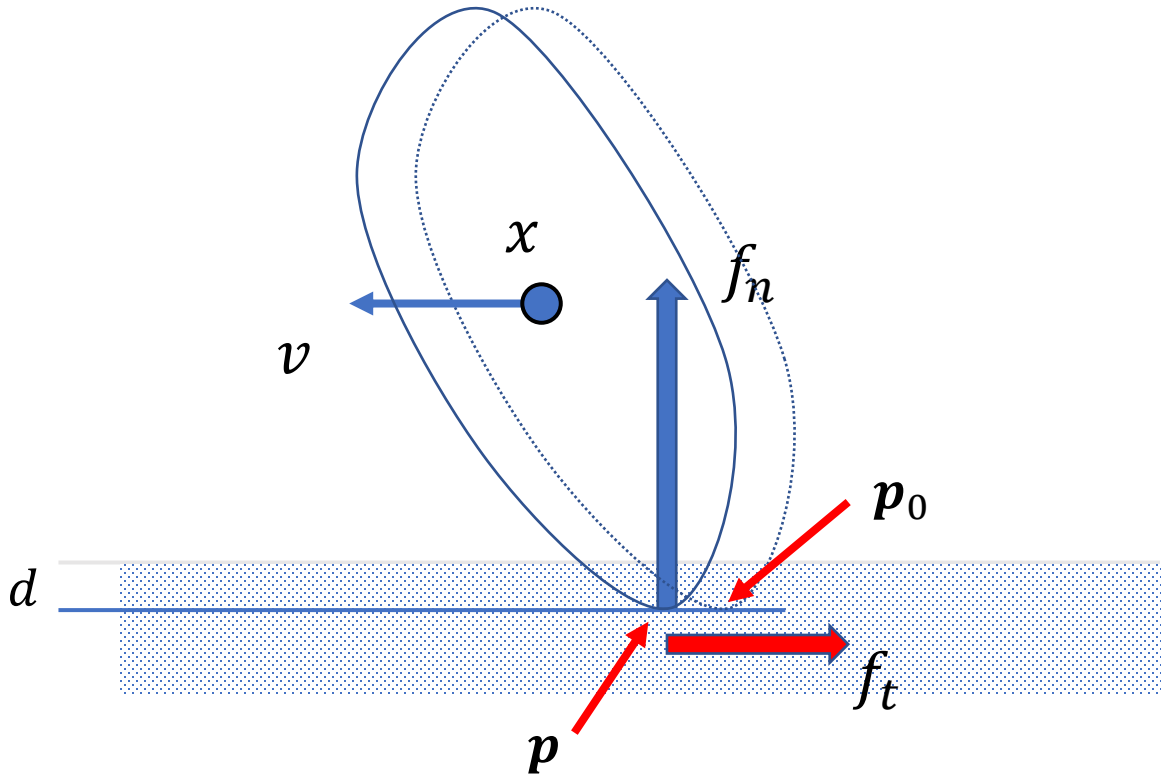
Frictional Contact



Coulomb's law of friction: $|f_t| \leq \mu f_n$

How to model static friction???

Frictional Contact



$$f_n = -k_p d - k_d v_{c,\perp}$$

Static friction: 静摩擦力

$$\bar{f}_t = k_p^f(\boldsymbol{p}_0 - \boldsymbol{p}) - k_d^f v_{c,\parallel}$$

Dynamic friction:

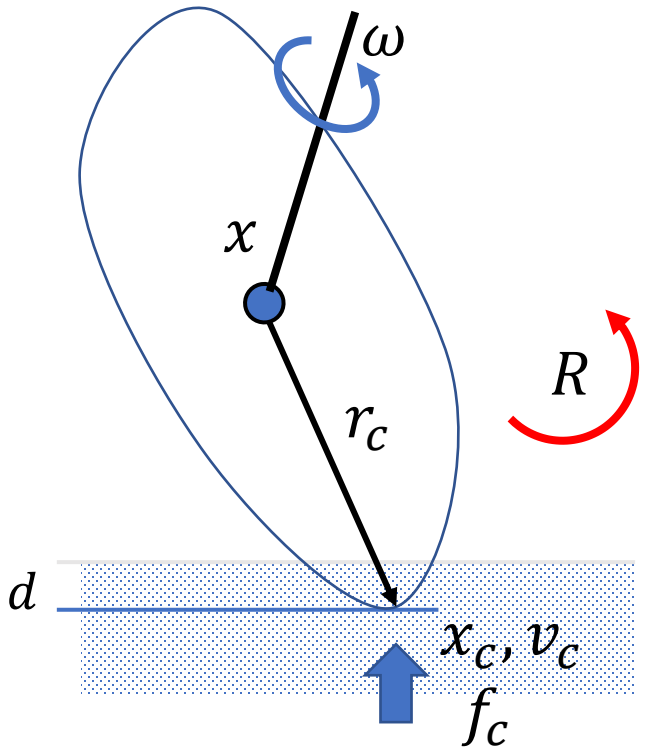
$$f_t = -\mu f_n \frac{v_{c,\parallel}}{\|v_{c,\parallel}\|} \qquad \text{if } \bar{f}_t \geq \mu f_n$$

Coulomb's law of friction: $|f_t| \leq \mu f_n$

[Yamane and Nakamura 2006 - Stable Penalty-Based Model of Frictional Contacts]

Contact as a Constraint

用约束而非罚项建模

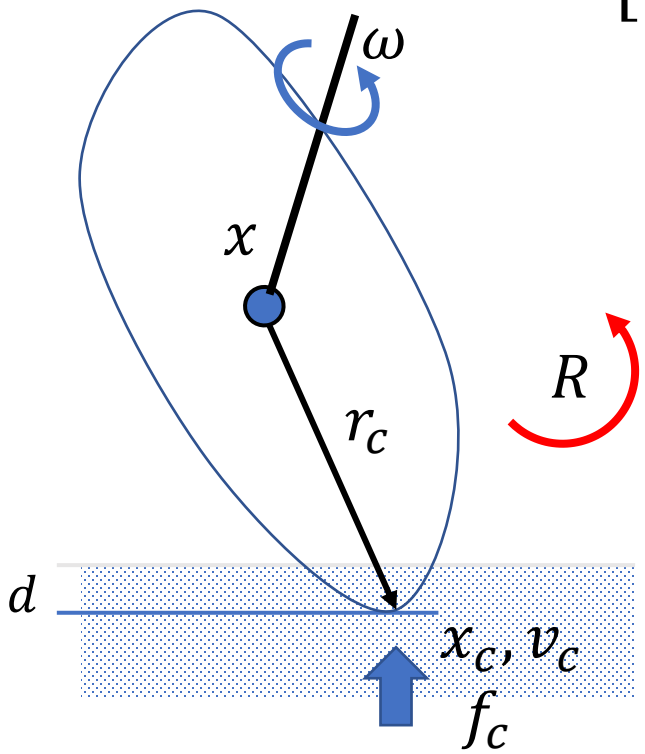


$$x_c = x + r_c$$

$$v_c = v + \omega \times r_c = J_c \begin{bmatrix} v \\ \omega \end{bmatrix}$$

$$v_{c,\perp} = v + \omega \times r_c = J_{c,\perp} \begin{bmatrix} v \\ \omega \end{bmatrix}$$

Contact as a Constraint

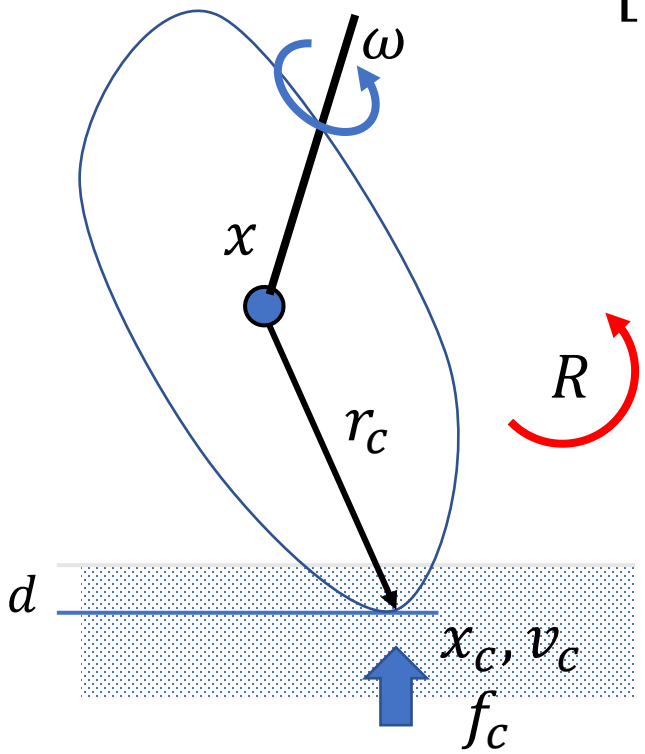


$$\begin{bmatrix} m\mathrm{I}_3 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \dot{v} \\ \dot{\omega} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega \times I\omega \end{bmatrix} = \begin{bmatrix} f \\ \tau \end{bmatrix} + J_{c,\perp}^T \lambda$$

$$v_{c,\perp} = J_{c,\perp} \begin{bmatrix} v \\ \omega \end{bmatrix} \geq 0$$

$$\lambda \geq 0$$

Contact as a Constraint



$$\begin{bmatrix} m\mathrm{I}_3 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \dot{v} \\ \dot{\omega} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega \times I\omega \end{bmatrix} = \begin{bmatrix} f \\ \tau \end{bmatrix} + J_{c,\perp}^T \lambda$$

$$v_{c,\perp} = J_{c,\perp} \begin{bmatrix} v \\ \omega \end{bmatrix} \geq 0$$

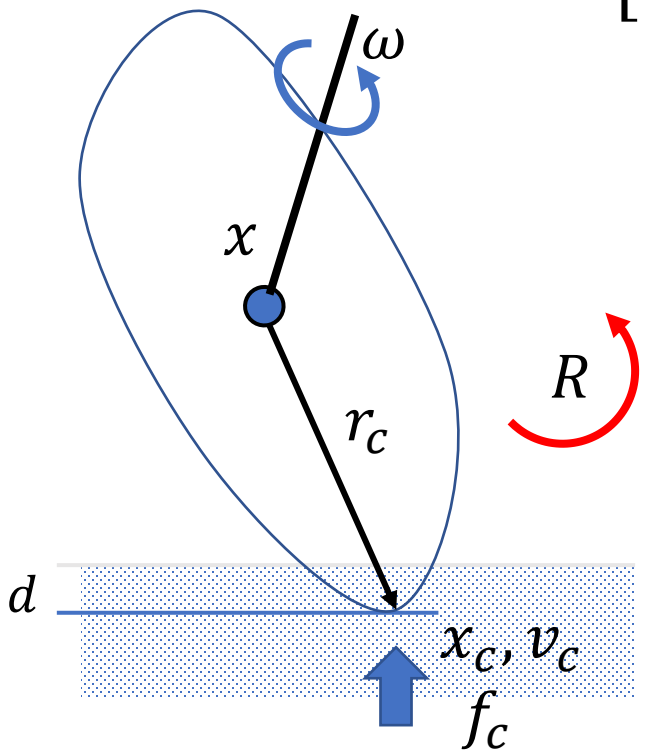
$$\lambda \geq 0$$

$$v_{c,\perp,j} > 0 \Rightarrow \lambda_j = 0$$

$$\lambda_j > 0 \Rightarrow v_{c,\perp,j} = 0$$

约束力不能做功

Contact as a Linear Complementary Problem



$$\begin{bmatrix} m\mathrm{I}_3 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \dot{v} \\ \dot{\omega} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega \times I\omega \end{bmatrix} = \begin{bmatrix} f \\ \tau \end{bmatrix} + J_{c,\perp}^T \lambda$$

$$v_{c,\perp} = J_{c,\perp} \begin{bmatrix} v \\ \omega \end{bmatrix} \geq 0$$

$$\lambda \geq 0$$

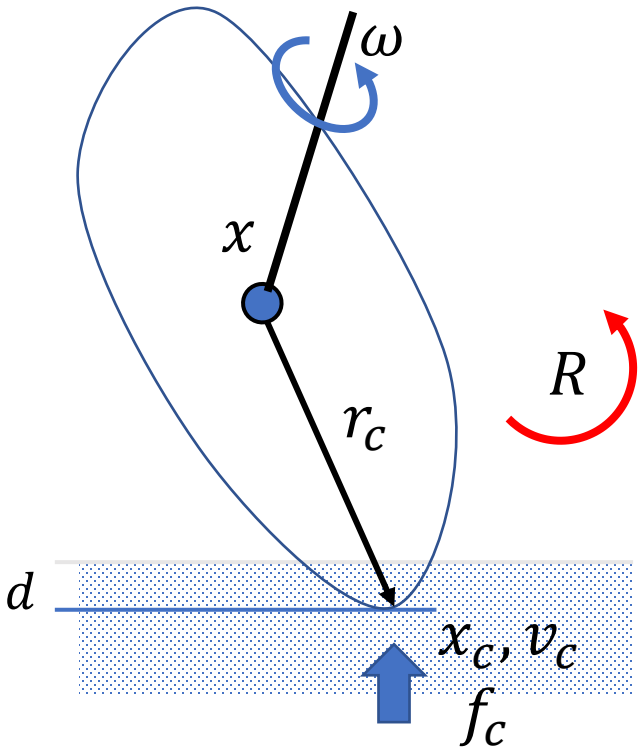
$$v_{c,\perp} \perp \lambda = 0 \quad \Longleftrightarrow \quad v_{c,\perp}^T \lambda = 0$$

(Mixed) Linear Complementary Problem (LCP)

To solve an LCP:
e.g. Lemke's algorithm – a simplex algorithm
Projected Gauss-Seidel method

LCP常用于建模刚体碰撞的问题

How to handle friction?



Contact and Friction Simulation for Computer Graphics

SHELDON ANDREWS, École de technologie supérieure, Canada
KENNY ERLEBEN, University of Copenhagen, Denmark

Efficient simulation of contact is of interest for numerous physics-based animation applications. For instance, virtual reality training, video games, rapid digital prototyping, and robotics simulation are all examples of applications that involve contact modeling and simulation. However, despite its extensive use in modern computer graphics, contact simulation remains one of the most challenging problems in physics-based animation.

This course covers fundamental topics on the nature of contact modeling and simulation for computer graphics. Specifically, we provide mathematical details about formulating contact as a complementarity problem in rigid body and soft body animations. We briefly cover several approaches for contact generation using discrete collision detection. Then, we present a range of numerical techniques for solving the associated LCPs and NCPs. The advantages and disadvantages of each technique are further discussed in a practical manner, and best practices for implementation are discussed. Finally, we conclude the course with several advanced topics, such as anisotropic friction modeling and proximal operators. Programming examples are provided on the course website to accompany the course notes.

CCS Concepts: • **Computing methodologies** → **Simulation by animation**; **Physical simulation**; *Real-time simulation*; • **Applied computing** → *Physics*.

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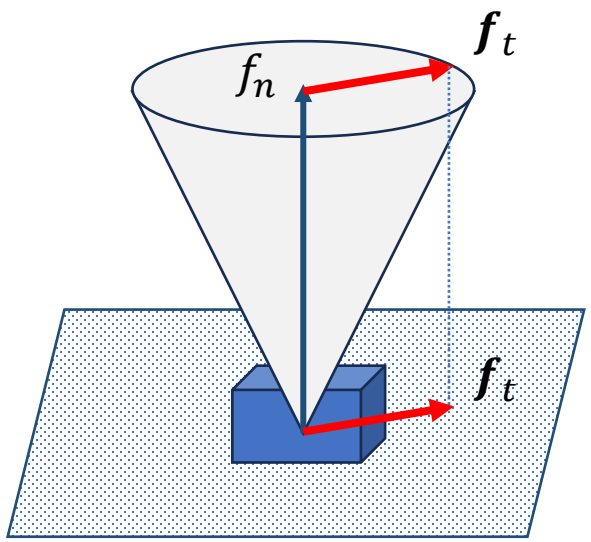
https://doi.org/10.1145/3450508.3464571

1

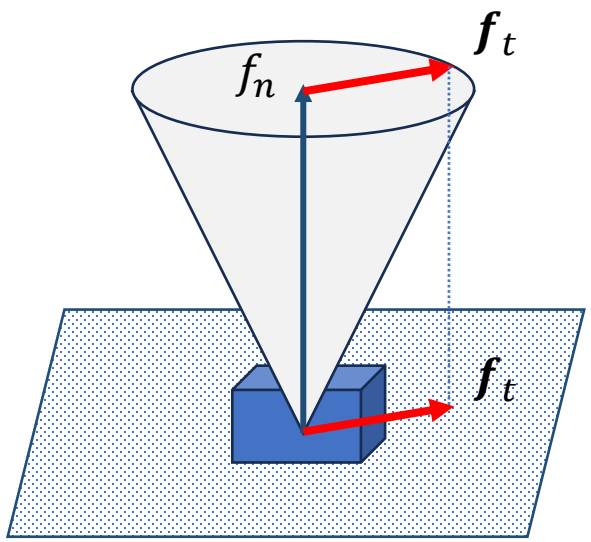
Friction Cones

Coulomb's law of friction: $|\boldsymbol{f}_t| \leq \mu f_n$

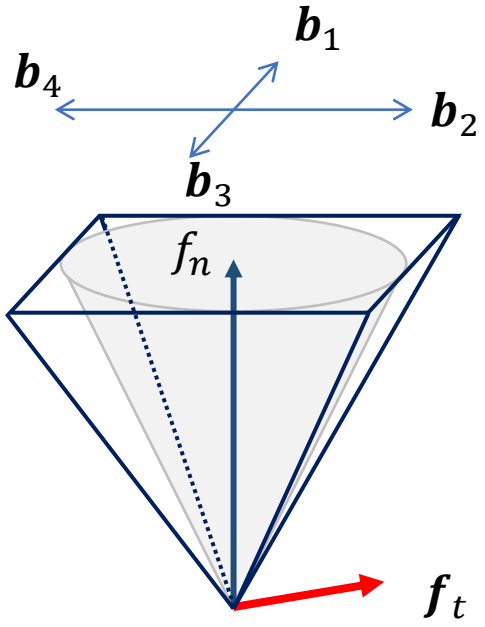


Friction Cones

Coulomb's law of friction: $|\boldsymbol{f}_t| \leq \mu f_n$

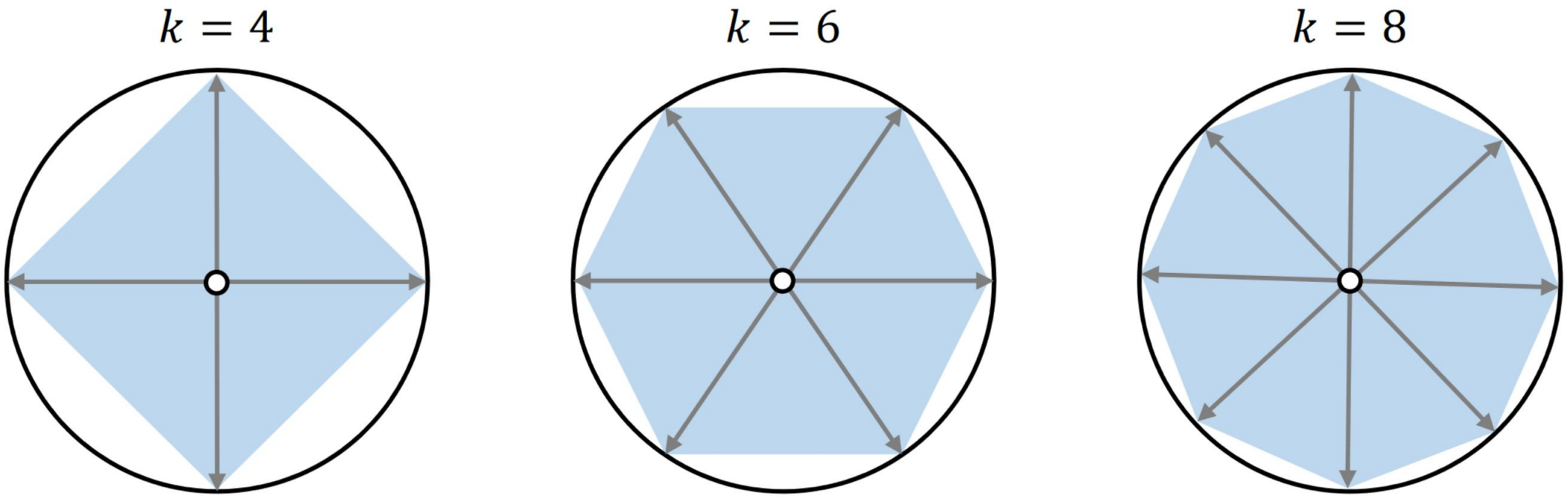


Linearize



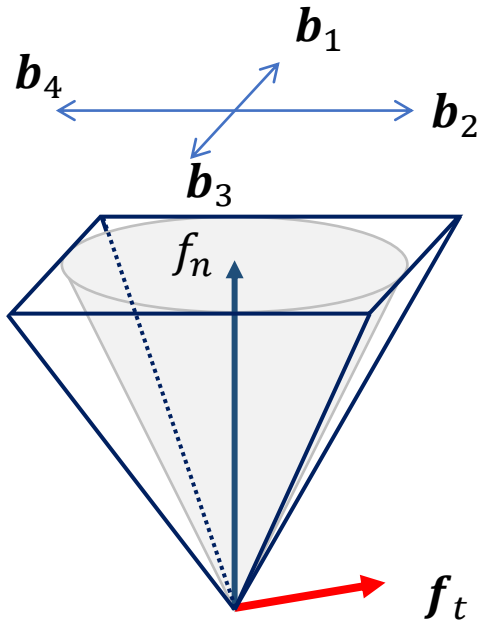
$$\boldsymbol{f}_t = \sum_k \lambda_k \boldsymbol{b}_k \qquad \lambda_k \geq 0$$

Friction Cones



$$\boldsymbol{f}_t = \sum_k \lambda_k \boldsymbol{b}_k \qquad \lambda_k \geq 0$$

Friction Cones



$$\boldsymbol{f}_t = \sum_k \lambda_k \boldsymbol{b}_k \qquad \lambda_k \geq 0$$

Coulomb's law of friction:

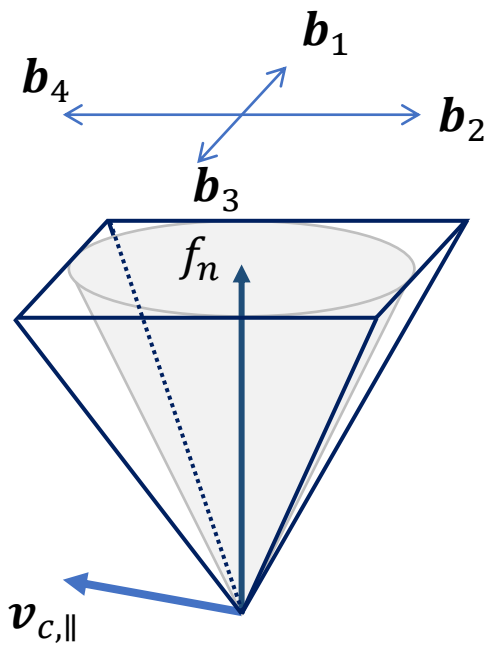
$$|\boldsymbol{f}_t| \leq \mu f_n$$

$$0 \leq \mu\lambda_n - \sum_k \lambda_k$$

Note: $\mu\lambda_n - \sum_k \lambda_k > 0$ if and only if the contact is not sliding

How to formulate this?

Sliding



$$\beta \boldsymbol{e} + \bar{\boldsymbol{v}}_c \geq \boldsymbol{0}$$

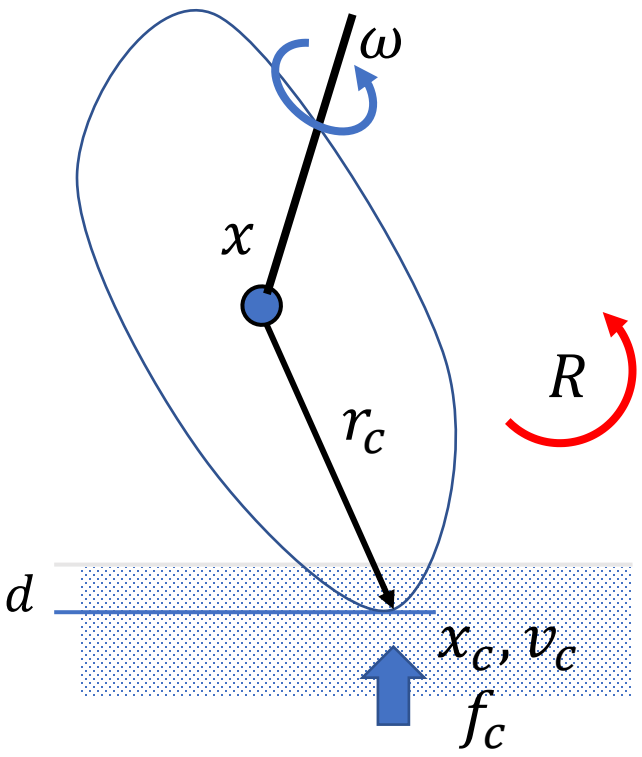
$$\beta \geq \boldsymbol{0}$$

$$\bar{\boldsymbol{v}}_c = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix}, \qquad \text{where } v_k = \boldsymbol{v}_{c,\parallel} \cdot \boldsymbol{b}_k$$

$$\boldsymbol{e} = [1,1,1,1,\ldots]^T$$

$\beta > \boldsymbol{0}$ if and only if the contact is sliding

Put it all together



$$\begin{bmatrix} m\mathrm{I}_3 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \dot{v} \\ \dot{\omega} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega \times I\omega \end{bmatrix} = \begin{bmatrix} f \\ \tau \end{bmatrix} + J_{c,\perp}^T \lambda_n + J_{c,k}^T \lambda_k$$

$$0 \leq v_{c,\perp} \perp \lambda_n \geq 0$$

$$0 \leq (\beta \boldsymbol{e} + \bar{\boldsymbol{v}}_c) \perp \lambda_k \geq 0$$

$$0 \leq \left(\mu \lambda_n - \sum_k \lambda_k \right) \perp \beta \geq 0$$

$$0 \leq (A\boldsymbol{x} + b) \perp \boldsymbol{x} \geq 0 \qquad \boldsymbol{x} = (\lambda_n, \lambda_k, \beta)$$

Gauss-Seidel Method

Solve linear problem:

$$A\boldsymbol{x} = \boldsymbol{b}$$

- A is symmetric positive-definite, **or**
- A is strictly or irreducibly diagonally dominant

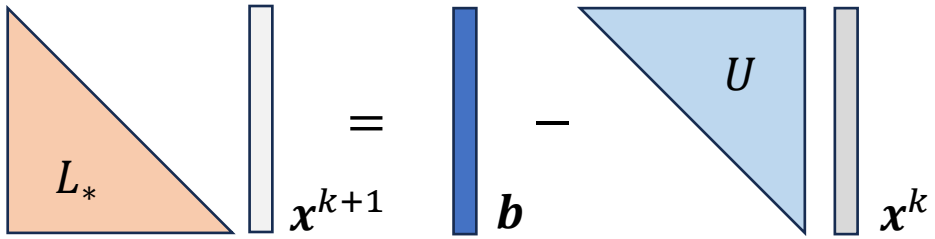
$$|a_{ii}| \geq \sum_{i \neq j} |a_{ij}|$$

$$A = \begin{bmatrix} a_{11} & a_{22} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} + \begin{bmatrix} 0 & a_{22} & \cdots & a_{1n} \\ 0 & 0 & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} = L_* + U$$

Start from an initial guess $\boldsymbol{x}^0$

Then, iteratively solve:

$$L_* \boldsymbol{x}^{k+1} = \boldsymbol{b} - U\boldsymbol{x}^k$$



Projected Gauss-Seidel Method

Solve linear problem: $0 \leq (A\boldsymbol{x} + \boldsymbol{b}) \perp \boldsymbol{x} \geq 0$

$$A\boldsymbol{x} + \boldsymbol{b} \geq \boldsymbol{0} \qquad A = L_* + U$$

$$\boldsymbol{x} \geq \boldsymbol{0}$$

$$\boldsymbol{x}^{\boldsymbol{T}}(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}) = \boldsymbol{0}$$

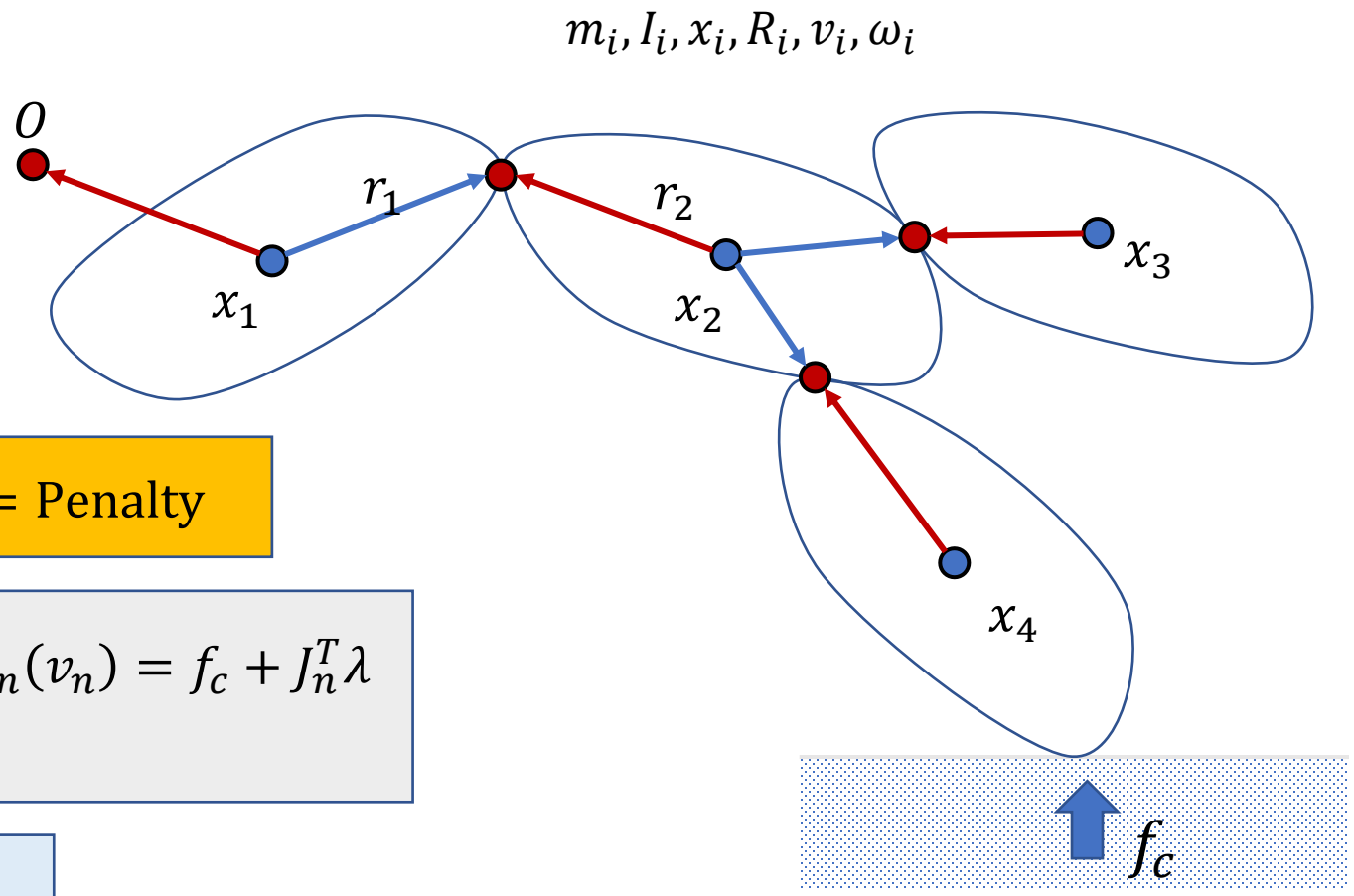
Start from an initial guess $\boldsymbol{x}^0 \geq 0$

Then, iteratively solve:

$$L_* \boldsymbol{z}^k = -\boldsymbol{b} - U\boldsymbol{x}^k$$

$$\boldsymbol{x}^{k+1} = \max\left(0, \boldsymbol{z}^k\right)$$

Simulation of a Rigid Body System



$I_n = R_n I_0 R_n^T$

$f_c = \text{Penalty}$

$$M_n(v_{n+1} - v_n)/_h + C_n(v_n) = f_c + J_n^T \lambda$$

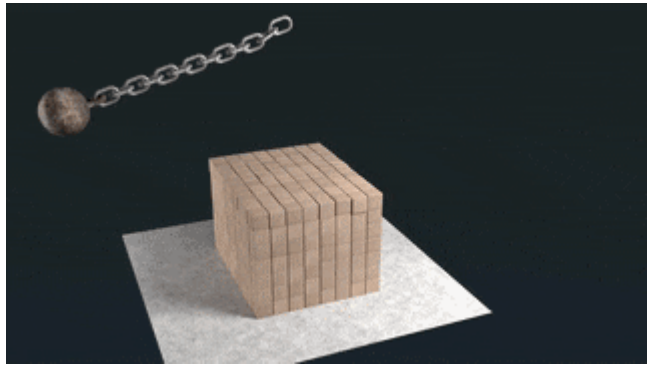
$$J_n v_{n+1} = c_n$$

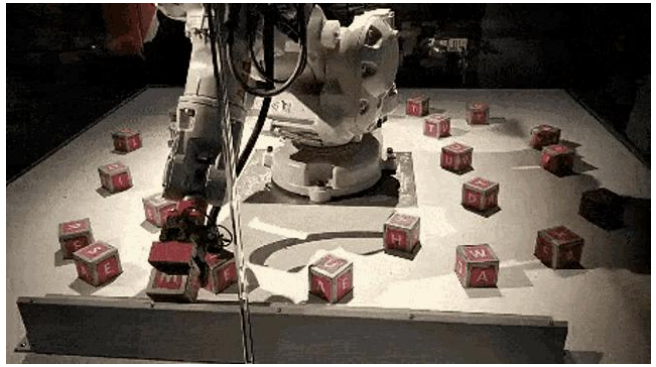
$$x_{n+1} = x_n + h v_{n+1}$$

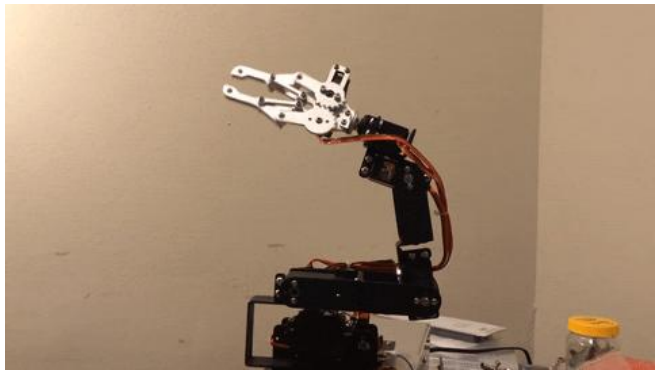
$$q_{n+1} = q_n + \frac{h}{2} \bar{\omega}_{n+1} q$$

Outline

- Simulation Basis
  - Numerical Integration: Euler methods
- Equations of Rigid Bodies
  - Rigid Body Kinematics
  - Newton-Euler equations
- Articulated Rigid Bodies
  - Joints and constraints
- Contact Models
  - Penalty-based contact
  - Constraint-based contact







https://www.cs.cmu.edu/~baraff/sigcourse/

Questions?